

Mus 250 Introduction to Music Composition

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MUS 250 • Samuel Andreyev

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Lecture 1: Foundations of Composition

At a glance — Foundations of composition.

- 1.1 It's a unique privilege to be here today, particularly for me, because I get a chance to talk about my life's work with you. I'm a composer. My name is Samuel Andreyev, and this is the central pursuit of my life. It has been for 35 years now. What we're going to talk about today is what I consider to be the most essential tools, techniques that are necessary to come to grips with the art of musical composition.
- 1.2 Musical composition is, to say the least, a complex art form. It's one that brings together a very large number of different skills. It's often considered to be somewhat daunting. People who are approaching composition for the first time have a sensation of vertigo. It's like, "Where do you even begin to write a piece of music? How would you break it down? What are the component skills that go into that?"
- 1.3 If you look at the way composition is typically taught, it very often comes at the end of a lengthy set of preliminary studies. Very often, composers start by learning an instrument to maybe a semi-professional, an amateur or a professional level, depending. They'll typically do a whole series of technical studies involving subjects such as harmony and counterpoint, ear training, and so on.
- 1.4 They'll often learn how to imitate historical styles, especially in continental Europe; that's a process that's quite common. A teacher will give you a piece by Mozart or a piece by Brahms, and they'll explain how it works, what the syntax is, how the piece is put together, what are the stylistic features, what are the technical features. You learn those things, you pull them apart, and you imitate them.
- 1.5 So it's a complex art form with a broad range of skills that tend to be applied towards it. But I also have to acknowledge that it's one that is under constant transformation. The way composition is taught today in 2025 is very different, profoundly different from the way it was taught even 50 years ago, let alone the way it was taught in the time of Bach, or the time of Mozart, or the time of Beethoven.
- 1.6 One of the things that any composition professor, including myself, has to deal with is the fact that students now have an enormously broad range of musical interests, and that's a new phenomenon. That wasn't the case 100 years ago. You were very unlikely to be exposed to all of world music. You were very unlikely to have whatever it is, 50 million pieces of music in your pocket. You didn't have access to that. Even if you did have access to it, you probably wouldn't be encountering it on a daily basis in your life. Whereas today, you can take nothing for granted, stylistically, technically, aesthetically. A composition student could be interested in absolutely anything.
- 1.7 My own origins are a testimony to that. Although I live in France and I'm now a French citizen, I grew up in Canada, and my exposure to music at a young age was marked by

randomness, something like that. So it was basically that things would enter your sphere and you would be interested in them and maybe pursue them and try to understand them a bit better. But there was no structure to it; it was haphazard. It was like you'd hear something interesting on the radio, or my sisters—they were a little bit older than me, and they grew up in the 1980s—they were really interested in British music from the '80s, pop music, so there was a lot of Thompson Twins, stuff like that in the house. My parents were interested in classical music, so there were some of that. It was any random thing.

- 1.8 That's increasingly the case. There's nothing you can really take for granted anymore about somebody's musical background or their technical level or anything like that. What you start with, essentially, is a desire to compose. Very simply, that. That has to be the fundamental motor if you're going to learn to study composition. So how did I manage to acquire something like a workable composition technique despite the haphazard nature of my musical education and the influences that I had when I was growing up?
- 1.9 Well, a lot of it was, if you're drawn to something, if you're interested in it, then you want to investigate it, naturally. You want to figure out how it's put together. You want to figure out what makes it work. I would do that with whatever it was, whatever caught my attention. Through a long process of doing that and continuously expanding the range of things that I was listening to, and very deliberately so, then gradually you start to notice, "I'm gravitating a little bit more towards this than that. This is where my interests really lie. This is a little bit more peripheral."
- 1.10 Then with time, you refine that, and you start to see some commonalities. That's the point at which I started to get really serious. By the time I was 17 or 18, it was very clear that I was going to be a composer and what sorts of music were of most interest to me. Then what happens is you start to gradually develop a clearer idea of what you're really interested in, what you want to do yourself, and what you need in order to do that.
- 1.11 The reason I'm telling this autobiographical preamble, if you like, is that musical technique, composition technique, has to be applied to an expressive project. There are some things in music that are elementary, that are rudiments that I think every musician should learn. But there are other aspects of technique that are always going to be deeply personal and that are going to have to reflect what your interests are. That's a process. That's something that you will discover as you go through this journey of writing your own pieces. Some things will not be relevant to you. Other things will be absolutely essential, and that will gradually become clear.
- 1.12 There are amazing stories of how people have managed to write pieces of music without having any standard musical education. The most successful composer of the 20th century is probably Paul McCartney, and he can't read or write music. I find that absolutely extraordinary. But of course, it's possible. Other people will study

counterpoint, harmony, form, analysis, orchestration for 20 years and never have a single original thought. So it all depends.

- 1.13** But what I can do today, what I can do, is to lay out what the terrain looks like so you can have a sense of, "Okay, this is what goes into making a piece. This is how you can make a piece out of almost nothing, and here are the steps." One of the things that I really want to emphasize in this course is: remember, we said earlier that people are daunted by the prospect of writing a piece; they don't know where to begin. They don't know what the first step is.
- 1.14** Well, the thing is, you can actually make a piece of music with almost nothing. This is one of the central pillars, if you like, of my composition teaching. This is something I've discovered through fifteen years now of teaching composition daily— I've had hundreds and hundreds of students from all over the world—and the most common problem by far is: how do you keep it going? It's rare to find a student that has no ideas and can't put them down on paper. Almost all students that I've worked with have the opposite problem: they have a profusion of ideas, but they don't know what to do with them; that's a much bigger problem than not having ideas.
- 1.15** So I like to look at that question from the other way around, which is, voluntarily imposing constraints, incredibly tight constraints, so that the range of what you can do is severely curtailed. That's not out of any sadistic impulses on my part. It's not out of any desire to limit creativity, to limit spontaneity or anything like that. But it comes out of a recognition that in any creative pursuit, constraints are necessary; they're vital.
- 1.16** What's very often happened with my students—if they're having a major problem composing a piece of music, if they're stuck, they don't know what to do, they've got 15 seconds of something really interesting, but what happens in 16 seconds? What happens in 20 seconds? What happens 30 seconds into the piece? Very often, what's happening is they don't have enough constraints to work with. Their range of options is far too broad, and that's paralyzing. If you are faced with the totality of all possible choices at every moment, it's impossible to make a decision.
- 1.17** You may have heard there's a parable about a farmer who gives two bales of hay to a goat and places them equidistant to the animal, and the animal starves to death because there's no rational basis upon which to choose left or right. So one of the things you do is you deliberately restrict the range of options as tightly as possible. That's extraordinarily effective, and we're going to look at some examples of how composers have done that in the past.
- 1.18** A lot of people, when they come to a creative discipline for the first time, they find that basic principle very counterintuitive because we're constantly told, "Creativity is everything. Creativity is the natural, unimpeded expression of your subjectivity. Whatever ideas go through your head while you're working, they come from you.

They're the spontaneous manifestation of your emotions, of your interior state, and that's valid."

- 1.19** We're also told that everybody's creative. The truth is somewhat more nuanced. Creative people, again, I have to emphasize this point: creative people work under intense constraints, intense constraints, and without those constraints, they wouldn't be able to produce anything. Some of those constraints are self-imposed, and some of them come from the outside.
- 1.20** An external constraint would be something like, imagine you're asked to write a piece of music and it has to fit a certain format. If you're in commercial music or if you're writing a pop song, then the format is very likely dictated by either physical media in the past, by the limitations of a seven-inch record, or the limitations of a radio format: a single can be between two and a half minutes and three and a half minutes long. That's a tight constraint: you can't write anything longer than three and a half minutes. That's not a constraint that songwriters necessarily choose, but that's how it goes.
- 1.21** Then there are constraints of instrumentation. What instruments do you have available? What do you know how to play? Who's going to be listening to it? Under what circumstances? What size of room will it be played in? All of these questions are the external constraints that composers have to work within.
- 1.22** Others include things like genre and form, each of which have their own set of expectations associated with them. For example, I'll bring up the idea of writing a song again—there are certain things that almost never happen in songs, and there are certain things that almost always happen. It's very likely that if you're writing a song, you're going to alternate verses and choruses. It's also very likely that you'll have a little middle break at some point, and you may have an introduction, and you may have an outro, as it's called, or a concluding section or a coda, but there's a basic archetypal form in almost all pop songs that is adhered to very, very, very broadly. So some of those things are external, some of them are unconscious—we don't necessarily think of them as constraints, but they're there.
- 1.23** Then there are the internal constraints that are self-imposed, the ones that don't come from without, but that come from within. That might have to do with something like deciding that you want to focus on this particular style. This is the piece you want to write, and in order to make a piece in that style that's coherent, that is structured, that can communicate to other people, then you need to make certain decisions about how that can go.
- 1.24** Now, in terms of just absolutely basic practical considerations: How do you start? How do you begin with a piece? What I always tell my students is, "Write really short pieces to begin with." By really short, I mean ridiculously short, like 30 seconds. Stick with 30 seconds to start with and see what you can do with that. The reason you do that is

because that takes away the whole problem of form. You don't really have to deal with form if you're writing a 30-second piece.

- 1.25** You do a little bit because even within a short space like that, 30 seconds, you can still have different elements, and you can still arrange them, and you can still have phenomena of contrast and so on. Just as I did when I was a student in Paris. I had a series of incredibly small flats that I lived in. Even if you've got, I don't know, 15 square meters to live in, you can still arrange it. It doesn't 100% take away the question of form, but nevertheless, it severely curtails your options, let's say. There's that.
- 1.26** Then I would impose additional constraints upon them. I would say things like, "Okay, so you're going to do a 30-second piece, but it has to be for piano, and you can't write chords. It's got to be a single line. No harmony, no counterpoint. Just write a 30-second line for piano." They'd think, "Oh, okay, I'll try that." You might even take it further. You might then say, "The 30-second line has to fit within the space of one octave." It's like, "What can I do now? I have almost no options left." That's the point, because when you have almost no options, that's when your personality, your artistic personality, starts to reveal itself.
- 1.27** The other thing about that is, no matter how ridiculously constraining or how technical the exercise might seem, every exercise you do is an opportunity to write something expressive. Anytime you write anything in music, there has to be an expressive intention behind it. I was reading a couple of weeks ago an article. It was an interview with the son of the writer, Hunter S. Thompson, an American writer, journalist. He was an interesting character. He has this reputation as being a very chaotic person, very out of control, and he was out of control in his personal life, but in terms of the attitude he took towards his writing, he was ferociously disciplined. Very interesting.
- 1.28** People are strange that way. It's not because you live a Bohemian lifestyle that the work that you do is equally undisciplined. The son of Hunter S. Thompson told an interesting anecdote about him. He said that above his work desk was a sign that he always kept in his room where he wrote. The sign said, "What is the intended effect?" That was his goalpost, the fundamental thing that oriented him when he was working: the effect.
- 1.29** I find that so interesting because we forget that when we're learning a complicated task, like writing a piece of music. It's very easy, especially if you study composition academically; it's very easy to get wrapped up in technique. Then the technique can easily become an end unto itself. You start thinking in terms of it.
- 1.30** A lot of composers that I work with, initially, are very wrapped up with the material they're working on. They get really obsessive about finding an interesting chord or writing an interesting line or finding a new sound or a new texture or a new timbre on an instrument. That's what I would call the material dimension of music. You need material to work. You need to turn it into something and make something out of it. But

my argument is that the material is not very important. It's not that it has no importance, but its importance is dwarfed in comparison to your handling of the material.

- 1.31 That's where the Hunter S. Thompson thing about "what is the intended effect" comes into it. Because no matter what you're starting with, you have to have an expressive intention, and you have to think in terms of an effect. Music is a psychotropic drug. It affects people. It has an effect on the world. It has an effect on listeners. You can be the architect of that effect. You can think in terms of—even if you're writing a little 30-second piece, you can think, "What's the effect? Is it going to be hypnotizing, relaxing, energizing, strongly kinetic? Is it going to be disorienting? Is it going to be strange? Is it going to be comforting? You can do that in 30 seconds in a piece of music.
- 1.32 In fact, there are songs that you'll hear on the radio that can create that effect in two seconds. I think we've all had this experience. Maybe there's a song you particularly like—two seconds into it, instantly you're in a different state; it's amazingly powerful. That doesn't happen by accident. It comes from the result of having an expressive intention. We're going to look at how that works.
- 1.33 I said that material has its importance, but the way you handle the material is far more important. I'm going to show you a couple of images that illustrate this idea. There we have two paintings of apples. Are those two paintings the same? They don't look that similar. The subject of the paintings, you could say, is the apple. "I'm going to make a painting of an apple." But those two paintings couldn't possibly be more different, not only visually, not only technically, but also in terms of the effect they create, the feeling that they generate. They're profoundly different.
- 1.34 The apple is a subject that has been treated by hundreds, if not thousands, of painters in an insanely broad variety of different ways. The point being that the apple is a pretext. The paintings are not really about apples. The point being that yes, they're both paintings of apples, but the effect that they create is radically different. That's, again, because the painter is going at it with the intent to create an effect.
- 1.35 It's exactly the same thing in composition. The materials of music, the chords, the rhythms, the melodies, textures, and so on, yes, they have their importance. But we're going to look at an example of how you can take something very simple, like the example of the apple, and turn it into something a little bit more interesting.
- 1.36 What I'd like to look at with you is a piece of music by a 17th-century German composer called Johann Pachelbel. You've probably heard the famous Pachelbel's Canon, which is a piece that is played thousands of times a day in every supermarket in the world. It's a very successful little piece of hold music, and it's become a shorthand for the Baroque for a lot of people through a series of historical accidents, essentially. It's a very little piece that was discovered in his papers, more or less by accident, by

musicologists. Music history is a strange thing. This piece had been completely forgotten for centuries, and now everybody's heard it basically on the entire planet.

- 1.37** Anyway, Pachelbel actually was a very important figure in his time. He was one of the main exponents of the Southern German school of organ playing and composition. At that time— this is in the late 1600s—there were two essential approaches: the Northern German school, which was exemplified by figures such as Dietrich Buxtehude, which exemplified complexity, subtlety, and strangeness of effect, and the Southern German school, which exemplified clarity, precision, symmetry, simplicity.
- 1.38** You can think of these as our archetypal attitudes that still exist today amongst creative people. It's partly a function of personality; it's partly a function of what your innate aesthetic interests are, how you see the world. Do you see the world as a place of boundless complexity and mystery and strangeness, or do you see it as a place where you'd like to impose order?
- 1.39** Pachelbel wrote a piece called Hexachordum Apollinis, which is a recondite title. I'll explain what that means: the title means The Six Strings of Apollo. What it consists of is a sequence of short pieces for organ. Each one of them is based on an aria or a short melody, followed by a set of variations, six variations for every piece. So there are six arias, and each aria is varied in six different ways or presented in six different ways.
- 1.40** So I think this is a good entry point into composition because, in some sense, these pieces are very simple. They're based on almost nothing. But it shows you how you can take a very, very, very simple idea and build something out of it. These pieces were written in 1699, and they still have a lot to teach us today. When I say that they're based on something simple, each of these pieces is based, actually, not on a melodic idea, per se, but actually on what is called a ground base, a bass line.
- 1.41** In some cases, a composer working at this time would have dealt with the existing body of very familiar bass lines. In the same sense as today, if you're writing a song, there's a certain number of set chord progressions that basically everybody knows and repeats. It's like public property in a sense. It doesn't really belong to you, but you can use that as a starting point.
- 1.42** So Pachelbel might have invented some of these bass lines himself, or they might have belonged to the store of bass lines that were familiar to every musician in his day. In a sense, it doesn't really matter whether he wrote them or not. The point is: What does he do with them?
- 1.43** We're going to start with one of these pieces. I'm going to write out the bass line. I'll play it for you, and then we'll talk about what he does with it. The little C here, for those who might not be familiar with this symbol, means common time, and common time is 4/4. The bass line looks like this [plays the bass line]. It's an eight-bar phrase, and it's completely symmetrical. It consists of two little phrases, each of which can be divided into two subphrases: it's two plus two plus two plus two, a very simple little

construction. Each phrase ends with what we call a cadence. A cadence simply means a close. It's a pause.

- 1.44** Music is mapped onto speech to a very large extent. A lot of the things that we're going to be talking about—phrase, cadence, and so on—these also apply to speech; they are extensions of it. Just as you would take a pause after a sentence in everyday speech, you would do that as well in a musical line. Let's look at how these little phrases line up [plays the first two bars]. That's the first two bars. It comes to a little pause, and then we start again [plays the next two bars]. That's the end of the first half.
- 1.45** If you look at the sheet music there on the screen, you'll see that there's what's called a double bar line at the end of the first system with two little dots there. That means that you then turn around and repeat the whole thing again. This is a very, very common structure in Baroque music, in classical music, and actually also in a lot of contemporary music as well. You repeat what you've just said in order to emphasize it and make it clearer.
- 1.46** A lot of what musical composition consists of is disguised repetition. You have to have enough redundancy in the material so that what you are saying can be comprehended. Too much redundancy without transformation becomes monotonous—that's where the disguised part comes into it. But whatever you're doing in a piece of music, there have to be some relatively stable aspects of it that the listener can hang on to so they don't feel helplessly adrift and confused when they're listening.
- 1.47** Let's hear that phrase again, just the first four bars [plays the four-bar phrase]. That's how it comes to a rest. Then he starts again. This is the second half [plays the next phrase]. You can also think of this as a rhyming couplet. The phrases seem to rhyme with each other. That's a very, very old, archetypal way of structuring a musical phrase. We're going to talk about that a little bit in the next course when I delve into the specifics of how melodies are put together.
- 1.48** But for now, let's look at what Pachelbel does with this. That's the interesting question. Here's the opening aria, which is based upon, again, not a melody, but on that bass line. He uses that to create a harmony and then a little line in the right-hand on top. Here's what it sounds like [plays the opening aria]. All of that is based on that simple bass progression that, again, he may or may not have written himself.
- 1.49** Let's pull this apart and look at what he actually does here. One of the first things you want to pay attention to—we talked about the importance of creating an effect intentionally when you're making a piece of music. You look at the rhythmic aspect. When music is taught in the Western world, nearly all the time, the emphasis is on pitch, on melody, on harmony. But the thing that I think sets a character faster than anything else, and that affects the listener most viscerally, is rhythm.
- 1.50** Before you even have a chance to process the rhythm—the melody, you're going to notice the rhythmic character, the rhythmic movement of a piece. This is why we say

that classical pieces are written in movements. The movement means physical movement, embodied movement. That can be quick, it can be slow, it can be walking pace. You can be skipping. You can be running. You can be riding a horse. These are all physical embodied rhythms that form the basis for all the rhythms that you hear in pieces of classical music.

- 1.51** There's not that many of them—I mentioned this a little bit in one of my other courses here: The History of Western Music—the point being that different musical cultures have different points of emphasis. Our rhythmic tradition is relatively simple compared to lots of other parts of the world and the rhythms that they've elaborated. In the case of this piece, it's a march. What is a march? A march is a rhythm in 4/4, so 4 beats per bar: 1, 2, 3, 4.
- 1.52** Why is it in 4 beats and not 3 or 5? Well, it's a march; it has to do with walking, fundamentally. Humans have bilateral symmetry. Left, right, left, right gives you one, two, three, four. If we had some completely different physical morphology, God knows what rhythms we'd have, but that's it. So one of the first things you notice is the regular rhythm of the bass [plays the bass line]. That gradual, continuous rhythm of quarter notes gives us a feeling of forward momentum that is akin to walking.
- 1.53** In the top, we have this melodic line [plays the melodic line]. Something to notice about that straight away is that a lot of what you're hearing there are passing tones or figuration. I'll explain what that means. If we were to boil that line down to its simplest possible presentation, we might have something like this [plays the simplified melodic line]. There's a skeleton in there. In order to make it flow, Pachelbel adds notes that connect between the main notes in the melody [plays the melodic line].
- 1.54** Some of that is figuration, and some of that is ornamentation. If you have something like a trill [plays a trill], that's an ornament. It's not necessarily a structural part of the melody itself, but it makes it more pleasing to the ear. If we take that phrase there, just take the bass line and then the top line, let's see what he does [plays the bass and melodic lines together]. What he's done is he's just added a little bit of figuration in the top. Because of the fact that the notes are very close together—that's what we call conjunct motion; there'll be a little bit more about that later in the course—it's very easy to sing.
- 1.55** There's a crucial distinction that any musician has to make between vocal music or music that has its roots in vocality in the voice and instrumental music. They're not the same. On an instrument, it's possible to play figurations that are impossible to sing. For example, you can have rapid arpeggios, but you can't sing that. It's impossible. Likewise, the nature of the human voice makes it amenable to melodies that are primarily conjunct, where the notes are closely spaced. They're adjacent to each other with occasional leaps to make it interesting.

- 1.56** Let's hear the entire thing because I only played the top voice there and the bottom. Let's hear the whole thing [plays the opening aria]. This is a fairly simple texture of accompanied melody for the most part. There's a clear melodic line that's singable in the right-hand [sings the melodic line]. You can sing it fairly easily. Then you have two voices on the lower staff that fill out the harmony and give us a little bit of figuration. They also give us a nice bass line.
- 1.57** That's the aria. That's what he starts with. It's very short. Remember, I said you can write a 30-second piece as long as you have expressive intention and you know what you're aiming for. That's what he starts with. Pretty simple.
- 1.58** Now, let's look at some of the variations. This is where we come to the parameters of musical composition. I mentioned right at the beginning that this is a complex art form. There's a lot of different skills that go into it, and there's a lot of different dimensions to a piece of music. Looking at this set of variations is a very good opportunity to see exactly how that works in practice.
- 1.59** The theme in variations was a form that became popular in the Baroque and also in the Classical era. It's built upon older forms, such as the Passacaglia. The basic thing to retain from that is that these are cyclical forms where the same thing keeps returning over and over again. In order to avoid monotony, the thing, whatever it is—it can be a melody, it can be a chord progression, it can be a bass line— is varied each time. It's presented in a different way. Let's look at what he does in his first variation [plays the first variation].
- 1.60** Okay, so fundamentally, he hasn't invented anything new in this first variation. It's the same harmony, it's the same bass line, and it's essentially the same melody. The only thing that's changed is that the figuration is more elaborate. Remember, in the opening aria [plays the melodic line of the opening aria], and this is then transformed into something a little bit more verbose [plays the embellished melodic line]. What he's doing is he's taking the same skeleton, the same basic sequence of tones, and elaborating on them [plays the first variation]. The rhythm is noticeably more fluent in this variation, and he moves from a texture mostly in quarter notes to a texture in 16th notes, a faster rhythmic movement.
- 1.61** Let's look at variation number two and see how this continues [plays the second variation]. Notice that here, the quarter-note rhythm that we had in the bass line previously has been flipped around. It's migrated into the right hand. The right hand is doing this [plays the bass line with his right hand]. That's what the left hand had been doing previously in terms of texture and rhythm [plays the bass line with his left hand]. So he flipped it around. The rapid figuration that we had in the right-hand in variation one is now in the left hand [plays rapid left-hand figuration], et cetera. So that gives this [plays the combined texture].

- 1.62** That's a simple case of permuting the elements a little bit. Instead of the figuration being in the right hand, it's in the left hand. Again, the scope of transformation here is not that great. It's really just finding a slightly different form, a slightly different presentation of essentially the same idea.
- 1.63** Let's look at variation number three [plays the third variation]. What does he do there? It's getting a little bit harder to recognize the melody in the right hand, partly because the figuration has now turned into something quite different [plays the melody line]. That's a skipping rhythm. We've moved away from a primarily march rhythm to this unequal rhythm, which is, again, a very different rhythmic character. And what the right hand is doing, especially at the beginning, is not exactly playing a line, but rather breaking up a chord [plays the broken-chord texture].
- 1.64** Now, those lines there could be represented this way, with the tones being played simultaneously rather than in alternation [plays the chords, then the broken-chord texture]. That's another type of figuration [plays the third variation]. The bass line is the same [plays the bass line]. He adds a little bit of ornamentation, a little bit of figuration in there, but essentially the same line.
- 1.65** By now, you're getting the idea that this whole book of pieces consists of just finding new ways to present the same idea over and over again. The success of a piece like that really lies with the composer's capacity to find sufficient variety in the material. That's where composition really gets interesting. That's where the skill comes into it. Finding the material itself is trivial; it's very easy to find. But what's not so simple is figuring out how to transform it, what you're going to do with it.
- 1.66** It continues. So variation four, we have yet another texture [plays the fourth variation]. There's basically no audible melody exactly anymore because the figuration has become more complex. Also, we have a principle of imitation between the two voices, where both the bass and the top line are equally elaborate. In the right hand, we have [plays the right-hand figuration]. And in the left hand, we have [plays the left-hand figuration].
- 1.67** What's happening is the voices are almost chasing each other. The left hand will play a rapid run like that, and then the right hand will imitate it and back and forth, and sometimes they'll play together, but it's continuously changing. That's one of the things that makes a texture interesting and dynamic is the fact that it's not exactly the same all the time. There's a liveliness and an unpredictability to it [plays the fourth variation]. You can hear this dialogue between the two hands, whereas in the previous pieces, it was pretty constant. Either the right hand was playing the melody, the left hand was playing the bass line, or you'd have a lot of figuration in the left hand, and you'd have chords in the right hand. But here, he starts to break that up a little bit.
- 1.68** In the following variation, we have something completely new once again [plays the fifth variation]. There, the texture has been reduced to just broken chords. There's a

latent melody in there. If you were to isolate the top notes that I play in my right hand and string them together without all that figuration, you would hear a continuous line. But the way it's presented is as a sequence of broken chords.

- 1.69** So you can present an idea in many, many, many different ways without ever needing a second idea. That's the amazing thing about composition: a lot of the time, what you're really doing is you're transforming an idea and extending it. Where a lot of composition students get tripped up is they think that they need a profusion of material. There'll be an idea and then another idea, and then another idea. But it's too many, and they aren't focusing on the transformation.
- 1.70** This is a nice example, I think, of how you can work within a very tight framework, very constricting, just an eight-bar baseline, and then focus on: What can I do with this? How can I present this? What are all the different facets that I can find? What's all the hidden potential in this line?
- 1.71** That's a little introduction to what I would call the parameters of musical composition. This is a multidimensional art form where you have to deal with rhythm, you have to deal with harmony: How are the chords constituted? How do they follow each other? How do you make a progression? They deal with melody, texture, dynamics, contrast, all of these essential aspects that make a piece interesting.
- 1.72** We're going to stop there for this part of the lecture. The second part is going to deal specifically with melody. How do you construct a melody? What goes into it? Before we stop for this part, are there any questions that anyone would like to ask about what we've looked at so far?
- 1.73** In the case of the Romantic, more late Romantic and modern era of composition, are we observing a disappearance of the skeleton for more of the orientation?
- 1.74** That's an interesting question. Not necessarily, but it depends on the composer. What you see in the late Romantic era is a dramatic expansion of the range of possibilities in music. That takes many, many forms. For one thing, the orchestra by that point had expanded into this gigantic dinosaur, sometimes featuring as many as 150 or 200 instruments. Forms had gotten more complex. Pieces had gotten a lot longer. There are Mahler symphonies that last an hour and a half, so a lot more than 30 seconds. So there was a dramatic expansion of the means of music that took place. Gradually, it was a process, but that took place over the course of the 19th century.
- 1.75** In the 20th century, it becomes exponential, partly due to the increasing sophistication and power of technology, the incredible range of different styles being practiced simultaneously, the ease of travel, the fact that people could now quite effortlessly go from country to country, hear all kinds of different musics from different parts of the world and create new styles and synthesis out of those different materials. It expands quite a lot.

- 1.76** But even in the Romantic era and the early modern era, you still see that there are different structural levels to a piece. There's something simple, maybe more abstract, a fundamental skeleton that underlies the piece. That has to do with basic formal considerations, such as the division into sections, the way the phrases are constituted. That's a pretty stable feature, I would say, throughout music history.
- 1.77** You can almost always take a piece of music and boil it down and look at what the constituent parts are, strip away the different layers, and see, "Here's what's happening on the level of the phrase, here's what's happening on the level of the section, here's how the movements are articulated, here's how this texture is constituted." You could say, "This is a contrapuntal texture, or this is a harmonic texture, or this is a polyphonic texture." You can boil it down, figure out what the constituent parts are, and start to understand how it's put together.
- 1.78** But as I said, as you get closer and closer to the present day, you have to cope with this dramatic expansion of possibility. In 1699, when that piece was written, the range of possibilities for a composer like Pachelbel, it wasn't nothing, but it was considerably more limited than it is today.
- 1.79** Yeah, thank you. My question is regarding when you were talking about repetition and variation. I wondered a little bit if you could explain a little bit more what you meant by redundancy in that context.
- 1.80** Okay, a very good question. Yeah, structural redundancy—that means that there's more repetition than is strictly needed in order to put your idea across. You can think of this in terms of other disciplines as well. It's also a concept in architecture. If you're designing a tall building, for example, you want there to be a certain degree of structural redundancy so that the building is stronger than it needs to be to withstand winds.
- 1.81** That's the case with the Empire State Building in New York City, for example. It's way stiffer than it needs to be. It's overbuilt. It's actually quite significantly overbuilt for a structure of that size. Engineering has gotten a little bit more sophisticated now with computers, of course, but the point is the same: it's stronger than it needs to be.
- 1.82** In a piece of music, you will repeat things more than is strictly necessary in order to make sure that the listener doesn't feel lost. Because music is difficult, not just in terms of writing it, but also listening. Listening properly to a piece of music makes incredible demands on you cognitively. There's a lot to pay attention to if you're really focusing on a piece. We are creatures of patterns. We need there to be a certain degree of consistency in what we're hearing in order to be able to apprehend the movement that is going on in the piece.
- 1.83** That can take many different forms. The rhythms might be repetitive. The melody might consist of many repetitions of a simple formula over and over again with slight variations. You might repeat entire sections of a piece, as was the case here, where

there's a double bar line. But if that repetition wasn't there, then, well, something else would have to carry the piece forward. But there has to be something that is permanent enough so that we can relate it to what came before and what might come next. We have a sense of trajectory, a sense of progression in the piece.

- 1.84 Now, there are some modern pieces that intentionally don't want to do that. They want to focus on the individual moment as it's occurring. That can be a very arresting experience where you're taken out of that continuum and into an eternal present. There are some composers that do that extraordinarily skillfully. But in a language like that one, there has to be a feeling of both continuity and forward movement, forward motion, that carries the piece along. That redundancy, if you like, is part of that.

Lecture 2: Building Melody

At a glance — Building melody.

- 2.1 So earlier, we were talking about the parameters of composition. We looked at how Johann Pachelbel takes a very simple little bass line and elaborates upon it, transforms it, and presents it in a variety of different ways, and how, really, you don't need anything at all. You can just start with something very, very elementary and spin a piece out of that.
- 2.2 We've looked at the different ways that you can present an idea. We talked about questions of texture, questions of harmony, of different ways of presenting a melody, figuration, rhythm, all of these different parameters. One of the fascinating things about composition is that you can isolate one of those parameters and choose to base your piece on that.
- 2.3 You can say, "In this piece, I'm going to focus on rhythm. I'm going to focus on expanding my rhythmic vocabulary." Or you can say, "I want to write an extraordinary, memorable melody, and that's going to be the focus in this piece." Or you can say, "I want to come up with a new type of chord progression that no one's ever heard before," or whatever it is. But you can isolate different aspects of musical composition and focus on them in a piece. Odds are, all of those parameters are not of equal significance in your piece. They all have to work together. They all have to be present to some degree, but they're not all going to be equally important in the piece.
- 2.4 Today, we're going to isolate one of those parameters and really get into detail about how it works. That parameter, in this case, is melody. In and of itself, the fact of sitting down and talking about melody for an hour or so is somewhat challenging. The reason is that when composition is taught, melody is not usually one of the subjects that's covered, which is strange in and of itself, because for a lot of people, that's the first thing that they relate to.
- 2.5 I mentioned that rhythm is primordial, rhythm is physical, rhythm strongly conditions our impression of the character of a piece, but melody is just as striking in its way. It's slightly different. You can think of melody as being an extension of speech, fundamentally. The patterns of melodies, the structures of melodies, are very similar to the patterns and structures of everyday speech. But strangely enough, we don't really have a theory of melody. It's not something that's really taught.
- 2.6 If you're studying composition, you're not typically going to get someone saying, "Here's how a melody is put together; here's how it's structured." What we do have a lot of is harmony teaching—that's very often one of the first things that beginning composition students will have to master, making simple chord progressions, building a chord on a bass line, or how do you combine multiple different voices to make a polyphonic or a contrapuntal texture in a piece of music. But the creation of a melody

itself is almost—I wouldn't say it's left up to chance, but it's not typically taught with the same degree of precision.

- 2.7** I'm not exactly sure why that is. It could be that it's much harder to codify or to describe what makes a melody interesting, maybe. But nevertheless, there are certain basic properties of melodic lines that you can learn, that you can come to grips with, and you can deploy them consciously when you're writing, and it can actually make it a little bit easier sometimes to invent a successful melody.
- 2.8** A lot of people will say, "Melody is something innate." It's like there's a gift for melody, and you either have it or you don't. Some people have a faculty for coming up with spectacularly beautiful melodies, and others don't. A very common assertion is that the composers of Italian opera, 19th-century Italian opera, such as Puccini, Rossini, Verdi, and so on, had that, and that it was partly a matter of the Italian language itself having something melodic about it, which is true. You can spend a few hours in Italy, and you'll very rapidly see that the quality of spoken Italian is incredibly melodious. There's no denying that.
- 2.9** Then they would compare that with someone like Beethoven and say, "Beethoven is not a great melodist." His themes are short, they're not very interesting, they're not particularly—they're memorable, but they're not singable tunes exactly in the same sense as what you might get in an aria by Puccini.
- 2.10** So there's this idea that melody is an inbuilt faculty. It's maybe a result of speaking a particular language or just having a facility with inventing melodies. Some of that is undoubtedly true. I think there are composers, musicians for whom melody comes more easily than others. But nevertheless, I'm going to lay out some basic principles and show you how melodies are structured.
- 2.11** In order to be comprehensible, there does have to be a structure of some kind. So I was thinking about this question of how to present this course, how to talk about melody. It's not an easy thing to talk about. I thought, "What's the simplest melody I can think of that has a structure?" The simplest melody I can think of is Mary Had a Little Lamb, so we're going to look at that. We're going to look at how that tune is put together, why it works, what its constituent parts are, and I'm going to use that as a jumping-off point for a little melodic analysis.
- 2.12** I'm going to write down the tune, and I'll play it, although I think probably everybody is familiar with it, and then we'll look at how it's put together. So first things first—this is a short, simple melody that actually has some common points with the Pachelbel bass line that we looked at earlier. In both cases, we have two phrases that can be subdivided into groups of two bars. A lot of melodic or thematic construction is built that way: two plus two plus two plus two. A lot of them are symmetrical: two bars, two bars. That makes a phrase of four. Then we have a little pause. Then we start again, and we have four more bars. Very, very simple structure.

- 2.13** One of the things that makes that structure as effective as it is is the fact that it maps onto some very simple linguistic structures [plays the first two phrases]. The first phrase [plays the first phrase] ends with a suspension, like a question. It comes to a pause. The second phrase answers the question [plays the second phrase]. We're going to look at how that actually works, specifically.
- 2.14** This type of construction, with two phrases that have a pause in the middle, is called antecedent-consequent. The antecedent is not a complete statement. It's a statement that implies a continuation or a resolution of some kind. In the same sense as if you were to say to somebody, "Is he the right man for the job?" Now, that invites a response. Then the second phrase might be, "Yes, let's hire him." That gives you the second part. That's how a lot of these tunes are constructed; they divide basically into two. Now, there's a lot of other features in this melody, simple as it is, that will tell us a lot about how these phrases are put together and how they relate to language.
- 2.15** Remember when we looked at the Pachelbel bass line, I pointed out that the phrases end with what's called a cadence. A cadence, again, is nothing other than a pause or a close in a phrase. Just like in spoken language, the pause can be a full stop, or it can be a partial stop. It can be a comma. It can be a break in a phrase, but not necessarily an end. There are different degrees of that.
- 2.16** It's subtle. We don't necessarily think about it consciously when we're speaking. But the way you end a phrase or the way you pause implies either a continuation of your idea or a summation of your idea, and you finished it and you're going to move on to something else now. It works very much the same way in melodic phrases. I want to emphasize this point again that melody is an extension of speech—historically, this is certainly the case—and an elaboration upon speech patterns.
- 2.17** The cadence in this case comes at the end of the fourth bar on the tone G [plays G]. Now, let's talk about range for a minute. One of the most archetypal fundamental properties of any melody is its range. This is very important because the human voice has a natural range to it, where almost anybody, if you figure out what tones can you sing with a reasonable degree of projection and reasonable force and clarity, what does that range correspond to?
- 2.18** You'll find that with most people, it's about an octave and a half, a solid octave and a half of usable vocal range for singing. Professional singers, of course, extend that, and they train for that, and they will extend their voice either upwards or downwards based on what their natural voice type is. In terms of singers, you have all these different voice types: soprano, mezzo-soprano, alto, tenor, countertenor, bass, baritone, and so on. There are other voice types as well, but those are the essential categories.
- 2.19** Those categories are partly physiologically determined; your voice will have a certain natural register that you can learn to emphasize. To a degree, you can learn to emphasize a particular part of your range and expand it a little bit at the edges and

develop a strong control of your chest voice or whatever it is. But by and large, most people will have about an octave and a half.

- 2.20** This melody is interesting because it's coherent and there's a clear structure to the phrase. It's easy to remember, but it's compressed into a very small space. The range of this melody covers a perfect fifth, just this space between middle C and the G, a fifth above it [plays C–G]. Everything that happens in the melody happens with these five notes [plays C, D, E, F, G, F, E, D, C]. That's it. That's a very, very limited range for a melody.
- 2.21** You can express that musically on a piece of staff paper by just writing the lowest note, which is C, and writing the highest note. That is the full extent of the range of the melody that you're writing. Other melodies will have a much broader range. If it's for an instrument as opposed to a voice, then the only limit is the range of your instrument. The piano has seven and one-third octaves, so there's a lot to choose from. You could conceivably write a single melodic line that takes you across the entire keyboard over seven octaves. You could do that. Nobody would be able to sing it, but an instrument, of course, can do it. But if you're writing for the human voice, of course, you have to deal with that basic limitation.
- 2.22** Why is the range of this melody so limited? Well, this is a children's song. Children's songs have to be simple. They have to be easy to remember. They have to be easy to sing, and they have to be mostly conjunct. That's a whole range of limitations. And despite all of that, they still have to be interesting or memorable enough so that you would want to sing them. How do you take such a limited space, only five notes, and make a line that has a shape, it has a direction, it has a structure, it's memorable, and it's interesting?
- 2.23** Well, we were talking earlier about how pieces of music have a skeleton, if you like. They have a basic structural armature that underpins everything in the piece. What you're listening to when you hear a finished piece of music is the flesh of the piece. You're not concerned with the skeleton. You're not concerned with the structure. You're hearing the fully clothed, if you like, piece with all of its elements: the texture, all the different lines that go into it. You're taking all of that in simultaneously, as it were. You're not probably paying too much attention to the structure, but every melody has a structure, so we're going to look at that.
- 2.24** Now the question is: How do you find sufficient interest and variety in such a limited space? [plays C, D, E, F, G, F, E, D, C]. That's like trying to write a story with three words, which you can do, too. You can look at Green Eggs and Ham. It has a very limited vocabulary that is permuted in every conceivable way. It's actually interesting to look at that book as an example of a creative constraint because it's all in rhyming couplets, and it all obeys a very simple meter from beginning to end. He uses an extremely limited vocabulary. I don't remember how many words, but it's a very small number. That's a little bit what's going on here.

- 2.25** I mentioned that there needs to be a skeleton, a boiled-down version of this melody that is coherent [plays the first phrase]. The first thing we notice is that this melody contains repeated notes [plays E, E, E, D, D, D, E, G, G]. The repeated notes add rhythm. They add rhythmic continuity, but they don't change the pitch. They don't affect the contour of the line. If we were to take out the repeated notes, we'd be left with this [plays E, D, C, D, E, D, E]. That's the skeleton, which you could actually reduce down even more. You could say [plays E, D, E, D, E, E, D, E, D, C]. You could say, well, actually, it's really only three notes, structurally.
- 2.26** Then the thing is, well, if you've got three notes, how do you work with that? Well, you might say, I'm going to start by emphasizing the E, which is the third note of the scale, C, D, E, F, G [plays C, D, E, F, G]. The melody starts on the note E [plays E]. Then it moves down to D [plays D]. Then it goes back up to E again [plays E, then the first phrase]. Then the next phrase also starts on E [plays the second phrase].
- 2.27** So simply by alternating three basic structural tones, E, D, and C, which comes right at the end [plays D, D, E, D, C]. There's our C. You can make a little story out of that. It's amazing. I say a story because what is a story, archetypically? What's the basis of it? It's something like you start out with a situation. The situation is disrupted somehow, and then some balance is found at the end. You go out into the world, you have an adventure, and then you come back, and you're transformed by your adventure. Music maps incredibly well onto that archetypal structure.
- 2.28** In this case, what's the story exactly? Well, it could be something as simple as we're starting out on one note, then we're going to move away from that note to one that is maybe a little bit less stable or one that has less of a sense of finality. Then we have to get back to where we started again [plays the first two bars of the first phrase]. We're establishing a center [plays E]. Then we move away from it [plays the next two bars of the first phrase]. That's the only time that the high note, G, appears in this line. You only hear it for one second [plays the first phrase]. That's a surprise.
- 2.29** It's a surprise for a couple of reasons. One of them is that it expands the range. So up to that point, we just had three notes [plays C, D, E, D, C, D, E]. Then suddenly it expands upwards [plays E, G, G]. Now, that's similar to an inflection in spoken language, an interrogatory inflection. If you're asking somebody a question, your voice probably goes up, and when you're answering it, your voice probably goes down. It's not a huge leap in terms of the range of the human voice, but it's enough to get the idea across.
- 2.30** That's a chance also to make a distinction between two types of melodic motion. The first is conjunct motion. Conjunct motion is when you're dealing with adjacent tones in a scale. Let's say that, as is the case here, we're going to make a piece in C major. In C major, you have seven diatonic tones to choose from [plays C, D, E, F, G, A, B, C]. One, two, three, four, five, six, seven, and then the octave at the end repeats the bottom note [plays C-C].

- 2.31** You have to find a way to move around within that space in a way that's interesting. You can do that either by having only conjunct motion [improvises a conjunct melodic line]. That little phrase I just played, there are no leaps at all. Every tone is adjacent to every other tone. I'm not skipping any keys on the piano, if you like.
- 2.32** But you can also introduce leaps into a melody. A leap is an interval larger than a second. In the case of Mary Had a Little Lamb, there is precisely one leap in the whole melody [plays the first phrase]. That one [plays E-G], where we have a leap of a third. That might seem like a trivial feature, but it only happens once. Why does it happen there? Remember the words? "Mary had a little lamb, little lamb, little lamb, little lamb." It repeats three times.
- 2.33** When you repeat something multiple times, it creates an exacerbation. It creates an intensification. That's reflected here on a very simple, very small scale by the move upwards to G with its associated leap. It's an intensification. Repeat something over and over and over again, and you'll start to feel exasperated. The expression intensifies due to the repetition in the text. Then, at that point, we come to a pause. Then it starts again, but not the same way [plays the second phrase].
- 2.34** What's different this time? Well, first of all, it doesn't end the same way. We go from the highest note at the end of the first phrase, and then we work our way back down to the lowest note. This little simple tune is a model of economy of means because he—the author, I don't know who the author is; I don't know if there is an author, whoever it is—they set out a floor and a ceiling. The floor is the lowest note, the ceiling is the highest note. You have to move around within that space in a way that conveys a little story.
- 2.35** We start actually in the middle. If we imagine this as a space with a floor and a ceiling, remember, the melody starts on the note E, which is right in the middle. It's here. Then it circulates around that middle a little bit, and then it comes to a close on the highest note. Then once that point has been reached, it circulates back down again to the lowest note. In so doing, you've defined a little space, and you've defined a movement within that space.
- 2.36** You'll notice also that in the first half of the phrase, remember we mentioned that the G, you only hear it for one second. It's just right at the end. That's the only time we hear it in the whole piece. That's the highest note. The lowest note, similarly, is only very briefly heard in passing in the first bar [plays the first two bars]. You hear it for a split second [plays the first two bars], and it's immediately left, and it's on a weak beat. Again, your ear is going to be drawn to the notes that are found on strong beats, the notes that are repeated. Also, you will be instinctively drawn to the extremes, the highest note and the lowest note. Those tones will be placed strategically by the composer.

- 2.37** Why? Why does it matter? Because it allies with the narrative structure of the melody that you're writing. It's a signal to your listener. This is a high point, and that translates to a dramatic intention. This is the low point, which could be a dip in the dramaturgy, or it could be the music coming to a close, coming to a rest. In this case, we just have the briefest passing notes to hear the tone C until the end, when the tone is finally emphasized, and we hear it for four beats [plays the second phrase]. That's how the piece comes to a close.
- 2.38** So within this tiny little space, we have a coherent phrase structure. We have cadences. There's a range being laid out. There's a narrative. There's a story being told. There's repetition. There are melodic figures, and all within five notes. It's amazing. Remember, we said that this tune has mostly conjunct motion, and there are repeated notes in the first phrase [plays E, E, E, D, D, D, E, G, G]. The repeated notes—again, remember, this is a children's song. It's very easy to repeat tones. It's one of the easiest things to sing. You just stay on the same note, and you rearticulate it. Remember that there was a phenomenon of repetition in the first phrase with the text "little lamb, little lamb, little lamb," which brings it to a close.
- 2.39** The second part gives us this same figure again [plays E, D, C, D, E]. That's like a little—what would you say? It's like a little curlicue. It's a note spinning around on itself [plays E, D, C, D, E]. It goes down, and it goes up again [plays E, D, C, D, E], a little symmetrical movement. The repeated note idea contrasts with that because it's static. It's not moving [plays E, E, E]. So even within this simple little line, there's a phenomenon of contrast of different shapes and different melodic compartments, if you like, or ways of imagining a phrase, conjunct motion, this symmetrical up-down movement, and repeated notes [E, E, E, D, D, D].
- 2.40** In the second half, that principle of repetition is extended even farther. In bar six here [plays E, E, E, E], instead of being repeated three times, it's repeated four times. In the following bar [plays D, D, E, D, C]. Those two bars there are not strictly identical.
- 2.41** Now, the author of this song, which is probably a composite, could have just done [plays E, E, E, E, D, D, D, D, C]. That would have worked, too. That would have gotten the idea across. But they didn't do that [plays E, E, E, E, D, D, D, D, C]. Why not? Well, because it's boring, and it doesn't take much to make it more interesting. What you do is you slightly break the pattern [plays D, D, E, D, C]. It's the same phrase profile as the previous bar, but with a slight twist. It goes up [plays D, D, E, D], and it goes down again to C [plays C]. The other thing that it does is that it intensifies the sensation of melodic downward movement, which gives us a stronger feeling of finality [plays E, E, E, E, D, D, E, D, C]. That's a falling motion.
- 2.42** What I'm trying to demonstrate with this simple tune is that any melody, no matter how simple, has a structure. The structure is actually, and can be, remarkably complicated if you break it down into its component parts. The way that it works, which is largely unconscious, nobody would ever listen to this and think, "Oh, yes,

there's the cadence, there's the repeated note motif, there's the leap at the end," and so on. We don't really listen that way, typically. But all of that is in there, working unconsciously to create a structure that we can intrinsically understand and map onto as a listener so that we don't need to have it explained to us. It slots into our linguistic capacity very easily.

- 2.43 I'm not suggesting that when you sit down to write a piece, you have to think in the way that I just described, where you're consciously coming up with rhythmic motifs. You're thinking, "Do I want conjunct motion here? Do I want to leap?" You're not necessarily going to be thinking analytically all the time when you're writing a piece of music, but it can be very useful to bear in mind what the parameters are, how they map onto speech, and how you can design a line that is coherent.
- 2.44 One of the things that I very often find with students is that they're not always paying attention right away to the highest and the lowest notes of a melody, and that is extraordinarily important. If your line doesn't work with that consciously on some level as a vector of expression, then you're going to be left with lines that feel aimless.
- 2.45 That might sometimes be what you want. Music is an ocean of possibilities. Some music is static, fundamentally, where it's not goal-oriented. It's not moving towards an obvious close. It's not necessarily going to take you from A to B, but it might just circulate in place. Some music is very dynamic. Some music is always pushing forward. It's always moving. It's intensely agitated, and it's strongly driving towards some ultimate point of resolution. That's partly a question of the personality of the composer. It's partly a question of style; different musical genres and forms, of course, will fall in different places along that spectrum.
- 2.46 But if you want to write a line, melodic line, and have it be, let's say, absorbing so that the listener can follow it with interest and be drawn along by it, then at least on some level, it really does help to pay attention to that. So you could write a line where there's no particular shape to it, and it might sound something like this [improvises a melodic line lacking forward motion]. You could just do that forever. You could do that forever. It would go nowhere. Maybe that would be appropriate for a certain style, but it might not necessarily be that interesting.
- 2.47 It can be actually quite interesting to listen to things like modal jazz. In modal jazz, you will typically stay on a single chord or maybe two chords or a two-chord vamp for 10 minutes. The harmony is completely static. An example of that would be the album *Kind of Blue* by Miles Davis, which is one of the great reference points in jazz. A lot of those pieces on the album are incredibly static: just one chord, two chords, maybe no modulation at all.
- 2.48 All of the interest in those pieces is in the rhythm and the spontaneous melodic invention of the players. When the chords are static, when the harmony isn't moving, then you have to chart an interesting pathway through whatever the range of your

instrument is. If it's five notes like in Mary Had a Little Lamb, then you have to find a way to turn that into a story. If you have two and a half octaves on a saxophone, then you will chart a path through that.

- 2.49** One of the ways composers will do that, incidentally, is by using what you might call guide tones or skeletal tones that give you an outline of movement [plays E, D, E, E, D, C]. That's the skeleton. It's the case with pretty much any melody. That's something that you can work with consciously.
- 2.50** I haven't written a melody at this point, but I've defined a space, and I've charted a little movement. Then I'm going to try to invent something [improvises a melodic line]. Then I'm going to try to invent a figure that allows me to get from note one [plays C, F, G#, B (low), C], and then all the way to the end [plays C (embellished with D-C-B-C), D, E (embellished similarly), F, F, G#], and so on.
- 2.51** Now, that can be more or less challenging depending on the material you're working with, the constraints of the piece you're writing, or your particular faculty for invention on a given day. Obviously, artists are not robots. We're not machines. We can't necessarily produce something perfect the first time. But it's also something you can condition yourself towards, to a certain extent, by maintaining a regular workflow, whatever that is. It can really help. So that's an example of how you can take a very, very simple little archetypal structure and then frame a melody on it.
- 2.52** I'd like to look at something a little bit more complex this time. We're going to take an actual composition, a piece of music that's more elaborate, and we're going to look at it for a few minutes. Now, it has all the same qualities as the Mary Had a Little Lamb tune that we've been looking at, but it's more elaborate.
- 2.53** Here's the Intermezzo in C-sharp minor, opus 117, number 3 by Brahms. [plays a recording of Brahms]
- 2.54** That's not the whole piece. This is what's called an ABA form, which is where you have an opening section, a contrasting middle section, and then a reprise of the opening section at the end. It's a very, very simple, very common form that you see in all kinds of different musical genres.
- 2.55** In what we just heard, there was quite a lot of repetition, more or less disguised repetition. You might be consciously aware of it. You might not be, but basically, there are two little phrase ideas in there, and he alternates them with varying degrees of transformation. The basic phrase itself is this [plays C#, D#, E, D#, E, D#, C#, D#, F#, E, D#, E, D#, E, F#, G#, F#, G#, A, G#, A, B, G#, A, F#].
- 2.56** That's an interesting phrase for a lot of reasons. One is, remember, I said you don't really need much material to make a piece of music. This phrase is redundant even on the level of the individual tones. There's a simple little shape in here [plays C#, D#, E, D#, E, D#, C#, D#, F#, E, D#, E, D#, E, F#, G#, F#, G#, A, G#, A, B]. A little rising shape

[plays C#, D#, E]. Like that. Three rising notes [plays C#, D#, E]. So you can see, even within the phrase itself, there are repeated elements that are very small, like little cells [plays C#, D#, E, D#, E, D#, C#, D#, F#, E, D#, E, D#].

- 2.57** And notice also how, just like in Mary Had a Little Lamb, the melody starts from a point and then expands outwards from there [plays C#, D#, E, D#, E, D#, C#, D#, F#, E, D#, E, D#, E, F#, G#, F#, G#, A, G#, A, B]. It goes up. It's an upward movement [C#, D#, E, D#, E, D#, E, F#, E, D#, E, F#, G#, F#, G#, A, G#, A, B]. Then it tumbles down again [plays G#, A, F#, G#, E, D#, C#, C].
- 2.58** It's a simple little symmetrical movement. It rises, and it falls, and it ends on a half cadence—we'll explain what that means later—which is a half pause. It's not a full pause, but you want that to resolve; it's not complete. That's basically the construction of that phrase. It's made out of almost nothing. A simple little figure, a rising figure, and then he has some sequences, bringing it higher and higher. He expands the range upwards. Then, when he reaches the highest point, he moves back downwards again.
- 2.59** The expression in this piece, its sophistication as a composition, is light years beyond Mary Had a Little Lamb, but the basic principles of melodic construction are exactly the same. I would argue that regardless of what style you're working in, what genre, how exploratory or conventional the piece is that you're writing, you're going to be confronted with these basic questions of melodic construction at some point.
- 2.60** I'll close with this. There's another aspect of that Brahms' theme that I think is quite interesting. Remember how in the Pachelbel piece, everything divided up neatly into four-bar phrases: two plus two plus two plus two. Mary Had a Little Lamb had an identical construction. If you take a phrase that ends with a half pause or a half cadence, and then add it to another phrase that ends with a full cadence, those two phrases together make what is called a period or a complete musical thematic statement.
- 2.61** The Brahms is interesting in the sense that it's not two plus two plus two plus two. The phrases are five bars long. They're asymmetrical, and you don't even notice it when you're listening [plays C#, D#, E, D#, E, D#, C#, D#, F#, E, D#, E, D#, E, F#, G#, F#, G#, A, G#, A, B, G#, A, F#, G#, E, D#, C#, C] It has a sense of escaping from gravity somehow. It's very strange. It's a very strange little phrase, incredibly carefully crafted.
- 2.62** It has another feature that is also different from Mary Had a Little Lamb. It has an asymmetrical phrase structure, five-bar units instead of four or two. It also has another aspect which is very, very common, particularly in Eastern European music, Austrian music. It has to do with the way the phrase begins and the way it ends. Mary Had a Little Lamb starts on a strong beat right away [plays the first phrase of Mary Had a Little Lamb]. This line does not. It starts on an upbeat. It starts before the first beat of the bar, and that's called an upbeat or an anacrusis.

- 2.63** The anacrusis is [plays C#, D#] that. Just those two notes. It comes to an accent on the third note, which is this [plays E, then C#, D#, E, D#, E, D#, C#, D#, F#, E, D#, E, D#]. If you were thinking in terms of a 4/4 meter, then the two notes that you hear at the beginning would actually come before the downbeat. They would come before the first strong beat in the piece. That's an anacrusis. An anacrusis is something that comes before the emphasized beat. The way it ends mirrors that.
- 2.64** The piece has an anacrusis, a rise, an accent on the downbeat, and then it falls away and comes to a close. That is a very typical Viennese phrase. You don't necessarily see it in every musical tradition. But that idea—it's a very elegant form of phrase construction because you don't start right away. You lead up to it, you hit an accent, and then you fall away again [sings the phrase structure]. That phrase structure. It's a very elegant form of construction. It doesn't fit with all styles.
- 2.65** If you look at the music of Bartók, a great Hungarian composer, or Stravinsky, the Russian composer, they don't have phrase structures like that. They start right on the downbeat. It's direct. The Viennese construction is elegant, but it's also not quite as direct.
- 2.66** Those are some of the essential parameters of what goes into making a phrase. We've talked about structural elements, we've talked about the importance of range, the importance of working within a range consciously to make a shape, to make a narrative dramatic arc. We've talked about aspects of phrase construction, such as the use of an anacrusis or starting straight on the downbeat. We've talked about structural repetition, where a little unit within a melody can be repeated multiple times to make a sequence, as was the case in Brahms. We've talked a little bit about how very, very simple ideas can be repeated and transformed. We're going to stop there as far as melody goes. If you have any questions, let me know.
- 2.67** Yeah, so a quick, simple question. What did you mean by diatonic? What does that term mean?
- 2.68** A diatonic scale is a seven-note scale. If you look at the piano keyboard here, you can see there are white keys and there are black keys. I can start on any tone and play all the white notes and all the black notes [plays a chromatic scale]. That gives me a chromatic scale—chromatic from chroma, which means color [plays a chromatic scale].
- 2.69** But generally speaking, in traditional harmony, you're not going to use every single chromatic tone to make a scale or to produce a melody. You'll use them occasionally, and you need them for—we'll get into that later because it's a vast topic. But most of the time, what you're dealing with is a seven-note scale, which is called a diatonic scale [plays a diatonic scale]. That's just the white notes on the keyboard. C, D, E, F, G, A, B, C [plays C, D, E, F, G, A, B]. That's seven. Then the eighth note, again, is a repetition of the first one [plays C], an octave higher [plays C–C, then C, D, E, F, G, A, B, A, G, F, E, D, C]. That's a diatonic scale. That forms the basis of Western melody and harmony for the

most part until well into the 20th century, at which point things get a little bit complicated, but we'll talk about that in a subsequent lecture.

- 2.70** At some point, you discussed how classical music would influence the way people talk, or was it the other way around, namely, with the Italians, the way they speak in their tone? I'd like to know about how perhaps the language in the tone or how music influenced each other?
- 2.71** Yes. I don't think that the music influences the way people speak. I think it's the other way around. There's a very interesting historical shift that happens around 1600. Quite fascinating. There's a major dramatic break that occurs. Up until that point, European music was primarily polyphonic in nature, which is to say that it was almost all vocal. Even when people played instruments, they were typically playing vocal music, but without the words. The voice was absolutely primal.
- 2.72** The settings of these texts were, for hundreds of years, polyphonic in nature, which is to say that you'll have a voice and another voice on top of it and another voice and the other voice. Because of the way that these pieces were composed, because of the styles that were dominant at the time, the words in those pieces would not all be heard simultaneously by every voice. It would be staggered. You would have imitative counterpoint. A word might be repeated. It might be stretched out like a melisma, which is where you take a syllable and hold it across many different tones.
- 2.73** When you're listening to a piece like that, it's not at all like listening to someone speak. It's a totally abstracted form of language in the sense that you'll hear the perfume of the text if you like. You'll get the atmosphere of the text. It will be expressed by the music, but you won't understand it word to word. You might understand almost nothing of the text, in fact. You can have this experience very easily, actually. You just have to go to hear an opera in a language you don't know. You'll get something out of it, of course, but you won't understand every single word.
- 2.74** What happened in 1600 was a shift away from that approach, where the language is there, but it's buried in a sense, or it's transformed into something else, but it has very little to do with how people actually talk. At that point, a group of mostly Italians, who were interested in antiquity—incidentally, this comes from the revival of an interest in ancient Greek theater, among other things—they wanted to have musical forms where the voice was comprehensible. You could understand every word.
- 2.75** In order to do that, there had to be a simplification of the polyphonic texture. There had to be a primary voice and then a very, very simple, sparse harmonic accompaniment. That's when we have what starts to resemble a contemporary song, the idea that there's a single voice that is the primary focus of the piece, and the other elements are essentially there to support it.
- 2.76** At various different points in the history of European art music, there have been different points of emphasis. Sometimes it's important that the words be understood,

that they be naturalistic, that it correspond to speech. Sometimes you're painting with sound, but you're not really so concerned with a direct relationship with speech as such. But all of that, it does map on to everyday speech. Certainly, you will see differences between the cadences, the rhythm, the melody, if you like, of Italian, of French, of German, of the English language, all of which have different accent patterns, all of which have different tonalities to them. It's a vast subject to talk about how specifically that might influence a musical style, and it's not a straightforward question, but it's a very interesting one. Thank you for the question.

Lecture 3: Shaping Harmony

At a glance — Shaping harmony.

- 3.1 We've looked at the parameters of composition. We've looked at how you can take a simple thing, elaborate upon it. We've looked at melody as an independent subject in and of itself. Now I want to look at harmony. Of course, to a certain extent, it's difficult to separate out all of these different aspects of the art of music because they go together. They create a whole. Rhythm is indissociable from harmony. Harmony is indissociable from melody. All of these things are strongly interrelated.
- 3.2 But I'm going to start this off with a personal anecdote. When I was a composition student, very young—I was maybe 17 or 18 years old, and living in Toronto at the time—I went and saw a Canadian composer who had been recommended to me by one of my teachers, and went and showed him my work. He was looking through it, and towards the end of the meeting, he made a comment that was really striking to me. He said, "Harmony is your life's blood."
- 3.3 I didn't exactly understand what he meant by that at the time. Unfortunately, I only met him—his name was Srul Irving-Glick, G-L-I-C-K, very interesting, eminent Canadian composer. Unfortunately, I didn't realize at the time he was very ill, and he died of cancer very shortly afterwards, so I only got to meet him a couple of times. But that line, "Harmony is your life's blood." I never forgot that. It took me a while to unpack it.
- 3.4 I think what he meant—what that means to me now—is that harmony is one of the primary means in music through which you can create a sense of movement. Music is about movement, fundamentally. Music is an enormously elaborate way of expressing movement in incredible complexity that is not possible in everyday life that we can experience vicariously through music. And harmony is one of the principal agents of that sense of movement. And harmony is also a topic that is often very poorly understood, especially by younger musicians.
- 3.5 So if you're learning to play the guitar, for example, what you're typically going to start with is tablature, learning chords. And in tablature, what you have is not staff notation—a five-line staff, and then you put the notes on the staff and so on—but what it shows you is actually a physical position. That's the difference between tablature and staff notation. It shows you what to do with your hands. It doesn't show you what the result will sound like. Just, "This is how you do an E-major chord, this is how you do an A-major chord, and so on."
- 3.6 A lot of musicians will learn how to play chords through these ergonomic principles. This is the shape that your hand needs to be in to make this type of chord on this instrument. You learn these patterns, and they become instinctive after a while. If you're picking up a guitar, you don't have to think. You just find the position for an E-major chord, or an A minor, or whatever it is. You can learn these progressions that

become second nature to you. But you might not necessarily be thinking about the actual constitution of the chords you're playing, how they're joined together, how you create a progression with that, and what are the different ways that you can use that chord to make a different sound, to present the harmony in a different way.

- 3.7** We're going to look in detail at some of the most basic aspects of harmony. How do you actually make a chord? What is it constituted of? How can you alter it? How can you create a progression, a chord progression, as opposed to a mere succession of chords? It's not the same thing. What are all the ways you can alter a chord to change its character or its quality?
- 3.8** The things I'm going to tell you here are going to be based around a very simple starting point, much like the Mary Had a Little Lamb example. I'm going to start with this C-major triad. That being said, it's very important that these lectures, in general, not imply any specific aesthetic or style. The point of all of these things that I'm showing is that they're broadly adaptable to a very, very wide range of different possible styles. There's music that is only minimally preoccupied with harmony. There's music that's only minimally preoccupied with rhythm. There are styles that are intensely melodic, others that are much less so. But what I'm trying to present are things that are primordial enough so that they can be abstracted and adapted to a wide range of different circumstances.
- 3.9** Let's start at the very beginning. What is a chord? Well, a chord is two or more notes sounding simultaneously. A chord can be constituted of any two tones, absolutely any two tones. It doesn't matter. If I do this, [plays A-C] that's a chord. That's a two-note chord. I can do this [plays G-A-B-C]. That's a four-note chord [plays C-E-G]. That's a chord. There are an infinite number of possibilities. But we're going to start with this one [plays C-E-G].
- 3.10** This is a C-major triad, and it is built on three tones. The tones are what's called the fundamental or the root of the chord—and the root is the bottom note of the chord, and that's C [plays C]. What you do is—the triad is built on stacked thirds—you add a third on top of the bottom note, and that gives you E [plays E, C, then C-E]. Now, that's not a complete chord yet, or not a complete triad, more specifically. The complete triad consists of the root [plays C], third [plays E], and the fifth [plays G]. And that gives you a triad [plays C-E-G].
- 3.11** Okay, so I'm presenting this as though this were an eternal feature of nature. It is and it isn't. It's a little bit more nuanced than that. In order to describe this chord structurally and explain why it is the way it is, we have to look at something called the harmonic series. I'm going to write that down, and you can see what it looks like. The harmonic series is a simple physical property of a vibrating string. This has absolutely nothing to do with culture or aesthetics, or even art for that matter. It's a physical property.

- 3.12** This was sorted out by Pythagoras thousands of years ago. What he discovered was that if you take a length of string and suspend it between two points, and then you have a movable bridge that you can place at any point along the string, then you can obtain a series of proportions that give you intervals. This device that he invented, incidentally, is called a monochord. The monochord, as its name suggests, is a single string. The monochord looks something like this. There's your string. It's fixed at both ends. Then you have a bridge, and the bridge can be positioned anywhere. It can be positioned here, it can be positioned here, and so on.
- 3.13** Now, if you take your monochord and put the bridge exactly in the middle, you're cutting the string exactly in half, that gives you a ratio between the two halves of one to one. Sorry, two to one. And that gives you an octave, an interval called an octave. Now, if you keep going and divide the string into increasingly smaller bits, you keep going with the series, 3:2, 4:3, 5:4, etc. You get a sequence of different intervals. So 2:1 gives you the octave, 3:2 gives you the perfect fifth, 4:3—now, I should play these actually so that it's clear what I'm doing. So 2:1 [plays C–C] gives you an octave. So that means that the upper of the two tones I just played is exactly twice the frequency of the lower one [plays C–C]. It's an exact multiple.
- 3.14** That gives you a very simple consonance because it's a simple ratio, and it's very easy to relate the sound of the upper tone to the lower one [plays C–C]. If you continue the series, 3:2, we put the bridge on our monochord in such a manner as to divide the string into three equal parts, and the bridge is that one-third of the way along. Then we obtain this interval [plays C–G], which is a perfect fifth. That's also quite a stable interval. It's very easy to relate that interval to the lower tone [plays C–G, then C–C]. This is a physical phenomenon. You have a fundamental tone. You can obtain a sequence of different intervals based on this sequence of ratios. This also pertains to what's called the harmonic series.
- 3.15** The harmonic series, again, is a product of any regular harmonic periodicity that you can obtain with a—it can be an air column inside a flute, or it can be a vibrating string on a violin. What it is, is that you can play a fundamental tone, and you'll obtain a series of overtones. When you hear a single tone on the piano [plays C], your ear thinks that you're just hearing one sound [plays C]. But if you were to do a sonogram of that sound, and analyze it, and look at its frequency composition, you would find that it consists of a very large number of overtones of varying degrees of intensity.
- 3.16** The sequence of overtones is as follows: you have the fundamental [plays C], octave [plays C], fifth [plays G], fourth [plays C], third [plays E, G, B $\flat$, C, D, E, F $\sharp$, G, A, B $\flat$, B, C], and all the way up the series. Now, that's not an exact representation of the overtone series by any means, because the 11th partial, which is this [plays C, then F, F $\sharp$], is actually in between those two notes—it's between F and F $\sharp$ [plays F, F $\sharp$]. It's impossible to play it exactly in tune on a piano. But by and large, that gives you an idea, the notes,

the intervals, the overtones, sorry, in the overtone spectrum are far apart at the bottom, and they get closer together as you go up.

- 3.17** The overtones that are closest to the fundamental are the strongest. They're the ones that we hear the most clearly and that we can most easily relate to the fundamental. As you get higher and higher up the series, it becomes harder to relate it to the fundamental because it's more distant. Just as the effects of gravity become weaker the farther you get from an object in physics, it's the same with this.
- 3.18** Remember that the order of overtones in the spectrum in this series is [plays C, C, G, C, E, G, B $\flat$, C, D, E]. Now, that's a slightly idealized, flattened-out version of it because, as I mentioned, the 11th partial is a little bit wonky. The seventh is also flat relative to what I just played. So the natural harmonics are not going to be tuned in exactly the way that you hear them on a piano. But nevertheless, these first overtones constitute the basis of the triad that I played earlier. If you start on the fundamental, the C [plays C], go up an octave [plays C]. There's your fifth [plays G], octave again [plays C], then there's your third [plays E]. Now you can keep going. But again, those are the strongest overtones.
- 3.19** Now, the exact spectral composition of a sound varies depending on the instrument. There are instruments, for example, that will only produce an odd overtone series; they won't have the even partials. There are instruments that have a great deal of energy in the spectrum, in the upper end, let's say, of that spectrum. Others that will have a very strong fundamental and octave. That's one of the things that constitutes timbre, incidentally. It's one of the things that allows us to hear the difference between a clarinet, and a flute, and an oboe, and a guitar, and a banjo.
- 3.20** The reason that they sound different to us is that they have very different attack, resonance, and decay characteristics, and also very different overtone spectra. They're going to emphasize certain partials, and other partials may be inaudible, not present at all, or weaker. But nevertheless, that's a physical reality, and it has to do with the properties of vibrating strings, the system of proportions that was discovered by Pythagoras.
- 3.21** And that leads to this [plays C-E-G], this simple chord, which actually is not so simple [plays C-E-G]. So that leaves you with a physical fact, but physical facts don't necessarily result in interesting art. So the task of the artist is not simply to take what's there, but of course, to shape it and mold it and turn it into something interesting. With something like a triad [plays C-E-G], a composer is not going to be content to just take simple chords like that and repeat them in unvarying sequences or in very simple progressions all the time. You're going to want to elaborate upon them and transform them in various ways and present them in a variety of different ways, depending on the expressive context.
- 3.22** What can you do with a simple chord? Well, we're going to take you through some of the possibilities. Okay, so here's what triad looks like in musical notation. There's your

C, E, and G. If I play a chord like that, that's called a root position triad. A root position because the root of the chord is the bottom note of the chord in the way I've got it here. Now, that's also a chord in what's called close spacing.

- 3.23** Close spacing means that you cannot have the three tones of the chord be any closer to each other than they are. Now, there's also open spacing where you might not have the tones immediately adjacent to each other, but you might space it like this [plays C–G–E]. In this case, I've taken the third of the chord and put it an octave higher. The chord now consists of the same tones, but in what's called a different voicing. Now we have this [plays C–G–E]. You can take any chord and voice it in a huge variety of different ways, depending on which tones you place on the bottom, which ones you double or omit, as the case may be, and what octave you place them in. I can present this chord as a simple root position triad in close spacing [plays C–E–G] or open spacing [plays C–G–E].
- 3.24** Okay, so here's my open spacing triad. You can see that in the first chord, the G is on top, and the tones are immediately adjacent to each other—they can't get any closer together. In this second version of the chord, the E is on top, and that gives us an open-spaced chord, open spacing. That's a first, what would you say, way of transforming a chord, presenting it in different ways. Of course, I can also place it in any register of the piano [plays C-major chords across the piano] and in any spacing I want [plays open and closed C-major chords].
- 3.25** Now, the other thing I can do is I can change the order of the tones from bottom to top. In the case of the root position triad, remember that the root of the chord goes in the bottom. But you can also put the third as the bottom tone of the chord, and that would look like this [plays E–C–G]. We have the same set of three tones, C, E, and G, but I've got the E as the bottom note [plays E–C–G]. Now, this is interesting. I want you to pay attention to the sound of this chord because it's a little bit less stable than the previous one [plays C–E–G]. If I just play a C-major triad like this [plays C–E–G], there's no particular need for movement. It's stable. It just sits there. If I put the third in the bass [plays E–C–G], then that's a chord that's starting to want to move. It's less stable. You can also put the fifth as the bottom note, and that looks like this [plays G–C–E].
- 3.26** Okay, so root position [plays C–E–G], first inversion [plays E–G–C], second inversion [plays G–C–E]. The reason these are called inversions is that you can take the triad and simply permute the order of the notes. The C is the bottom note here. In first inversion, you put the third in the bottom, and in second inversion, you to put the fifth on the bottom. Those are called inversions. Any chord can be inverted. So you can take a chord [plays C–E–G, E–G–C, G–C–E] and present it in a different order, if you like, with a different note on the bottom [plays C-major chord inversions].
- 3.27** Now, the different inversions, again, have different degrees of stability and different sonic characteristics also. That opens up a lot of possibilities. So very often in music, the bottom note of the triad is going to be your bass note. So if I take the first inversion

triad, which was this [plays E-G-C], I put the E an octave lower [plays E-G-C], and I have a nice little bass there [plays E-G-C, then F-A-C]. Now, you might want to use those inversions in order to create a smoother sense of continuity when you're joining one chord with another [plays E-G-C, then F-A-C]. But there's a lot more you can do with a chord. We've talked about the root position, first inversion, second inversion. We've talked about how you can have a chord in close spacing or open spacing.

- 3.28** But you can also double tones. When you double a tone, what you're doing is you're taking one of the three notes of the triad and placing it in an octave, or two octaves, or more, higher or lower. That tone is included twice in the chord in different octaves. That is a great resource also because you can have sonorities that are a lot more full, that are richer, and that have more variety to them. A simple doubling, and the most common one by far is to double the root. I take my C-major triad, and I double the root at the octave. That gives me this [plays C-E-G-C]. Now, we have an octave in the chord [plays C-C, then C-E-G-C]. I haven't added any distinct tones. I've just doubled the root [plays C-E-G-C]. Of course, you can do that freely. You can do that as many times as you want.
- 3.29** One of the things that you'll find with composers is as you're developing more of a personal sonic signature, as your music is taking on more and more individual characteristics, you're going to be hyper-attentive to the way that you present chords. It makes a world of difference if you just play a triad in the middle of the keyboard [plays C-E-G in the middle of the piano], or if you come up with a spacing that's maybe a little bit more [plays C-E-G spread across octaves]. You can come up with sounds that are really quite different.
- 3.30** So that's a function of chord voicing: doubling, and inversions [plays E-G-C-G-G]. And even with something as simple as a C a major triad, I can come up with hundreds and hundreds and hundreds of them [plays C-major inversions across the keyboard]. And what I'm doing, there's no harmonic motion at all. It's just the same chord played hundreds of different ways. So what I'm doing is harmonically static, but it's not sonically static because the sound of each chord is, of course, different.
- 3.31** You can decide when you're writing, first of all, what tone do you want to have on top? What tone do you want to have in the bass? Maybe you want to put the E on top, and maybe you want to put the G in the bass [plays E-G, then E-G-C-E]. You get a particular sound that way. Maybe you want to put the G on top and the E in the bass [plays E-C-E-G]. Maybe you want C on top and E in the bass [plays E-E-G-C]. There's endless possibilities. You become really attentive to this. Beethoven will often start his piano pieces with very thick chords in inhibital registers [plays dense chords in the low register]. Like that. There's a lot of doubled tones. We're in the bass where the individual tones are a little bit harder to distinguish. It makes almost a percussive sound, very, very full, very rich.

- 3.32** One of the exercises I'll often do with my students is I'll just get them to take one chord or two chords and just say, "Sit at the piano for an hour and write as many variations as you can possibly think of on that one chord. Listen for the ones that really strike you, that catch your ear, and pay attention to those because that's going to give you a clue as to what kind of sonic world you're interested in." If you don't do that, if you don't experiment and don't try things out, then you're going to be left with the automatic, first-choice, almost mechanical versions of these chords, let's say. Which can be fine, it can be serviceable, but you might not be going as far as you might.
- 3.33** That's just a simple C-major triad. We've looked at how it's derived from acoustics, from natural properties of vibrating strings. But you can, of course, do more than that. You can alter chords, you can add tones, and we're going to look at how that works.
- 3.34** So triads can also have extensions. So remember, I said that the triad is based on stacked thirds. There's a major third on the bottom [plays C-E], in the case of a major triad, and a minor third on top [plays E-G]. You can keep going with that. You can keep adding thirds. So if I add another third on top of the G there, that gives me a B [plays C-E-G-B]. That's called a seventh chord [plays C-E-G-B]. That looks like this. Likewise, as with any chord, of course, you can invert a seventh chord, too [plays C-E-G-B, E-G-B-C, G-B-C-E, B-C-E-G, C-E-G-B]. The range of possible permutations and different presentations of a seventh chord is, of course, considerably greater because there's an additional tone.
- 3.35** You can keep going [plays C-E-G-B-D]. You can add a D on top of the B, and then you get a ninth chord [plays C-E-G-B-D]. That looks like this. Now we're up to five tones. It doesn't stop there. Jazz musicians will write ninth chords, 11th chords, 13th chords. You can just keep going.
- 3.36** Now, you can also alter a chord. You can take one of the tones and raise it or lower it by a semitone and obtain a wide variety of other sounds. Remember, all of this just pertains to the C-major triad. I'm not talking about any other triad for now. We're not looking at minor chords, or augmented chords, or diminished chords, or anything like that. Just a simple C-major triad. I'm going to cover some of this in the Introduction to Music Theory course, and I'm trying not to duplicate the material too much, but I will get into more specifics of the different types and qualities of chords in that course.
- 3.37** But for now, we're just going to look at how you can take a simple C-major triad and what are all the different ways that you can alter it? What are all the different ways you can present it? And what are the different potentials that lie within that set of pitches? Okay, so there's my C-major triad again. I'm going to add a seventh on top. There's the B.
- 3.38** This is an interesting exercise for composers. If you just take a triad like that [plays C-E-G-B] or with a seventh on top [plays C-E-G-B], just try this. Take any one of the tones of the chord [plays C-E-G-B]. Alter it by raising it a semitone above or below, and

listen to the difference in the sound [plays C-E-G-B, C-F-G-B, C-E-G-B, C-E-G-B $\flat$, C-E $\flat$ -G-B, C-E $\flat$ -G $\flat$ -B, C-E $\flat$ -G $\flat$ -B $\flat$, C-E-G $\flat$ -B $\flat$, C-E-G-B $\flat$, C-E $\flat$ -G-B $\flat$]. This is an interesting improvisation exercise that musicians can try. So let's start with a C-major triad [plays C-E-G], and then alter one tone at a time and see where it goes [plays C-E-G; then improvises a series of alterations, raising or lowering one tone at a time]. I go back to C at the end. You can do that forever—you just go over the keyboard—and pay attention to how the sound changes when you just move one note at a time. See if you can do that for 10 minutes. See where it takes you and try to become habituated to all the possibilities. You can take a chord and move it absolutely anywhere, any direction.

- 3.39** In a sense, a chord in a piece of music [plays C-E-G], if you just play a C-major triad like that, it's like if you just had an isolated word on a page. It doesn't really convey much meaning. It certainly doesn't convey a sense of movement because there's no context. A C-major chord [plays C-E-G], it doesn't tell us really anything. It doesn't even really tell us what key we're in. We could be in F major, we could be in A minor, we could be in C major, we could be in A $\flat$ major and it's a median. There's all kinds of possibilities.
- 3.40** In order to create a sense of movement and context, then there needs to be a chord progression. Now, I said earlier in this lecture that there's a distinction to be made between a chord succession and a chord progression. If you just take a series of unrelated chords and juxtapose them, then each new chord is effectively going to erase the previous one [plays a series of unrelated triads]. You can do that. Actually, there are composers like Philip Glass, a New York minimalist composer who works this way sometimes with triads, where the music is consonant—it consists of simple chords like that—but there's no particular harmonic connection between them. What happens is you're paying attention to the sound of each chord, but they don't necessarily link up to form a progression.
- 3.41** A progression is defined by a sequence of chords that conveys a sense of movement towards a goal. You're starting in one place and you end up somewhere else [plays C-E-G, C-F-A, B-D-G, C-E-G]. Now, instead of just an isolated chord floating in a void, I have the suggestion of a key. I have the sense of a movement towards somewhere. Then I can play with phenomena such as, well: How far am I from my starting point? Do I want to resolve that tension? Do I want to keep it going? Do I want to amplify the tension? Where am I going to go next?
- 3.42** Now we've looked at the constitution of a simple triad. We've looked at the ways that you can extend a triad by adding additional tones beyond the root, the third, and the fifth. We've looked at inversions. We've looked at chord voicing and spacing.
- 3.43** There are other things you can do as well. You're not obligated to have a complete triad every time you write a chord. For example, you can just write chords that consist of thirds [plays C-E] with no fifth [plays C-E]. Or you can have chords with no third [plays C-G]. And chords with no third, just a root and a fifth [plays C-G], are harmonically somewhat ambiguous. We don't really know if they're major or minor, but they have

this [plays G-C] very open, stable sound without a third [plays C-G-C]. That's a sound that's very common in heavy metal [plays C-G, D#-A#, F-C, C-G, D#-A#, F#-C#, F-C]. This power chord sound. Those are chords that lack a third.

- 3.44** And of course, you can double any tone you want as many times as you want to create the sound that you want. In terms of the order of priorities in traditional harmony and voice leading, if you're going to double a tone, the tone that you would typically prioritize would be the root. You would typically double that tone as a first priority, simply because it's the strongest tone in the chord. It's the most stable. If you double that tone, then it gives you a very nice balanced sound [plays C-E-G-C]. But you can also double the fifth [plays C-E-G-G]. If you're going to compose music, you can spend hours and hours listening to the subtle distinctions between these different voicings and listening to all the tones in the chord [plays C-E-G-C] and how they relate to each other.
- 3.45** The last priority, and it's usually avoided, is doubling the third. Doubling the third is typically not done for a variety of reasons. One of them is that it makes the intonation of the chord a lot more difficult. Well, on a piano, I suppose that's a relative concern. But in any other situation, with voices, for example, or with string instruments, or with woodwind instruments, it's much harder to tune thirds than it is to tune fifths and octaves. You can double the third like this [plays C-E-G-E]. Again, on a piano, it's not exactly disturbing [plays C-E-G-E].
- 3.46** But again, the question is to pay attention to all these different voicings, their different sonic qualities, and use them intentionally to create the effect that you're after. Take a simple chord like that, come up with hundreds of different variants, sit down at a keyboard, put your ear really close to the soundboard, and listen very carefully to every single tone until you can really hear them. So that you're not just hearing the chord as an undifferentiated object, but you're really leaning into the sound and noticing exactly what its quality is [plays various inversions of C-major triad across the piano]. You can actually make a piece with that, believe it or not. We're going to look at this a little bit later.
- 3.47** But there are, in fact, quite a few pieces that are 100% harmonically static. The only thing that's changing is the voicing of the chord, the register. You can do that for five minutes, provided that you do it with skill. You don't need 50,000 chords to make a piece of music, but you do need to be attentive to the quality of the chord you're using, and the way that you're shaping it in order to create a different acoustic effect. I'm going to stop there for now with triads. There's obviously a lot to say about harmony. This is only scratching the surface. A harmony course would take countless hours to go into sufficient detail with, but that's a good start for now.

Lecture 4: Layering Sound

At a glance — Layering sound — musical texture.

- 4.1 Now I want to focus on musical texture. This is perhaps the most difficult to define of the concepts that we've looked at so far. We've talked about harmony, about all the many different ways you can present a single chord. We've looked at melody and phrase construction. We've looked at the different parameters of music. What is texture exactly?
- 4.2 Well, texture has to do with the way that the elements in a musical composition are laid out and what the priority is, what you want the listener to focus on the most. It also refers to the broad qualities, the broad sonic qualities of the piece that you're listening to. Just as you can present a chord in 100 different ways, and just as you can take a ground bass and compose a sequence of variations on it, you can also present a piece in multiple different textures.
- 4.3 Texture has to do with: What are the number of different elements in the piece? Are they heard simultaneously or not? Are there elements that are more important than others? Are they all equal? There are many different types of textures that at various points in music history have been more common than others.
- 4.4 The simplest type of texture that I can describe is monody. Monody is just, as its name suggests, a single line with no accompaniment, no harmony, nothing. And monody is what characterizes Gregorian chant, plain chant. In monody, you might have a line that looks something like this. Now, for the sake of this example, I'm not even going to indicate any rhythmic values. I'm just going to write whole notes. This is just a very spare, unaccompanied line [plays C, D, E, A, G, F, G, A, B, G, F, E, D, C, D]. That's the simplest and most straightforward presentation of a melodic idea.
- 4.5 Now, this was not only common, but this was the primary way that music was invented 1,500 years ago. The idea that you would more than one tone sounding at the same time was something that had to be invented. Then you had to deal with the problem of, well, how do you combine tones? How do you have them sounding simultaneously?
- 4.6 The simplest way of doing that is to have a second voice that sings along with the first at a fixed interval. The fixed interval can be something very simple, like an octave. I'll just take this line, and I'll add another line that's doing the same thing, but one octave higher [plays the same melodic line in parallel octaves].
- 4.7 Then you obtain something that's called organum, O-R-G-A-N-U-M. Organum is simply the parallel motion of voices that are proceeding along the same intervals at a fixed distance from each other. It can be an octave, or it can be a fifth like this [plays the melodic line in parallel fifths]. That's a very rudimentary form of harmonization, if you like. It's not really harmonization, but it's a parallel voice that runs along with the first

one. Those are the two very rudimentary, if you like, forms of texture. Organum and monody.

- 4.8** But of course, there are many other possibilities as well. You can also have counterpoint. And counterpoint is a term you've probably heard that strikes fear into the hearts of musicians everywhere because it's associated with something abstract, difficult, intellectually demanding. But in fact, counterpoint is one of the most fundamental building blocks of the Western musical tradition, and the concept is not so difficult to understand.
- 4.9** In counterpoint, you have two independent voices, and they're not moving at a fixed all the time, and they're not simply repeating each other at the unison, as would be the case in monody, but rather they have a certain independence. One voice can go up, the other can go down. They're not always at a fixed interval. Let's look at a couple of simple examples of a contrapuntal texture. Again, I'm going to keep it very, very simple for now.
- 4.10** Here I've got an upper voice that's going to do this [plays A-G-C-B], and a lower voice that's going to do this [plays F-E-D-C-D]. I take those two voices and play them together. Then I get this [plays the two voices together]. Now I've got a degree of independence between two voices—this, of course, can be sung. When I say voices, I literally mean voices. This is intended for singers. The degree of independence between the different voices is also an important parameter and counterpoint.
- 4.11** When you're learning how to write it, you learn that there are four basic ways that two tones can move or two voices can move relative to each other. We're going to look at what those four types of motion are. They all have different degrees of independence from each other, which is a crucial point. Remember that in organum, there is no independence. There's no rhythmic independence, and there's no melodic independence. It's just a fixed interval. But in counterpoint, the independence can be quite significant.
- 4.12** Here we have a top line that's doing this [plays G-G-G-G-G]. Okay, it's not the melody of the century. It's static. The top line isn't moving; it's staying on a fixed pitch. The bottom line is going to do this [plays C-D-F-E-D]. We have one line that isn't moving and one that is [plays the two voices together]. That's called oblique motion.
- 4.13** Now, we also have the case where both voices are moving in the same direction. Here's my top voice [plays C-D-F-E-D]. My bottom voice [plays E-F-A-G-F, then the two voices together]. And that's called parallel motion. You might be thinking, "That's the same thing as organum." Well, there's a slight nuance there, but for beginners like me and you, we'll stop there for now.
- 4.14** So you might be able to guess what the others are. There are only so many different ways that two voices can move relative to each other. So then we have this [plays E-F-

A-G-F]. My bottom voice is going to be [plays A-G-C-D, then the two voices together]. That is contrary motion, where the voices are moving in opposite directions.

- 4.15** And finally, we have similar motion [plays E-F-G-C-D], there's my top voice. [Plays G-D-E-G-G], it's my bottom voice. Both voices are moving up [plays the two voices together]. But they're not moving up at a fixed interval. In the first beat there, the voices are at an interval of a sixth [plays G-E], then a third [plays D-F], another third [plays E-G], fourth [plays G-C], and a fifth [plays G-D]. The interval between the tones is changing, but the direction of the voices is the same. That's similar motion.
- 4.16** As you can see, parallel motion has a very low degree of independence [demonstrates the two voices in parallel motion]. The voices seem to fuse together. In contrary motion, you have a much stronger sense of independence of the voices [demonstrates the two voices in contrary motion]. In similar motion, there's an intermediary degree of independence between the voices [demonstrates the two voices in similar motion]. Not quite the same, but very similar. And in oblique motion, there's a strong degree of dissimilarity or independence [demonstrates the two voices in oblique motion].
- 4.17** So counterpoint is a subject that you could devote your entire life to. It gets extraordinarily complex. But those are at least the basic ways that you can have two voices interact with each other in a contrapuntal texture. We're looking at different textures, different ways of presenting musical materials. In counterpoint, one of the basic principles is that the voices are of equal importance in a texture. If you have two voices, there isn't one voice that is the main one, let's say. They're of equal significance. Both voices are moving, both are independent, and both have distinct melodic trajectories.
- 4.18** Next, we come to accompanied melody. Accompanied melody is a simplification relative to contrapuntal textures. If you're writing two independent voices, then, well, you have to have two complete lines that are independent from each other. Of course, you can write four-part counterpoint. You can even write eight-part counterpoint. There are even 40 voice motets that have been written with 40 independent parts. Obviously, that gets quite complicated, but you might want a much simpler texture. Probably the most common one in the broadest range of music that we actually hear from day to day is the accompanied melody.
- 4.19** Accompanied melody differs from counterpoint in that it establishes a strict hierarchy between the melody, which is presented in a very clear and straightforward fashion, and it is clearly the primary element in the piece, and the accompaniment—the purpose of the accompaniment is to support the melody. That can, of course, happen in a variety of different ways. The support could be rhythmic in nature, could provide a propulsive rhythm, it could be harmonic, it could provide a harmonic filling out that enriches the sound, or it could be something else entirely.

- 4.20 But in any case, the melody is without any ambiguity the thing that the ear is drawn towards the most. So you could have a little line like this [plays the melodic phrase]. That's a little phrase from the C-major Piano Sonata by Mozart. And the left hand is doing this [plays the accompaniment figure]. That little figure in my left hand is called an Alberti bass. The reason that's an accompanied melody type of texture here, and not counterpoint, is that this little figure is the same, really, as a chord, but just broken up. We're just taking a chord and creating a little figure out of it by alternating the tones rhythmically [plays the accompaniment figure]. You can just keep going like that [plays the melody and accompaniment together]. There, you have a clear hierarchy between a melody and an accompaniment.
- 4.21 What tends to happen, perceptually, is that the accompaniment is there, but you're not necessarily listening to it consciously. It's not necessarily soliciting your active focus when you're hearing. Instead, it provides a substratum that supports what's going on in the right-hand [plays the melody and accompaniment together] so that you're going to be drawn to what's happening in the melody. That's a very simple and effective presentation of a melody because it puts it on a pedestal, so to speak, and there's nothing else really to hinder our direct perception of it.
- 4.22 There are many, many other types of musical texture, of course, that you can explore. Those are some of the most basic ones. There's also the chorale. The chorale is one in which you have multiple different lines, as is the case with counterpoint, but typically, you don't have a great deal of rhythmic independence between the voices. In other words, the voices are all pretty much singing in the same rhythm. That gives you a texture that's like this [plays a chorale-style texture].
- 4.23 Every voice in what I just played is, for the most part, with a couple of exceptions, playing the same rhythm. It's not got the same degree of independence as a completely contrapuntal texture, but it also doesn't have the clear division of melody and accompaniment that you would have in, for example, that Mozart Sonata. Those are some basic texture types: accompanied melody, chorale, counterpoint, monody, and organum.
- 4.24 I want to play a piece that illustrates this idea a little bit. We've already seen examples of different textures in the Pachelbel variations that we looked at, the Hexachordum Apollinis. I want to show you what Bach does in the C-major Prelude to Book One of the Well-Tempered Clavier. This is an extremely famous piece which serves as a particularly clear and straightforward presentation of this principle. In this piece, there's ambiguity or a permanent interrelationship between the vertical and the horizontal. I'll explain what I mean by that.
- 4.25 Remember that to create a chord, you have to play two or more tones simultaneously [plays C-E-G]. In order to have a melody, you have to play tones successively [plays C, E, G, E, A, F, D, E, F, G, E, C], et cetera. The melodic dimension is the horizontal dimension, and the harmonic dimension is the vertical dimension in music, tones being

heard, tones being heard simultaneously. In Bach, a lot of the time, it's not so straightforward to pick out which is which. Let's listen to this piece as an example [plays Prelude and Fugue in C major; BWV 846].

- 4.26 The texture in that piece is quite interesting for a couple of reasons. The first is that the rhythm is completely repetitive. It's one rhythmic figure repeated over and over and over and over again, more than 30 times with zero variation. Every single measure has the same pattern. That has a couple of effects. The first is that it's hypnotic because it's a pattern that keeps repeating over and over. The second is that, in a sense, it almost takes rhythm out of the equation. You're not really paying attention to it because it's not changing.
- 4.27 In a piece of music, whenever there's a parameter that isn't changing, it tends to slip to the background of your perception. You tend to be focusing more on the things that are changing in the piece. We very quickly understand after two or three bars that this is a pattern that is going to keep repeating, and it's not going to change until the piece stops. But what is changing is the harmony. The harmony changes in every single bar, and it changes in ways that are not predictable.
- 4.28 As far as the texture goes in this piece, you'll notice that it seems to consist only of arpeggios. An arpeggio is simply a chord that's been broken up and played from bottom to top or top to bottom, like this [plays C-E-G-C]. That's an arpeggio. You can take a chord [plays C-major chord], present it like that, or play the notes successively, roll them from bottom to top [arpeggiates C-major chord], and that gives you an arpeggio. That seems to be the primary texture of this piece: a sequence of arpeggios. You can, of course, play that as a chord [plays C-major chord]. Then you'd get that. This piece could, in fact, have been presented to us simply as a sequence of chords [plays the opening of BWV 846 in block chords].
- 4.29 Is it the same? Well, no. I mean, it's not quite the same. It's like the two paintings of apples that we looked at earlier. It's the same exact progression. It's the same tones, but it's presented in a different way as a sequence of chords as opposed to a sequence of arpeggios. But I mentioned also that this piece has the additional feature that the horizontal and vertical are indissociable in it. So what does that mean exactly?
- 4.30 When we're listening to it, we're hearing a sequence of harmonies, and we're hearing a simple rhythmic pattern [plays the opening of the Prelude in its normal arpeggiated pattern]. But Bach is extremely careful about the way that the chords or the harmonies move from one to another. If you listen carefully, you'll see that there's the outline of a melody if you just take the top note of each chord [plays the opening arpeggiated pattern in abbreviated form]. And that gives you [plays the implied melodic line formed by the top notes], and so on.
- 4.31 There are hidden voices in all of those sequences. And likewise, if you take the bottom tone, you get this [plays the implied bass line formed by the bottom notes], and so on.

You notice that there's a shape to that. This is incredibly crucial to creating a sense of movement in a piece of music. The shape is, it's a descending line. The bass starts here on middle C [plays C, B, C, B, A, D, G, G, F, F, E, E, D, G, C, F], and it goes down.

- 4.32 One of the reasons that this piece works is because the ear is carried forward by this strong sense of movement towards the bass. It's difficult to make a piece like that work because, again, the rhythm is reduced to almost nothing. The rhythm is an almost mechanical feature in the piece. The melody is latent. It's present, but you're not necessarily consciously aware of it when you listen to it because you're probably focusing more on the figuration and the different twists and turns that the harmony takes. But what makes it work as a piece, what orients our perception, what orients our ear, is the large-scale movement of these voices, the direction that they take, the journey that they take.
- 4.33 Now, it could have been very easy for a piece like this in lesser hands to produce something banal and boring. A piece where we seem to be going absolutely nowhere, where we're just turning around in circles, and you're just waiting there, waiting for it to finish. Why does that not happen? When you're listening to any piece of music, and you're trying to get a handle on it, a very good question to ask, if you like the piece, is: Why isn't this boring?
- 4.34 That's also a question that you can apply to your own creative work. If you're trying to write a piece, is it boring? Yes, no. This is the little decision tree. If it's not, why is it not boring? Then you try to come up with an answer for that. Because you could say, Well, it's just an endless sequence of arpeggios, the rhythm never changes, the texture never changes, and the register hardly changes at all. That sounds like a recipe for a pretty boring piece. Why isn't it? Then you have to look at the direction of it, the way that it moves, and also the way that Bach creates surprises.
- 4.35 You might notice when you're listening to this that there are times where the music seems to be taking a very almost inevitable path, where we're carried along by the momentum of the harmony. There are other times where there will be a tone, maybe in the bass, maybe in the top, maybe in one of the vocal voices that you're not expecting, and that throws everything off, and suddenly, you no longer know where you are. That effect of surprise is one of the key tools in your arsenal as a composer. You want it to be that the listener is engaged, they're following you, they don't quite know where you're going next, and you can throw little surprises at them.
- 4.36 I'm going to play this, and I want you to pay attention to the spots where the harmony does something that we're not quite expecting. Those are the moments that you want to focus in on and look at them and see, "Why am I surprised? [Plays Prelude and Fugue in C major; BWV 846].
- 4.37 You may have noticed that there are chords in there that really throw you for a loop. For example, when he moves from this [plays a stable consonant chord] to this [plays a

surprising, less expected chord]. Now, that's a possible progression, but it's not an obvious one [plays the progression]. And part of the reason that it's surprising, well, first of all, it's a diminished seventh chord, which is a fairly dissonant chord [plays G-B $\flat$ -E-G-C $\sharp$]. It's a chord that is restless. It wants to go somewhere, but you don't know where exactly, and you won't know until you hear the following chord after it.

- 4.38 But it also takes us from C major to D minor [plays F-A-D-A-D, then F-A $\flat$ -D-F-B]. Then back to C major again [plays E-G-C-G-C]. Now, this chord here [plays E-F-A-C-F]. Again, that's a perfectly possible chord. It's a little bit surprising, though, because he puts this dissonant interval in the bass between the bass and the tenor [plays E-F-A-C-F]. Now, that's a chord. When I say dissonant, what I mean technically is that it's a chord that has a strong incentive to move; it's not a stable chord [plays E-F-A-C-F]. That E wants to move downwards [plays E] to D [plays D, E-F-A-C-F, D-F-A-C-F, E, D, and then continues the Prelude].
- 4.39 Now, notice here, this ties in a little bit with what we were talking about earlier with respect to harmony. If I just look at the top voice in these few bars I've just played, the top note is not moving very much. That's often the case if you want to have smooth voice leading, which is to say, when you're connecting chords together, you may want to make it so that a minimum of movement is required to move from one chord to the next.
- 4.40 For example, if you want to move from C to F, you could just play block triads like this [plays C-E-G, then F-A-C], where the entire hand has to move in order to play that. Or you could just move two fingers [plays C-E-G, then C-F-A]. Whenever there's a common tone between two chords that you want to play, you keep the tone in the same position, and the other tones will move as little as possible [plays C-E-G, then C-F-A].
- 4.41 Bach adheres to that principle throughout this piece. The voice leading is quite smooth. If you look at the top voice in these bars that we just looked at [plays C-E-G-C-E-G-C-E, C-E-G-C-E, then C-G-B $\flat$ -C-E], that voice isn't moving. It stays on E [plays E, then C-E-G-C-E-G-C-E twice, C-G-B $\flat$ -C-E-B $\flat$ -C-E twice]. It stays on E even here [plays F-F-A-C-E-A-C-E twice]. The harmonic implication and the quality of a note can change dramatically based on the context [plays E]. So this E can be the third of a C major chord [plays G-C-E]. It can be the third of a C major dominant seventh chord [plays G-B $\flat$ -C-E]. And it can also be the seventh of an F chord [plays F-A-C-E]. It's the same note every time, but it has a different harmonic function in every chord [plays chords from the Prelude].
- 4.42 I think what Bach is telling us in this piece is that he's giving us a vision of the different components of music being indissolubly linked together, a total perfect fusion of the vertical and the horizontal of harmony in melody. He does this with a piece that has basically no material. There's no theme as such. The figures are reduced down to almost nothing. The entire piece is just movement itself. That's all that's happening.

You start with a figure, you take it on a walk, you take it somewhere else, and you create, hopefully, an interesting and compelling form of movement out of that.

- 4.43 Texture is something that you can work with consciously as a composer, and that can be something that you can start with when you're planning to write a piece. You could say to yourself, "What do I want this piece to be constituted of? Do I want it to be primarily melodic? Maybe I want to make a rhythmic texture with lots of different interlocking rhythms." From the world of popular music, there's an entire spectrum of different possibilities.
- 4.44 You can listen to things like *Fear of Music* by Talking Heads, and there's extremely little harmony on that album. The songs are static. Most of them are one-chord songs. The thing that animates a record like that is the rhythm and the texture and the vocal performance, obviously, but the harmony, there's almost nothing happening. That can be a conscious decision.
- 4.45 You could choose to focus on a melody that's placed front and center—that's almost all pop music does with a simple accompaniment. Or you could choose to write a piece where there's multiple different things going on all at the same time, as in a contrapuntal texture, and none of them, particularly, stands at the top of your hierarchy of attention. They're all equally present, and you can get completely absorbed in textures like that. Or you can write a piece like this, where there's a latent melody, there's a counterpoint, there's harmony, obviously, but they're completely fused together. If Bach placed this piece at the head of a mammoth collection of pieces called the *Well-Tempered Clavier*, which occupied him for years, it's not an accident. This is a vision of a totally integrated musical world, and a very orderly one as well, must be said.
- 4.46 So we've looked at some of the basic components of musical textures. This is something, incidentally, that you can try for yourself when you're listening to anything. Just ask a series of questions. It could be a song you're hearing on the radio, it could be an album you're playing, it could be a piece of music that you're learning at the keyboard. What is the texture? What is this piece made up of?
- 4.47 For composers, there are really three essential pillars to becoming a better composer, to getting a handle on this very difficult art form. One of them is learning traditional techniques like counterpoint and harmony, understanding form, analysis, and so on. Some people will say, "I'm concerned that through studying those disciplines, I'm going to lose my essential spontaneity." That's not true. There is no way that knowing more can hinder you creatively. It can help you. It's very unlikely to hinder you.
- 4.48 I think a good rule of thumb for composers is: assume that those techniques are going to be useful to you because they probably will be, even if you end up writing a type of music that bears almost no resemblance to anything that we've heard so far. You'd be

surprised. A lot of these concepts are very broadly and universally applicable. That's the first pillar: getting a grounding on traditional techniques.

4.49 The second is expanding your awareness of the possibilities of music through listening. The listening is something that you can build into your life. Even if you're very busy, you can always find ways to do this. It's amazing what you can discover if you were to take, for example, just even 15 minutes a day to listen to new things that you haven't heard before and do that ritualistically. Do that every day for five years. It'll drastically expand your musical consciousness.

4.50 Of course, you can do what most people do, which is to listen to the same pieces over and over and over again—I do that, too. I'm certainly guilty of that. Not that there's anything to be discouraged. But you can also stay with the things you know to the detriment of the expansion of your musical universe. It can be very useful to take some time and just sit down and say, "Here's a bunch of things I've never heard before. I'm going to give them five minutes," so continuously expanding and deepening your musical horizons through listening.

4.51 Then thirdly, of course, is your creative practice, your individual personal creative practice, developing that. You learn through doing. There's no other way, really, to achieve mastery of this art form than sitting down and making pieces. It's the same with anything. If you want to be a filmmaker, you have to make films. If you want to be a photographer, you have to go out and shoot. If you want to be a composer, then you have to sit down and write lots and lots and lots of bad pieces. You'll be frustrated, and it's not going to go the way you want, and you'll get stuck. But that's how you figure out what the problems are.

4.52 You solve those problems by listening, because it's extraordinarily unlikely that you, as a composer, are ever going to encounter a problem that no other composer has faced before. That's almost certainly not going to happen. It's like you don't know how to set up a texture. Look at how somebody else did it. You're not sure how to conduct a melody? Look at examples of melodies that are successful. Look at how they're built. We looked at some of the ways you can do and so on and so forth.

4.53 Continuously expanding your knowledge of what exists already. A careful and patient study of counterpoint, harmony, and so on. Then, of course, the development of your creative practice. The thing is, you can do all of these things simultaneously. When I was a student learning to get a grip on composition, I started making pieces even when my means were very—my technical means, my knowledge of music, was very limited. But I would just think, "Well, this is what I know how to do. It's not very much, but I'm going to try to make an expressive project that is coherent with what I already know."

4.54 Maybe that means, again, maybe that means you write 30-second long pieces for a while, but you try to make the best 30-second piece that you can possibly manage. Then you move the target back and try something else. Expand it a little bit. "I did a 30-

second piece. Can I do a 45-second piece? Can I do a one-minute long piece?" Then you target specific problems every time you sit down to write. You could say, "Well, my rhythm is getting a bit square, and maybe I need to focus on rhythm for a bit." So you can design a project that is all about rhythm. Or you can imagine an exercise similar to this where you try to make a piece that consists of not one figure repeating over and over and over again, but make it so that you are completely fascinated by it. You can't stop listening.

- 4.55** How do you do that? In my 30-odd years of experience, I don't know a better way to make progress as a musician in those three pillars. I think they're all necessary. I'll stop there for now. Are there any questions?
- 4.56** It was expressed that music is very important, especially in the Western world. How does one who's not primarily interested in music get to pay due respect and get knowledgeable enough in this field?
- 4.57** I wouldn't say that it's especially important in the Western world. I would say it's important virtually everywhere. In fact, it's very hard to think of any culture anywhere in the world that isn't heavily invested in music in some manner or another, which is interesting in and of itself because different cultures have different priorities and different areas of focus. But music seems to be pretty universal, which is quite interesting.
- 4.58** How do you invest in this art form and give due credit to it? Well, that's never been easier than it is today. There used to be quite substantial barriers, barriers of cost, of availability of materials, and that's gone. We have the opposite problem now. We have the problem of an excess of availability. We can hear any piece of music that's ever been written instantly, without any effort whatsoever, and at no cost. I think the defining challenge today is not so much: How do you find music to listen to? How do you pay attention to it? But rather: What do you focus on? That's a difficult problem.
- 4.59** What I would suggest is you can always start with the pieces that are broadly recognized as being great pieces. It's not terribly difficult to find out what those are. It's the same in popular music. It's the same in classical music. There's a core, if you like, of artists that are recognized as being important historically, aesthetically, because of the breadth and the quality of their contributions. You can start with those.
- 4.60** Imagine that's a core, and then you can expand outwards from there and explore. One artist will lead to another. If you're interested in Mozart, then you might be interested in Haydn as a consequence, and then you keep going down the chain. But again, that's a process, and it takes concerted effort, and focus. You have to set aside time to do that. But I believe the rewards are immense if you do it.
- 4.61** When composing a piece of music, do you see texture as something that is added maybe after you've got the other elements in place, such as harmony and melody, or do you consider everything all at once?

- 4.62 My life would be a lot easier if I could consider everything all at once because then it would just be clear. But no, it doesn't work that way. The initial stages, at least for me—and this is a very personal thing—are marked by confusion because I'll be oriented by instinct. I compose as though I'm sleepwalking in a sense. You have a sense of something, but you're not necessarily consciously directing it yet. You move blindly towards it and you're going to bump into things and make mistakes and go in wrong directions and so on. But you feel your way through it. For me, there's an extended process at the beginning where I don't know where I'm going. I'm heading towards something by instinct, because I'm drawn towards it, but you don't really have a map of the territory yet. It takes a while for that to happen.
- 4.63 Some composers will put together a highly functioning and generalized improvisable technique that they can apply to a wide range of circumstances brilliantly. That can be an extremely effective way of working because you're not constantly torturing yourself all the time with, "How do I make a piece?" You've got a technique and it allows you to do certain things.
- 4.64 But there are other types of composers, I think, who are very restless, perhaps, or who are driven to explore, driven to try things they haven't tried before. When that's the case, it's much less easy to settle on a technique and then deploy that technique. Instead, you're going to be constantly searching, constantly trying things. When that's a part of you or a creative personality, you're not going to have necessarily a set method. You're going to discover things as you go along. You're going to make mistakes. You're going to go down byways and alleys that lead nowhere. But sometimes you're going to discover things that are really surprising to you, and then you find a little thread, and you pull it, and you see where it leads.
- 4.65 In some respects, I wish that I could start a piece and have a very clear method like that and say, "Okay, here's what the texture will be, here's what the duration will be." Sometimes it is like that, but often you get just a glimpse of something, you're not sure what it is, and you're trying to get closer to it, understand what it is, and then you can start to work with the materials a little bit. That's how it tends to go for me.

Lecture 5: Extending Ideas

At a glance — Extending ideas.

- 5.1 Ladies and gentlemen, today we come to one of the most difficult aspects of composition to get a handle on. This is something that I've probably had to work on with my students more than anything else. If there's one thing I can draw from the now more than 20 years of experience I have teaching composition, it's that nearly everybody struggles with this. It's a tricky problem, but there are ways to deal with it. This is form and trajectory in music.
- 5.2 Now, these are two related but distinct questions. The reason this is so difficult, I think, is—we talked about this a little bit yesterday—musical material, in a certain sense, is not particularly difficult to find. We showed how, in painting, you can take a simple subject like an apple or whatever it might be, or a bowl of fruit—there are many different motifs that you see in old classical paintings—and you do something with that. What the artist is really doing is deploying their particular way of handling that material. That's the essential thing.
- 5.3 That's important in music, but there's also an additional dimension which is probably unique to this art form, which is that music, of course, unfolds over time. So we have to deal with the problem of extending an idea. We have to deal with the problem of formal continuity and also cohesion. Coming up with some ideas in and of itself is not the hardest part. A lot of students will get stuck there. They'll get stuck because they'll be hyper-focused on finding a sound that they like, finding a tune that they like, but then they won't know what to do next.
- 5.4 What you see very often is that instead of doing something with it, instead of having it transform over time, they'll introduce a second idea. Then, of course, the problem keeps recurring. Then you need a third idea and a fourth idea, and a fifth idea. Soon, the piece starts to look like this disordered collection of fragments, essentially.
- 5.5 There's a joke about this, a running joke in the Peanuts cartoon, where Snoopy is trying to write his famous novel. He's sitting on his dog house with his typewriter and writing, "It was a dark and stormy night. Suddenly, a shot rang out! A door slammed. The maid screamed. Suddenly, a pirate ship appeared on the horizon." He's writing this ridiculous novel where every sentence is basically a non sequitur. He can't actually tell his story, so he just keeps adding unrelated phrase after unrelated phrase. At one point, he says something like, "I may have written myself into a corner" because there are so many unrelated threads in his story that it becomes totally absurd. It's fun to laugh at that, but it's actually a problem that many, many musicians struggle with.
- 5.6 I'm going to try to simplify this question somewhat, break it down, and lay out what some of the options are for extending an idea in a musical composition. I said we're going to talk about form and trajectory. It's not exactly the same thing. Maybe I'll start

by laying out the distinction between those two things. If you take a course in musical analysis, you'll find that the teacher will almost always begin by laying out some of the archetypal forms that you see in Baroque and Classical music. These forms will often be described in terms of a diagram that can be written down on a blackboard.

- 5.7 These diagrams can actually be very helpful for a number of reasons. One of them is that you start to understand that a piece of music has a formal articulation. That is to say, it's not an undifferentiated stream of ideas, but rather, just like in a piece of prose, there are paragraphs and sentences and individual words, and you can partition these and see how the piece is actually put together. You also learn to recognize things like repetition, transformation of ideas, development of ideas. You start to recognize all of these different ways of extending something.
- 5.8 But the limitation of these diagrams, in a sense, is that, yes, it helps to see how a piece can be laid out, but it's not how we experience things when we're actually listening. So these diagrams, I can show you what they look like. You can see that they can be helpful in terms of giving you a bird's-eye view of a piece very rapidly.
- 5.9 There's a fairly limited number of what you might call archetypal forms or ways of arranging different sections of a piece in any piece of music—this can be a symphony, it can be a two and a half minute song. The diagrams that I was talking about might look something like this. You'll point out that a piece has a first section—we'll call this A. It might consist of, let's say, a first theme. Then you might write down, if it's a piece of tonal music written in the classical tradition, you might point out that it's in D major. Then there might be a little transition that allows us to modulate to another key, usually the dominant. Then when we get to there, we'll call that B, and there might be some other theme, and so on and so forth.
- 5.10 The piece is articulated in this manner as a series of little boxes. A, transition, B. There might be a reprise of A, either an identical reprise or a varied presentation of the same material. Then who knows? Maybe there's a C-section, or maybe there's a reprise of B, and maybe it keeps alternating in B. But you can articulate that, again, as a diagram, a sequence of individual discrete sections.
- 5.11 That does actually work relatively well for some musical forms. If you're trying to get a handle on music, on a particular form or on the works of a particular composer, and you're trying to understand how they put a piece together, it can be useful to make these kinds of notes and pay attention to when something's repeated. And when it's repeated: Is it transformed? Is it restated in identical fashion? Or is there something about it that's changed? And you pay attention to that. This can be of enormous help if you're writing your own pieces and you're trying to figure out what to do with your ideas. You pay attention to how other composers transform their ideas.
- 5.12 In a lot of classical pieces, a theme or a melodic idea might be restated three or four times. But usually, not identically. Then you pay attention to what precisely is changing.

In the case of the Pachelbel pieces that we looked at yesterday, the harmony doesn't change, the bass line doesn't change. What often changes in those pieces is the texture, the amount of figuration, and so on. You start to become sensitized to those, and that will start giving you some strategies for how you might approach this question in your own work. So that's form. You can think of form as the proportion of different sections relative to each other and the order in which they appear.

- 5.13** There are two ways that you can approach this question as a creative person. You can say, "I'm going to start with a top-down approach to my piece. I'm going to start by laying out a formal template before I begin writing." You could say, "I'm going to write a piece in three big sections. It's going to be an A-section, a B-section that contrasts with the A-section, and then a reprise of the A section at the end." That's one of the oldest forms in existence: ABA.
- 5.14** Then you could say, "Now that I've got my A section, my contrasting B section, and so on, I'll go into slightly more detail and start elaborating on the material." But you might start with a form and then invent material—some people work that way, and that suits them just fine. Others will start with a kernel of an idea, like a seed, without necessarily thinking about how that's going to play out over the duration of the piece. It will start with a little idea, and it could be something as small as one chord, it could be something as small as three notes, and then you allow it to grow, or rather, you see what you can do with it, and you discover the form through the process of working with that material.
- 5.15** This is entirely a question of artistic personality. You discover these things through trial and error by trying out different approaches. It might turn out that you're the sort of artist who really needs the security of working out a form ahead of time. I know composers who are extraordinarily precise about that. They will even work out the form down to the last second before they write a note of the piece. They know exactly what the proportions are in the sections, they know how long they're going to be, and they often have a very precise idea of what the material will be, even at that stage, before they start writing the score. Others are happy to just take an idea on a walk as it were, and see where it leads. Think of that as a continuum, of course. It's not that you're on one or the other, but you'll find that the approach that works best for you will fall somewhere along that line.
- 5.16** The difference between form and trajectory: that's something that I want to talk about here because it would be easy to think that form is all you need to think about, that that basically solves the problem. You've got your A, you've got your B, you've got a reprise, you've got maybe a transition, you've got an introduction, you've got a conclusion or a coda and so on, and you're good.
- 5.17** But there's a little bit more to it than that. The reason is that when we listen to a piece of music, of course, we're not thinking in terms of these little boxes. The message of your piece, in a certain sense, is the way that you move from one thing to the next.

That's not exactly the same thing as form. That's why I like to use the term trajectory. You can also think of it in terms of a dramaturgy, a dramaturgical arc in a piece. This has something to do with the way that ideas are transformed in your piece and the nature of the journey as we move through it.

- 5.18** Of course, as there are many ways to move from point A to point B in real life, it's the same in music as well. A musical trajectory can be smooth and continuous with very few surprises, or it can be completely unpredictable, where we have no idea where we're going to go from one minute to the next. There are many other possibilities as well. You could have a situation in which you regularly alternate between two or three ideas. They keep returning, possibly in a different order.
- 5.19** You've probably had this experience. If you're in a new city for the first time, you can, of course, turn your GPS on and figure out exactly how to get from your hotel to the museum or wherever it is you're going. Or you can just walk out and start moving around at random and seeing, "Oh, there's an interesting street. I'm going to go down there. Now I'm going to turn left. Now I'm going to do this." You circulate around and you don't know where you're going to end up.
- 5.20** It's the same with a musical composition. That is to say, you want to think about what the experience of the listener is going to be as they move through it. Are they going to be steadily guided by a predictable and even an inevitable structure that holds everything together? Again, that's the case with a lot of songs. Songs use one of the oldest formal principles there is, which is the refrain idea, the idea that you have something fixed that keeps recurring over the course of the piece, alternating with something unfixed that changes each time.
- 5.21** In songs, if you want to schematize this—of course, this is an abstraction, and individual songs inevitably will deviate from this norm—but you'll typically have a verse, a chorus, a verse, a chorus, and the choruses are repeated. They're repeated as many times as the songwriter needs to tell the story that they're telling or paint the picture that they're painting. The verses typically are different each time. They have different words, the arrangement might change from verse to verse, but the choruses are, generally speaking, pretty much the same when they return.
- 5.22** That is an extraordinarily effective and economical way of planning a form because one of the basic problems that any musician has to deal with is: How do you make the piece comprehensible by using repetition, the redundancy that we talked about the other day, but how do you avoid monotony? Because you need both. You need variety so that the piece is interesting, it's not completely predictable. But you also need some measure of consistency so that we don't feel like we're spinning off into infinite space with no idea where we are. We have no bearings.
- 5.23** That form, where you alternate a fixed element with little episodes that can change, is one of the most efficient ways to do that. You have consistency that comes from the

repetition of the chorus, and you have variety that comes from this simple principle of alternating verses and choruses. That's a very simple little form, and you can describe it visually quite easily.

- 5.24** It could look something like this. You might start with a little introduction. The introduction is usually instrumental. If it's in a song, you might have a few bars of piano playing or something. What that does is it establishes a character, a mood, a tempo. It lets the listener know where they are. Then you might begin with a first verse. You'll also notice, if you're listening to songs, you'll notice that very often, they ease you into it. The introduction in the first verse will often have a more stripped-down production style. There won't be a lot of elements necessarily. Part of the reason for that is to present a clear space for the vocalist, so that the personality of the vocalist can immediately make its presence known.
- 5.25** Then we'll have a first verse followed by a chorus. The chorus typically is a little bit more built up than the first verse. There will be instruments added. The dynamics might change. There's a cliché about grunge music that they have two dynamics, which are soft and loud, and they just alternate them with the choruses being loud and the verses being as quiet as possible. It's a caricature, although it's somewhat true. Then you'll keep going from there: verse two, chorus, et cetera, et cetera.
- 5.26** You can elaborate upon this simple form, of course, if you want. You can add additional features, but that's the core of it. You notice that all you really need, in order to keep things both coherent and with sufficient variety to stay interested, is two ideas. That's enough. You need a first thing and a contrasting thing, and then you can keep your listener engaged with that for a while.
- 5.27** The nature of that trajectory, if you like, is a fairly limited one because there's not going to be a huge span of transformation over the course of a two- or three-minute piece, typically—unless you're Queen and you're writing Bohemian Rhapsody, but that's a whole other category. There are very few examples of pop songs that do things like that, by the way, partly because it's enormously costly. It's very difficult to do, and you're taking a massive artistic risk if you do it.
- 5.28** Good Vibrations by the Beach Boys is another example of that, of a multi-section piece that is completely unpredictable in its form, and that takes you through many different texture types in many different sections over the course of it while still offering a coherent trajectory. It's actually very, very difficult to do that. But that's the basis of that approach to form.
- 5.29** Again, typically, there's not a very significant range of transformations within that form. You can think of that as an episodic form—episodic meaning that we have a discrete section, followed by a contrasting discrete section, followed by a contrasting discrete section. Maybe some of them return cyclically, and maybe they don't, but each section is distinct from the others, and they might contrast with each other in a number of

different ways. You might modulate from the tonic to the dominant, for example. You might change the instrumentation. You might change the tempo. You might change the character, the mode of expression, etc. There are all of these different parameters, again, as in the example of the Pachelbel, that you could imagine changing. That's an episodic form.

- 5.30 But there's another approach that you can also take. This approach is the dialectical form. The dialectical form is something that is strongly associated with the high classical style. This is the style of Haydn, Mozart, Beethoven. It's a very difficult form, in a sense, to get a handle on, but the basic principle of it is not that complicated.
- 5.31 As its name suggests, it's a dialogic form in which, like the episodic form of songs and other related forms, you have distinct contrasting ideas. The difference is that in an episodic form, these little sections are partitioned off. They're distinct from each other. They contrast with one another, and they don't necessarily change each other. To a certain degree, you could even permute the order of sections within an episodic form without necessarily changing the comprehensibility of the piece.
- 5.32 If I were to take those six variations by Pachelbel, permute them, and play variation number three before variation number two, the piece would be just as comprehensible; it wouldn't make any difference. In a dialectical form, that's not the case. The reason is that the different materials in a dialectical form are talking with each other, so to speak.
- 5.33 This is really modeled on an idea from antiquity, the platonic ideal, where in the Socratic dialogs, you'll have a position, A, and then a contrasting position B. Two people will have a discussion. They'll start out by laying out their positions and saying, "I think this," and the other person says, "I think that." Through a process of dialog, they'll gradually come to understand each other's positions and possibly modify their own. They might say, "Features of what you just said are actually interesting, and they complete and correct features of what I've just said." Person B might also say the same thing if it's a productive conversation. Then at the end of it, there will be a synthesis of those two points of view.
- 5.34 This is essentially what happens in dialectical forms: you have an idea, a contrasting idea, and initially, you lay them out as a pure opposition. Once you've exposed the ideas—this is called an exposition—then you can start to pull them apart and have them dialogue with each other. Over the course of this, the ideas change; they change because they're interacting with each other.
- 5.35 It's a formal principle, but it's also maybe better to think of that as a process. It's a way of imagining a musical continuity. Just as in a conversation, you've probably all been in a situation where you're talking to someone who's not really listening to you. You say your thing and then they say something unrelated, and then you're trying to bring it back to what you were saying, and it doesn't really work. In a true dialectic form, the

ideas are initially distinct from each other and gradually start to shift and interact. There's many, many different ways you can do that.

- 5.36** One of the questions you might ask as a composer is: What do I do next? What do I do with these ideas, with A and B, or A, B, and C, or whatever it is? It could be, "I could present them simultaneously." Or, "I could combine them and have a polyphonic texture where I have two ideas going on at the same time." So, do you hear them at the same time? Or do you hear them in a sequence? Maybe it's A and then B. What happens when they come together? Do they blend together harmoniously? Or is there a distinct tension that occurs when that happens, where they seem to be pushing apart from with each other. These are essentially dramaturgical considerations. What is the nature of the continuity in this piece? Again, is it full of contrast, is it full of opposition, is it full of tension? Or is it pretty much smooth sailing all the way through?
- 5.37** There are pieces of music, in which the range of transformation of the ideas is very slow and gradual, so that when you're listening to it from one moment to the next, you're not necessarily aware of any particular change or any particular contrast in the material. It's not until you've heard four or five minutes that you can start to see that there's a gradual change happening.
- 5.38** In other pieces, you're constantly being jolted upright by a sequence of totally unexpected contrasts that give the piece a very strong degree of dynamism. Musical forms and trajectories, again, can be smooth and continuous or sharply dynamic. That's one of the main features that gives a piece its particular character: How do you join ideas together? What is the process by which you transform them?
- 5.39** The biggest secret in composition is the fact that you hardly need anything to make a piece. All of the skill of the composer resides in their capacity to see the possible scope of transformations of an idea and to lead us through that.
- 5.40** Take, for example, the American minimalist composer, Steve Reich. Steve Reich became very famous in the 1970s, along with Philip Glass, for giving us what was at the time quite a new experience of music, one in which there is no dialectical principle at all. That's not there. But there's also no episodic structure either. Instead, he made these gleaming continuous pieces where simple rhythmic figures are gradually transformed over the course of a long duration. It could be as much as 20 minutes or more.
- 5.41** What he would do would be to set up a rhythmic pattern and then gradually transform that pattern. He might add a beat to a particular line in the texture or subtract a beat and then keep doing that until it shifts into something else. When that's happening, it's difficult to speak of form exactly because there are no discrete sections in the piece, but you can speak of a trajectory. You can speak of a trajectory in the sense that we start here, and over this amount of time, it gradually transforms into this. There's a piece by Steve Reich called Six Marimbas. As is typical of a lot of his music from this

era, he takes six of the same instrument and uses it to build up these very absorbing and complex rhythmic textures. Just remember this, that this is a piece with almost no contrast in it.

- 5.42** Contrast is one of the most fundamental aspects of articulating a musical form. Typically, if you want to have more than one thing in a piece, then you make it so that they're distinct from each other. You can easily tell them apart; you can recognize it when you switch from one to the other. Contrast is also a way of injecting a certain dynamism into a composition. When you completely eliminate that, then you're forced to listen in a very different manner.
- 5.43** You can become completely hypnotized by this very gradual and slow process of transformation. In fact, the composer doesn't even really need to do that much. It's like he sets a process in motion. Then he is very careful, of course, about the timing, the duration of each little transformation, but he has an idea of where he's eventually going to end up. It's a piece with no discrete sections, no real contrast, only a process of transformation.
- 5.44** If you were to make a little schematic of something like that, it might look something like this. It's a continuous curve of transformation. You can, of course, plan that out over as long a duration as you want. Maybe this is one minute, two minutes, three minutes, four, et cetera, until we get to 20. Some artists find this actually a very helpful way to imagine a piece because they have the proportions plotted out. They know how long the process is going to take. They know where they're starting from, and they know where they're going to end up. That's one approach you can take.
- 5.45** The reason that this took off and was very popular at the time, and actually still is, is that it gives the piece an automatic legibility, an automatic comprehensibility. You can see where you're going. You're carried along by this process, which gives it a certain degree of predictability, but you also don't exactly know what way it's going to change and where you're going to end up at the end of it. A lot of listeners found that enormously reassuring, hypnotizing, and exciting.
- 5.46** In a piece like that, you might ask: What is the idea exactly? What's the material in a piece like that? There's not really anything in the way of melodic invention. The rhythm is this steady pulse. The pulse doesn't really change. The articulation of the pulse and the exact figures gradually transform over the course of the piece. But what exactly is the material? Well, the material is the transformation itself. It's the process of transformation. You don't need very much.
- 5.47** This always reminds me of a quote from Hamlet where he says, "Even if I were enclosed within a nutshell, I would still be a king of limitless space." What that means is, even if you were confined to something infinitely small, you could still transcend it. Even if you had almost nothing to work with, you could use the power of your imagination to

transcend the limitations that you're working within. That is a really important lesson for an artist.

- 5.48** Whenever you're tempted to keep adding, keep adding another idea, stop and first think, "What could I do through the power of my imagination to transform what I already have? Have I really considered every possibility of transformation? Can I take an idea and transform it dynamically?" There are all sorts of trajectories you can imagine. A simple dynamic trajectory is just a crescendo that can build up over time [plays a piano figure, increasing volume to illustrate a crescendo]. You just keep going. You start with something very discrete, just a single note, and gradually expand it outwards. I could keep that going for 10 minutes. That's a type of process or a type of trajectory that you can have in a piece.
- 5.49** So it's useful to think of those two things together. The form is something like the articulation of a piece into discrete sections and the way that those sections are organized relative to each other. Then the trajectory would be the way in which you move, the way in which your materials are transformed over the course of the duration of the piece. I want to emphasize once again that you need very little to do that, because what's really holding your listeners' attention is the movement, the nature of the movement.
- 5.50** In an example from the operatic repertoire, the opera *Das Rheingold* from Richard Wagner has a very interesting prelude. Prelude is about four or five minutes long, and it's completely static along almost every dimension, but it's totally bewitching and haunting.
- 5.51** The harmony is 100% static. Nothing changes. He stays on an E $\flat$ major chord for minutes at a time. One chord. The rhythm, well, there's basically no rhythm to speak of at the beginning of the piece, just a held sound, a held chord that gradually constitutes itself. As the piece progresses, the sound of it gradually changes. It changes through orchestration, through adding instruments of the orchestra, gradually building up the texture.
- 5.52** There are a lot of songs that do this, too. I mentioned this earlier that you can start with a very, very stripped-down instrumentation, perhaps just a guitar and a voice or a piano, and then you add a bass, and then you add a keyboard, and then you add drums, and then you add a synthesizer, and so you keep building it up in layers. That's an additive process or an additive structure in a piece of music. It's very effective. It gives you a built-in trajectory that you can follow when you're listening.
- 5.53** What Wagner does here is that at least one of the parameters in this piece is frozen. It's just out of the equation. It's not moving. You can do this with any parameter in a piece. You can say, as in the Bach piece we heard yesterday, the *Well-Tempered Clavier* prelude, the rhythm is static; it doesn't evolve. It's the same figure over and over and over again. No movement there, no transformation. In the case of Wagner, that's the

case in the harmony. The Bach piece was constantly moving around in unpredictable ways harmonically, but the timbre, the texture, and the rhythm were static. In the Wagner, there's zero harmonic motion, but the texture, the timbre, the orchestration, and the dynamics are all evolving.

- 5.54 You can do that over as long as you want, essentially, provided you have the skill to keep your listener engaged. Let's listen to the beginning of the prelude to Das Rheingold by Wagner.
- 5.55 [Das Rheingold by Wagner]
- 5.56 That's a pretty simple form. It's like a wedge. It starts from a point and gradually amplifies until it reaches a paroxysm at the end, a point beyond which it cannot go. It's reached a point of maximum intensity. But it's extraordinarily effective. The reason is that form has a very strong degree of directionality built into it. That's another very significant parameter to bear in mind when you're working on a piece of music, or for that matter, any story: What is the degree of directionality in the piece?
- 5.57 A piece of music can be goal-oriented. It can have a sharply teleological structure in which you're being drawn forward towards an ultimate endpoint. The endpoint is already implied in the beginning. When you start listening to the Wagner, it takes you a few seconds to realize what's going on. But once you do, you get the picture. The piece is a long, gradual crescendo. The orchestra is gradually building up layer by layer, and it's gradually getting louder and louder, and thicker, and more intense. That's a strongly directional form.
- 5.58 At no point does Wagner deviate from that essential trajectory. Other forms are much less directional in the sense that you cannot predict at the outset where it is you're going to end up, not with any reasonable degree of certainty. The stronger the directionality is in a piece, in a sense, the easier it is for the listener to follow what's going on.
- 5.59 Those are a couple of examples of what you might consider to be process forms, processes of linear transformation, where the material, again, in the case of the Wagner, the material is really his consummate skill with orchestration, his capacity to build up an extraordinary texture, layer upon layer upon layer, without needing to really do anything with respect to harmony. Of course, he was also one of the great masters of harmony, but in this particular example, it wasn't necessary.
- 5.60 I'm going to stop there with form for now. There's a lot more that I could say about it, but I think I would just emphasize that the way you think about form, the way you plot the course of your piece, is in many respects more important than some of the individual elements that go into it. Are there any questions?
- 5.61 How does one get to explore a vast array of ideas like Schönberg without falling into complete unrelatedness?

- 5.62** You're referring to a piece that I presented in one of my previous Peterson Academy lectures. I believe it was the Great Composers, and I talked about Arnold Schönberg, who was an Austrian composer who was active from the late part of the 19th century all the way up to his death in 1951. So Schönberg enacted quite a radical break with many traditional forms of harmony in his music, and also traditional forms in general, in the sense that he began writing pieces around 1909, 1910, that broke entirely with the classical models of how you would organize a piece into discrete sections.
- 5.63** He started writing these pieces that, in many respects, were almost a musical analog for what was happening in abstract painting at the time. Painters such as Kandinsky, for example, where you no longer have traditional subjects in recognizable forms. Instead, you have arrays of shapes and colors that communicate directly through the play of intensities, the plays of inherent contrasts and tensions that can exist within elements in a picture.
- 5.64** When you look at pieces like this, and the piece that I played as an example, I believe, was the first piano piece, Opus 11, by Schönberg, you're confronted with this question: If you have this total liberty in working out a piece, what's grounding you? How do you orient yourself? How do you make a choice to go here or to go there? The first thing I would say is that's probably not something you want to attempt if you're a beginner in composition because you're putting yourself very much in danger in the sense that you'll have nothing to anchor you. What that's going to do is to make the process exponentially more difficult than it needs to be for somebody who's trying to get a handle on an art form.
- 5.65** How did Schönberg get away with it? How was he able to do that? There's a couple of reasons. First of all, Schönberg had an extraordinarily profound understanding of classical forms, structures, processes, and also of harmony. If you listen to his earlier music, the things he was writing towards the end of the 19th century, *Verklärte Nacht*, for example, which is a string sextet, they're written in a late-Romantic idiom that is pushing at the edges of what you can do in that style, but still very much recognizable as a late 19th-century piece.
- 5.66** He absorbed those forms thoroughly into his fingertips. He could work within them the way that you speak a native language. It's not something you have to think about. You're not reaching for syntax. You're not reaching for words when you're speaking your native tongue. He was thoroughly enmeshed in that way of imagining music, which is essentially the Germanic tradition, particularly the 19th-century Germanic tradition of development. He got that essentially from a mixture of Brahms and Wagner, but also from many other composers in that lineage.
- 5.67** When Schönberg enters this creative phase, that's somewhat akin to a crisis where he's pushing at the edge of what you can do in this language. Eventually, some of the old supporting structures just become irrelevant. It becomes no longer helpful at all to think in terms of classical forms because they've become so transformed, so free, that

they don't really apply. They no longer exert the same gravitational force, the same directional force that they once did. At a certain point, he just decides, "It's not helpful anymore." In a sense, he unmoors his boat and heads out into the waters and sees what will happen.

- 5.68** The first thing to recognize is that that was the product of a crisis in a very, very particular time, not only in human history, but also in artistic history, musical history, certainly, where he was compelled to see what lay around the corner, what's over the edge of that cliff, if I look, and putting himself into a very real form of existential danger as an artist, and also professionally and as a human being in doing so. Many of his colleagues at the time thought he was mad. This often happens when people experiment, when they try things that haven't been tried before. Obviously, the outcome is not known in advance, and especially in a conservative milieu like Vienna at that time, where audience goers really did not like artworks that pushed too hard against established conventions, that was a very difficult thing to do.
- 5.69** As it happens, he was successful at it. The pieces are extraordinary, and they show us entirely new continents of meaning and sensation and emotion and sound. These are amazing pieces. But to do that, to get on a tight rope without a net, you have to be in complete command of your resources. You have to have a virtuosic command of harmony, of structure, of form. You can still fail; it can still not work out. But he was supremely confident in his abilities to do that, and that's the result. It's not as though every time he picked up his pen, he produced a masterpiece. But he had that confidence and that solid grounding in traditional techniques that allowed him to make those kinds of explorations without his creative faculties abandoning him completely.
- 5.70** For most creative people, working in that way is intensely costly, cognitively speaking, and also in terms of the amount of risk you're taking on. There again, that's a question of temperament. Schönberg, by temperament, was drawn to experimentation. He was drawn to explore. He was one of these musicians who always had to see how things worked and then try to come up with a new model, try and come up with a new way of doing things and see what happens. That's a question of temperament.
- 5.71** Others are going to be much more interested in sticking with forms that they know and understand well. There's an example of the Italian composer, Vivaldi, who wrote hundreds, about 500 concertos. There's a rather snide remark by Stravinsky who said, "I can't imagine a more dull person writing the same concerto 500 times." That's a little unfair to Vivaldi. Those concertos are a high achievement. But it's true that he kept reusing the same basic template over and over and over again in these three movement concertos. You can say the same with the keyboard sonatas of Scarlatti, of which there are hundreds, or the dances of Schubert for solo piano, hundreds and hundreds of those as well, all following essentially the same template. That's a question of temperament.

5.72 I suppose the disadvantage of working as Schönberg did was, in a sense, it makes it much harder to produce a piece, and you can't repeat forms, really. Each form is completely unique. So that's both an advantage and a disadvantage.

Lecture 6: Shaping Expression

At a glance — Shaping expression.

- 6.1 One of the points we keep coming back to over the course of these lectures is the fact that no matter what type of a piece you're making—it can be a 30-second long piece or it can be a 10-minute long song or a half-hour long movement of a symphony—it's important to always work with an expressive intention.
- 6.2 Remember we talked about how Hunter S. Thompson had that sign above his desk? "What is the intended effect?" That is a question that must always be at the forefront of your mind, at all stages of the creative process. Because it's very easy, once again, to lapse into automatic habits or to think strictly in terms of the material you're working on, the processes that you're applying towards it, but not necessarily thinking in terms about what the listener is going to experience, what you want them to experience.
- 6.3 There's a wonderful anecdote about Alfred Hitchcock when he was working on the film Psycho. He made the remark to a journalist that he was playing with the audience's emotions the way an organist pulls stops on their instrument to change the sound. He was saying, "I pull this stop and they shriek with horror. I pull this stop and they laugh. I pull this stop and they relax." It was all intensely manipulative and very consciously so. Of course, that is plain to see. You can see any one of his films, and instantly there's an atmosphere that's created that has emotional resonance for the audience. That's never very far away.
- 6.4 It's the same in musical composition. Even the simplest piece, no matter how rudimentary, no matter how severe the constraints you're working within, even if you're in a nutshell, you can still write with an expressive intention. This is something that makes the process of learning to compose much easier because you can always be guided by the effect that you're trying to create. You're never going to be confronted with a totally blank page with no idea what to do, as long as you have some kind of an intention expressively.
- 6.5 I'm going to walk you through some of the ways that you can do that, some of the ways that you can shape material and give it a stronger sense of character. We're going to talk about expression, character, and movement. Expression, fundamentally, is working out what it is you want to say in the first place. Character is something like, how do you want to say it? What do you want that to be like? Movement is something like the process or trajectory through which that plays out over the course of the piece. That's something we talked about a little bit with respect to form, but it's relevant to this lecture as well.
- 6.6 What you can do when you're trying to make a piece of music, whatever it is, is start with the question, again, I'm going to keep repeating this: What is the intended effect? That's number one. Number two: What is the process, or trajectory, through which the

piece advances? Again, that takes you back to the question of the nature of the journey. Smooth and continuous, unpredictable, full of contrast, full of dynamism, minimally dynamic, very high predictability, very low predictability? Those sorts of questions. Then finally, the third question to ask is: How do I create this result technically? The answer to that might be extremely different depending on the type of piece you want to write and the duration of the score.

- 6.7 What I want to do is take you through some examples of how a composer can instantly project a very strong sense of character. This is where we come back once again to the parameters of music. That's where you want to think very carefully about, what sort of rhythmic sensation do I want to create? What sort of harmonic sensation? What should the melodic line be like? If there is one; maybe there isn't one. You can start asking these sorts of questions, and it really helps to simplify the process.
- 6.8 In Classical music, and in Baroque music before that, to some degree, these questions weren't things that composers had to torture themselves with because when you were writing a piece, it would be embedded within a stylistic formal framework, and you'd also be working within a set genre. All of those things had conventions attached to them.
- 6.9 For example, if you were writing an aria in a Baroque opera, there were formal, structural, and expressive conventions that went along with that as a matter of course. If you were writing a dance suite, then there wouldn't be an infinitude of options to choose from. You'd be writing maybe a gavotte, or a minuet, or a bourrée— there were all kinds of different dance styles that were popular in France and Germany and Italy, and so on.
- 6.10 There was a repertory of rhythmic figures that you could choose from, and those rhythmic figures would predictably create a kinetic result in the people listening to it. You might want to write something with a tragic form of expression. You might want to write something that was lively and dance-like and so on. There were conventions associated with all of these forms of expression.
- 6.11 The other thing that really strongly gives a piece a sense of character is its articulation. This is, again, where we come back to the connection between music and speech, which is a very, very strong one. Whenever we speak, of course, we're dealing with articulation, with the distinction between different sounds and the degree to which they're pronounced. It's exactly the same in music. Music also deals with articulation, and it's one of the ways that you can project a very strong sense of character.
- 6.12 There are languages that are very strongly articulated. The French language, for example, is very articulated. There are languages that fall on the opposite end of that spectrum. My wife is from Denmark. I don't know if any of you have ever heard Danish, but it sounds like they don't articulate anything. It's quite funny if you're not used to it, and it's inarticulate—it's very difficult to learn to speak Danish if you're not from there.

They can understand each other somehow, despite the fact that there's very little articulation going on. Different languages have different degrees of articulation that they possess as a natural property of them. It's the same with pieces of music.

- 6.13** Let's go through a very, very simple little musical idea, and I'm going to show you how you can shape it and give it a character and a sense of articulation through many different means. Notated music has evolved to carry a whole repertory of symbols and notations that convey to the performer an expressive and an articulatory intention. So that just by looking at the sheet music, the performer can see at a glance what the intended character is. You might ask: Why is this worth devoting a lecture to? Why is this so important?
- 6.14** The fact is, in the mid-18th century, or even earlier, for that matter, these sorts of articulation markings and expression markings would have been largely superfluous for the reason, once again, that if you were writing a piece, it was in all likelihood embedded already within a very specific expressive context that the audience knew, the performers knew, and the composer was intimately familiar with.
- 6.15** You didn't necessarily have to spell it out in the same way that when we write down written language, we don't put thousands of diacritical markings over every word to explain to the reader how it's meant to be pronounced, where the accent patterns lie, where the points of emphasis are, and so on. It's implied, partly through the context, through the nature of the text, and the reader can usually pick that up intuitively without too much difficulty.
- 6.16** For a long time, this was the case in music as well, where the performer would have instinctively understood the articulation patterns, the accent patterns, the rhythmic patterns, and how to perform them. This is where it gets a little bit tricky, though. If you're not within that context, if you don't live in Leipzig in 1723, then some of these things that would have been culturally obvious to people in that context at the time might not be obvious to you. You might not instinctively know how a particular phrase is meant to be played—with what articulations, with what dynamic marking, and so on.
- 6.17** With the rapid expansion of different musical possibilities, different musical languages, different esthetics throughout the 19th century, and the increasing ease of travel, and the increasing mix of populations, and the increasing amounts of music that were being published also and distributed all throughout the world, it started to be the case that this assumption, this shared assumption between performers and composers and audiences, started to become somewhat less obvious.
- 6.18** Composers started to become a lot more careful, a lot more precise, about their dynamic markings. If you open up a Baroque score, like this Pachelbel piece—just open the score up at random—you'll see that there are no dynamic markings, there are no articulation markings. There's no tempo marking. It doesn't say how fast to play. It doesn't say play forte or play piano or play mezzo piano. It doesn't say articulate the

notes in groups of fours. There are no hairpins to indicate a crescendo or a decrescendo. There are no accents. It's just notes and rhythms. From the standpoint of a contemporary composer working today, this appears to us to be quite minimal. There's almost no indication as to how this piece is meant to be played.

- 6.19 Some composers went even farther than that. The Art of Fugue, which is one of the late works of Johann Sebastian Bach, not only contains none of those markings, but it also doesn't even tell you what instrument to play it on. So the musical text of the Baroque was considerably more open in certain respects than is the case today. But remember, again, that these pieces would have been embedded with very specific performance traditions and performance practices, which would have made those markings completely superfluous at the time.
- 6.20 When you move into a situation, as the 19th century wears on—where you have a profusion of different styles, profusion of different playing techniques, a profusion of different esthetics, all happening at the same time—well, you have to be more specific in your notation because you can no longer assume that the performer is going to instinctively understand your intentions. Let's look at an example of what that can look like.
- 6.21 When you're teaching composition, one of the first things you have to establish, if you're looking at a score, is, "Do I see at a glance what the composer wants? Do I see at a glance what the intention is? And will the performer understand that intention, moreover?" Because if they don't, if they look at their part—maybe it's a harp player looking at what they have to play, or a tuba player, or a flutist—they've got their part in front of them, and it's not obvious from the context what the intention is, musically. They don't really know.
- 6.22 Then what they do is they sightread what they've got. People are incredibly good at sight-reading now—they can do this accurately without too much difficulty. But they're going to give you a very flat reading of what you've written because it won't be clear to them how to shape it, how to shape it into something expressive. This is where expression markings become really essential. I'll give you an example of that.
- 6.23 Okay, so let's say I'm writing a score, and I've got a little idea. It could be something like this. If I write my idea down like that, as you can see on the screen, it's just a succession of 16th notes. I haven't told the performer anything about how I want this to sound. So this is a kind of shape, I suppose, a little phrase shape. But if I'm an oboe player looking at this, I don't really know what to do with it. It could be something like this [plays the phrase with a slightly hesitant, springy rhythm], sort of a little bit hesitant with a springy rhythm like that, a medium tempo [repeats the phrase with the same expressive quality].
- 6.24 Now, you could give that a very tentative, hesitant quality, depending on the way you play it [plays the phrase with marked hesitations and elastic timing]. Notice that I'm not

playing it completely regularly. The rhythm is what we call a rubato, where it speeds up a little bit, slows down a little bit. There's a pulse, but it's very elastic. The spaces between the notes are not regular. This is where expression markings come into it, because another player might look at this and play it like this [plays the phrase evenly, without rubato]. It's completely straight, with maybe a more emphatic expression [repeats the phrase with the same expressive quality]. But we don't know which one it is.

- 6.25** Okay, so then you start engaging with the repertory of symbols that help to make your intention clear. In the case of the first, you might write something like this. And notice that this marking, *leggiero*, is in Italian. For reasons of convention, expression markings are typically given in Italian in musical scores. There are some exceptions to this. There are some composers who, because of their particular temperament, or their adherence to their country, their language, or whatever it is, choose to write markings in languages other than Italian. There are some amusing examples of this.
- 6.26** The composer, Percy Grainger, for example, insisted on writing everything in English, including very standard markings like *mezzo piano*, *mezzo forte*, and so on, or *accelerando*, which means to speed up, or *rallentando*, which means to slow down. These are very basic terms that are widely adopted by musicians throughout the world. But Percy Grainger thought that that was pretentious, and he wanted to put everything in English. So instead of writing "crescendo," he would write, "louden." And if he wanted a "molto crescendo," substantial crescendo, he would write, "louden lots," which is pretty silly, but that's part of what makes him an interesting character.
- 6.27** Anyway, in this example, I've used the Italian term *leggiero*, which indicates to the player that this is to be played with a lightness of touch, a light articulation. Then we have additional markings that tell me exactly how this should be articulated.
- 6.28** Now, let me give you an example from a Canadian who knows this. If you're from Toronto, which I am, you don't say "Toh-ron-toh," you say "Tuh-ruh-no." You kind of swallow it. You don't articulate every syllable. I've been overseas for long enough that I don't do that anymore, but I know how to do it. Pronouncing every syllable gives you a very different sound and character.
- 6.29** In music, it's the same. Now, I can have a legato articulation, in which the notes are, as it were, grouped together in a single continuous phrase without a separate articulation on every note. There's no gap between the tones, in other words. A legato articulation would sound like this [plays the phrase smoothly and connected]. If I were to articulate every single note distinctly, you would have this [plays the phrase with short, detached notes]. The way you notate this is either with what's called a slur marking or a phrase mark, which is like this. Or in fact, more likely you would put it underneath. That would look like this. That's a curved line that connects the phrase together from the first note to the last. It tells the performer, "Do not articulate every single note separately."

- 6.30** If you're an oboist, or a flugelhorn player, or a tubist, then you have many different possibilities to articulate notes, depending on whether or not you put your tongue to the embouchure between every note. If you do, then you get a "ta, ta, ta, ta, ta" kind of articulation with every single note being separately attacked. If you do not, then you set the air column in motion on the first tone you play, but you do not rearticulate between the notes. You don't use your tongue at all. Something very similar is possible on a keyboard as well. Here's the legato [plays the phrase smoothly and connected] and the staccato [plays the phrase with short, detached notes].
- 6.31** Now you can mix the two. This is where it can get very interesting because it doesn't have to be one or the other. In Baroque music, it was very common to do two notes slurred and then two notes articulated like this [plays the phrase with two notes smoothly slurred followed by two distinctly articulated notes]. That gives you a very distinctive [vocalizes the pattern: ta-da ta ta, ta-da ta-ta, ta-da ta ta] kind of rhythm. It's a light, more elastic approach to articulation.
- 6.32** But of course, you can do it many other ways as well. For example—now, these little dots that I'm putting underneath the tones are staccato markings, which mean that the note is to be played short and articulated. Start that again [plays the phrase with one staccato note followed by three legato notes]. There, I'm exaggerating this in order to make the point, but there we have one staccato note and three legato notes.
- 6.33** Again, every time I change that, I get a different character. I get a different musical sensation from what I'm playing, and it can really transform the material. Suddenly, it's gone from being this shapeless, overcooked piece of spaghetti, to something that has an articulation, an intention, and a shape. You are shaping the phrase.
- 6.34** Now, I still haven't put in a tempo or a dynamic indication. That tells the player how loudly or how softly you want the passage to be played. Now, some instruments—and we'll talk about this a little bit more in the orchestration lecture, which is coming up a little bit later—different instruments have a different dynamic range, which is to say that a different capacity to play a wide range of dynamics between soft and loud. If you have a small dynamic range, that means you can't make much of a difference between the two. It means that your quietest sound is not going to be that much quieter than your loudest sound. Conversely, other instruments have a very wide dynamic range.
- 6.35** That's one of the things that you learn as a composer when you're writing for this or that instrument is, you learn to instinctively grasp what the range is that you can work with. If you're writing for an oboe, for example—the oboe being a double reed woodwind instrument—the oboe has a comparatively limited dynamic range. It has one, but it's not that big. It's useless to write triple piano on a low note on the oboe. They simply can't do it technically. The instrument doesn't allow it due to its construction, due to the nature of the reed, due to the way that the instrument is played; you can't make those kinds of very, very large dynamic jumps on an oboe.

- 6.36** But an instrument like the snare drum, for example, has a colossal dynamic range. On a snare drum, you can make sounds that are almost inaudible using wire brushes—very, very soft. Then you can pick up a set of wooden sticks and play something loud enough to damage the hearing of someone standing next to you. That's an enormous dynamic range. A symphony orchestra is like that, too. The orchestra taken as a whole, can produce everything from something so quiet that you can barely hear it to something that is potentially damaging. That's one of the professional hazards of being an orchestral musician, incidentally, especially if you play in the brass section or the percussion.
- 6.37** You need to tell your performer what the dynamic marking is. Because we're not in 1680, unfortunately or fortunately, so this is no longer taken for granted. I'm going to have to decide. Now, a lot of composers will start their pieces with this. This marking that I just put is "mp," which stands for mezzo piano, or medium soft. Mezzo forte, "mf" equals medium loud. You might see a marking like this, triple piano, "ppp", or pianississimo. I think I have one S too many in there. Oh, well, you'll forgive me, I'm not Italian. Pianississimo, fortissimo, is "ff" and so on. There's a whole spectrum of dynamic markings you can write from very soft, pianississimo, through to mezzo piano, mezzo forte, forte, fortissimo, fortississimo, fortississississimo, et cetera.
- 6.38** It depends on how much you like to exaggerate, but you can keep going. The language permits it. That looks something like this, from very, very soft to very, very, very loud. Again, if you're writing for a harp, for example. It might not make sense to write these kinds of extremes of dynamic markings because the instrument might not be able to meaningfully distinguish between them.
- 6.39** On a piano, you can do a pretty good job of that. If you have a solid technique and a good instrument, you can make a pretty big crescendo. You can also play very, very subtle distinctions of dynamics. One of the things that you learn to do as a player, and that you become sensitive to also as a composer, are these very subtle dynamic shadings.
- 6.40** I'm not a pianist, so I can't do this with the skill of a trained pianist, but this will give you an idea of what's possible. You can start from [plays C softly, then gradually louder, then soft again], from something very, very subtle and gradually increase the intensity of the dynamics. That's a fundamental way of projecting a character in a piece of music. It also has another effect, which is the effect of proximity or intimacy with the listener. If you're playing so softly that it sounds something like this [plays very softly], the listener is invited to go like this, to get closer to the instrument, to get closer to you.
- 6.41** It's as though you were talking to somebody, you're right next to them, they're in the room with you, and they're an inch from you. You're not going to shout into their ear. You're going to talk softly. You might want to lean in closer so that you can hear distinctly what they're telling you. That's a very intimate form of expression. The listener has to go towards you and listen very carefully.

- 6.42** You can hear this in songs. It's something you have to train yourself to notice, but there are singers who have an extremely dynamic voice, very broad dynamic range. They'll sing in a way that's just a whisper—very, very, very close. If you were in the studio with them listening to them record, you would notice that they were speaking at a level that's much softer than a normal speaking voice and singing that way. Then they can bring it up to a deafening roar as well. They're able to do that.
- 6.43** Even if you were to equalize everything in the recording studio so that it has the same apparent energy output—you can compress an audio signal so that the difference between loud and soft is much less—but even if you were to do that, the tonality of the voice would still be different. The overtone composition of the sound would be different. If you play softly, if you speak softly, then you're not exciting the upper partials in the sound. The sound is going to be impoverished in terms of overtones. It's going to be softer, it's going to project less. It's going to have a different timbre.
- 6.44** Then if you're singing as loud as you possibly can, then there's going to be a much broader range of frequencies that will be present in that sound. Even if you were to then equalize everything in the studio, make it the same apparent loudness, you would still be able to tell which one was the very quiet version and which was the very loud one, because the timbre and the spectral composition of the sound is going to be quite different.
- 6.45** It's the same on a piano. If I play a note on the piano very softly [plays C softly], it's not just the dynamic that changes, it's the timbre as well. As I play more loudly, I'm exciting the string with more force, and it's going to cause more partials to come into vibration, thereby changing the timbre of the note [plays C softly, then loudly]. You're also changing the attack, resonance and decay characteristics of the sound when you do that.
- 6.46** If you play a note with that level of intensity, what you get is a rapid, very strong transient, which is the very beginning of the sound, which falls off extremely quickly. It looks like a spike. Within milliseconds, that spike immediately drops. But that transient gives you the character of the sound and a characteristic timbre. It also fills the spectrum with noise very, very briefly. The nature of an attack like that is it's percussive. You're getting more noise in the sound, and then it rapidly drops off, and then you get a long tail end that gradually falls off.
- 6.47** Sound is this complex living thing that you can shape as a performer and also as a composer. To return back to my example here, I could start this little phrase off with the dynamic marking mezzo piano. You see a lot of scores that do that. But let me tell you an anecdote that puts this into perspective a little bit.
- 6.48** There's an anecdote about the American composer Elliott Carter, who's from New York City, who taught at the Juilliard School for much of his career. He was looking at a score from one of his students. He opens it up, and the first thing he sees on the first page is

a snare drum marking, a little roll on the snare drum. The student had written mezzo piano under the snare drum.

- 6.49** Carter looks at this, he turns to the student and he says, "You can't do that." The student says, "What's the problem?" He says, "You can't start a piece with a marking of mezzo piano. It doesn't mean anything. It's medium, it's tepid, it doesn't tell the player what to do." Mezzo piano only makes sense as an intermediary marking between two other markings. If we haven't established what piano is and what forte is, then mezzo piano means nothing, and you're going to get this characterless, medium, lukewarm sound on the snare drum. Now, if you want to be a bit annoying, you can open up the literature for solo piano and find countless pieces that are great pieces that start mezzo piano or mezzo forte—they do exist—but point well taken nonetheless.
- 6.50** It's a good idea to exaggerate the characteristics of whatever it is you want to write and to not be stuck in this medium mode. If you want it to start softly, write piano or pianissimo. If you want something assertive and forceful, write something forte. Remember, it's not just a dynamic, it's a timbre, and it's also a level of intimacy with your listener.
- 6.51** If you've got eight horns and four trombones and two tubas blasting away at fortississimo, then it's not an invitation to go towards them and go like this. You'll go deaf. That's the sound that can fill a large room, and you can be sitting all the way in the back of the hall and hear it without any difficulty. It's going to project like crazy, and it's going to be very dynamic. You might want to make a dramatic announcement, and that might be how you want to do it. If you have something of urgent significance to tell somebody, and you have to tell them right now, you're not going to walk over in slow motion and whisper it in their ear.
- 6.52** Dynamics tell the player and the listener about proximity, how close you are, what the nature of the expression is, what the timbre is, and it's very, very important. Maybe you don't start with mezzo piano, maybe you start with something more like this. Pianissimo. But maybe you don't want the entire phrase to be pianissimo. If you look at this little line that I wrote [plays the original phrase introduced earlier], it's an arch. It goes up [plays the ascending portion of the phrase] from the C, middle C, to the E $\flat$ at the top [plays C–E $\flat$], and then it falls down again [plays the descending portion of the phrase].
- 6.53** Now, maybe you want the dynamic curve to follow the melodic curve. In that case, you need a crescendo marking. Now, some performers will just do this instinctively without being told because it follows the logic of the phrase. Others might not do it. Some performers will want to stick strictly to the text of what's written and not introduce anything additional to the piece.
- 6.54** Then you have this little hairpin which tells us to play louder until we reach another dynamic marking, which in this case might be mezzo piano. Now, in this case, mezzo

piano means something because we've started from a very, very low dynamic, and now we've reached a medium dynamic, and we're going to go back down at the end to pianissimo again. Here we have a hairpin that goes in the opposite direction. It opens up to mezzo piano, down again to pianissimo [plays the phrase, starting at mezzo piano and tapering down to pianissimo].

- 6.55** Now, this little phrase has a much stronger character. It has a dynamic marking, it has a dynamic curve, and it's articulated. This doesn't mean that you have to be obsessively covering your score with markings all the time. But you have to reflect carefully on what is the minimum necessary information to convey your intention to a player. This is assuming that you're writing a score that's going to be played by somebody else.
- 6.56** If you're writing a piece for yourself to play, or if you're working on a DAW, a digital audio workstation, and you're working with VSTs or sample libraries, then you're going to put all those indications in yourself directly in the MIDI or through performing it on a MIDI controller, and you're not going to have to worry about this notation. But if we're working within this written tradition and you're writing a score, then you have to think, "What is the minimum amount of detail and information that I need to convey to the player so that they'll understand my expressive intentions, they'll understand how to shape the phrase?"
- 6.57** You don't want to overdo it because the more signs you put in your score, the more difficult it becomes to read it, the more time it takes to read it. In certain instances, the more difficult it can be for the performer to integrate all of that information. Some things maybe don't need to be said. They're clear enough from the expressive context that it's not necessary to hit the performer over the head with it. But so those are a few of the markings that commonly go into scores.
- 6.58** On top of that, there's also tempo. Now, if you open up old Baroque scores, you'll see markings like allegro, moderato, largo, and so on. And these markings persisted throughout the Classical era, and they're still used today for that matter. Now, these are not exactly tempo markings. They don't explicitly tell you how quickly or how slowly to play something, but they're an expression of character. The expression of character combined with the context created by the piece itself, will tell the performer how quickly they should play it in order to create that character.
- 6.59** Allegro, for example, doesn't mean fast. There's a French word "allégresse"—you can do something "allègrement," which means joyfully. It's the same with allegro—with joy, lively. That tells the player that you probably want to pick things up a little bit. But then you'll have to look at the actual figures in the piece to see what is a reasonable speed that I could play this at, given the technique required to execute this passage, that will allow me to execute it clearly and precisely, but giving this impression.
- 6.60** So there are pieces that contain this kind of information already in their titles. The Baroque composer Jean-Philippe Rameau, French composer—who was an almost exact

contemporary of Bach—wrote a whole series of little character pieces called *Pièces de clavecin*, keyboard pieces, that have little characteristic titles, some of which are quite funny. He wrote a piece called *La Joyeuse*, "the joyful one." It's all there in the title. The figures that he writes in that piece have that almost frivolous character that makes it quite obvious how he wants it to be played [plays an excerpt from *La Joyeuse* to demonstrate its joyful character].

- 6.61** That's a piece where you don't have to necessarily think for two weeks about how to do it because the character is so clear in the music and also in the title. You could, of course, if you were exceptionally insensitive to musical material, play it in completely the wrong way and completely fail to give that impression of a joyful rapid movement, but I think most performers would instinctively do it in the way it's intended. You want to get that sense of springy articulation in the playing.
- 6.62** Of course, you could [play the passage more slowly and without the light, articulated character]. But it becomes insipid when you do that. It's got no shape. It's not [plays the passage in the same slow manner again]—the material becomes very banal, actually, when you do it that way. This is why it's so important to shape it, and when you're composing to think about this. Remember, the difference between [plays the passage first without its light, joyful character, then again with a lively, clearly articulated feel]. Completely different. So whenever you're writing something, think about that and try to convey it with as much force and character as you can possibly imagine. And if you need to, then you can make that explicit in the score.
- 6.63** We've looked at dynamic markings. We've looked at some articulation markings. We've looked at character indications. There's a couple of other things that we need to look at as well. These little dots here, which you can see on the first of each group of four 16th notes, indicate, again, a staccato marking, so a very short, detached articulation. But there are other articulations as well that composers will use to indicate to the player how they want something to be played.
- 6.64** Perhaps the most common is this symbol here, which is an accent. The accent means that the note is to be emphasized dynamically. To return to our example here, I might decide, instead of a little light springy staccato accent there, that maybe I want something a little bit more pronounced. I'm going to put an accent on the beginning of each note, and I'm going to change the dynamic from *pianissimo* to *forte*. Then our little passage looks like this [plays the original phrase again, now with each note accented and at a *forte* dynamic]. You might do that if you want to have a really strong delineated rhythmic profile.
- 6.65** You can make things more interesting by not having an absolutely regular pattern of accents, but by displacing them. That's something that can make something a little bit more lively. Here I'm going to put some accents in places other than the beginning of the beat. That's going to give me this [plays the phrase with accents displaced to unexpected points in the beat]. Now our passage becomes totally unpredictable

because you don't know when an accent is going to occur. It's irregular, and the accents fall on weak parts of the beat. Now you've got something that's going to jolt you a little bit because you don't know what's going to happen [plays the phrase again, emphasizing the surprise effect of the irregular accents].

- 6.66** Even with this simple little three-beat phrase, or four beats if we include the end, you can shape this an infinite number of ways. It's like a block of plasticine, molding putty, you can turn it into almost anything—infininitely elastic, depending on tempo, dynamics, articulations, expression, and you can give that a strong intention.
- 6.67** Of course, you can also place it in different registers. And there again, you can transform the character quite substantially. You could do something like this [plays the phrase in several different registers]... Hundreds and hundreds of ways you could present it. The musical idea here is: the material allied with an expressive intention and a shape. Just as how you can completely transform the meaning of a spoken phrase in English—or in any language for that matter—by changing the tonality, the emphasis, the pitch of your voice, it's the same with a musical idea.
- 6.68** So shape everything. Make sure that even if you were half asleep and sitting in the back row and completely distracted, you could instantly and effortlessly understand what the composer is telling you and how they're saying it. You need to exaggerate everything. Maybe we'll stop there for now. That's a lot of information on expression and articulation. Are there any questions?
- 6.69** Thank you. My question really was about the expressive notations that are written on the score of sheet music. Something I often notice is that they are a lot more dense towards the start of the piece of music when you're reading the score. Is that because the composer is—they've set their expressive intent, and then they expect the performer just to maintain that? Or are they inviting the performer then to introduce an element of their own interpretation as the piece progresses?
- 6.70** It depends very much on the piece. So in many cases—there are certain things that need to be established at the outset. You generally need to start with a tempo marking. Tempo markings are given in BPM or beats per minute. If you grab hold of a metronome, you'll see that there's all these numbers on it and a dial that you can turn. You go from, usually, largo markings, which are quarter note equals 40, so 40 beats per minute, all the way up to 240 beats per minute or sometimes more. Those are different metronomic speeds.
- 6.71** Contemporary composers will typically give a metronome marking at the beginning of the piece. But unless the tempo is modified later, or unless there's a *rallentando*, or a change of tempo, there's no reason to give that indication more than once. And likewise, with a time signature, you put that at the beginning of the score, so the performer immediately knows how many beats per bar and so on. You don't have to give that information again unless it changes. Same with dynamic markings. If your

piece doesn't change all that much in terms of its different sonic characteristics, then you put all that at the beginning and it establishes an interpretive context, interpretive framework, and then that's enough.

6.72 But also there are composers who are incredibly punctilious about those kinds of markings. You'll open their scores up and they'll just be absolutely filled to the brim with hundreds and hundreds of markings. Gustav Mahler, the Austrian composer, was one such figure. He did not trust orchestras to figure it out for themselves. He put every single marking in. The scores are incredibly detailed, to an extent that makes them easier to play, I suppose, because it's crystal clear what he wants at all times.

6.73 But other composers were far less precise and left it up to the performers, or assumed that the intention was clear enough from what they were doing that they would sort it out. It's also a question of individual priorities and how much time you want to spend putting those marks in.

Lecture 7: Blending Sound

At a glance — Blending sound — orchestration.

- 7.1 So today I want to talk about the art of orchestration. And orchestration is a complex task that is nevertheless an essential one if you want to write for more than one instrument. Orchestration is related to, but not quite the same thing as, arrangement. So if you're making a studio recording of a song or some other piece of music, you might arrange it first before you go in.
- 7.2 Arranging or instrumentation is the process of deciding what instruments are going to play what parts. If you're making an instrumentation of a piece, you might start with a little piano score or some form of tablature or maybe a jazz notation or something and decide, "The bass line is going to be played by this instrument, and then the chords are going to be played by two guitars, and then we're going to do this and so on." You're going to assign different instruments the role of filling out this or that part.
- 7.3 That's not exactly the same thing as orchestration. Orchestration has to do with taking a collection of instruments, not necessarily a symphonic orchestra, and considering that group of instruments as a single, gigantic instrument. In a symphony orchestra, you might have as many as 80, 100, or even more players. You're not scoring for a flute, plus a clarinet, plus a bassoon, plus a trombone, plus a viola. You're thinking in terms of the orchestra as a whole, complete, organic, sonic body. What you're doing is you're creating a sound that is the synthesis of all of these instruments together.
- 7.4 So orchestration is often taught as a separate discipline for the simple reason that it's extraordinarily hard to get your bearings on it if you don't have some specialized instruction in it. There's a lot that you need to learn if you want to learn how to orchestrate.
- 7.5 However, once again, you don't need a full symphonic orchestra in order to orchestrate. You can orchestrate a trio of instruments in the sense that you're going to be thinking not just in terms of, "I have this instrument, plus that instrument, plus that instrument," but you're thinking in terms of how you can blend them to create a sound that you're designing yourself in all of its dimensions.
- 7.6 Orchestration is a wonderful resource for composers because it allows you a seemingly infinite range of possibilities for presenting an idea. It also allows you the possibility of differentiating your ideas and establishing contrasts, establishing oppositions that you can make clear through your choice of instruments.
- 7.7 The point I want to emphasize here is that the orchestra is not a bunch of instruments, but it is an instrument. Once you start thinking of it that way, you start to see limitless possibilities for how you can work with it. This is something that you can pay attention to in pretty much any genre of music where there's multiple instruments playing

together. What you'll find is that there are, roughly speaking, two approaches that you can take.

- 7.8 There's the analytic approach, which is where you can distinctly hear individual instruments within the overall sound of the piece. You can pick out a guitar, or you can pick out a clarinet, or you can pick out whatever the instrument is, and you can hear the individual instruments relatively distinctly, even though they're working together to create a broader sonic tapestry.
- 7.9 Then there's the synthetic approach. The synthetic approach makes it so that you are not particularly aware of the individual instruments in the sonic texture, but you're hearing a globality in which the instruments blend together to a very high degree.
- 7.10 To some extent, these different approaches to orchestration are reflected in different national schools. There's very clearly a French lineage of orchestration that tends more towards the analytic variety. That school is also associated with the Russian School of Rimsky-Korsakov and Stravinsky and so on. Those textures tend to be transparent. Even if there's a very large number of instruments playing, you can clearly hear each line; you can clearly hear each instrument.
- 7.11 There's a Germanic school exemplified by figures such as Wagner, for example, or Richard Strauss, or the orchestra compositions of Schumann before them, which is much more synthetic. You're not necessarily thinking in terms of solos. They're present. You'll have the occasional flute solo or the occasional violin solo in those pieces. But much of the time, what you're hearing is like an organ in the sense that there are all these sounds blending together that transcend their individual characteristics.
- 7.12 There's an extent to which this is the result of national schools that have different preferences in terms of how orchestrations are done. But it's also a question of temperament. Some people are going to prefer that highly blended sound, and others will want to be able to pick out distinct colors as though you are listening to some extraordinary kaleidoscopic texture in which you have lots and lots of different subgroupings of instruments that's changing all the time. You can have a very beautiful sonic experience that way.
- 7.13 You can hear this also in pop music. There was a record producer called Phil Spector who pioneered something called the Wall of Sound. The Wall of Sound was a technique whereby he would have large multiples of a single instrument playing the same line in unison. Instead of getting one guitar to play in the rhythm section, you'd have 12 guitars all playing the same thing at the same time. Instead of one bass, there might be five or six basses and so on and so forth. You'd have multiple pianists all doubling the same parts.
- 7.14 The result, of course, is that if you have many people playing the same part simultaneously, there's a chorusing effect that takes place where the sound is thicker and it's richer. Of course, it's not exactly together because human beings aren't

machines, and so you get this thickness to the sound. That's akin to this more synthetic approach to orchestration.

- 7.15** Then you'll have other producers who will favor a much leaner, more transparent sound where everything that's there is distinctly audible and has its own individual character. If you listen to a song like Penny Lane by the Beatles, every element in that piece is clearly distinct and very easy to hear, including the piccolo trumpet solo, which is a very nice bit of orchestration. In fact, that was an unusual thing to do at the time. The point being that the instruments in that song don't really blend. Rather, they have a specific function, and they're clearly distinct, distinctly audible.
- 7.16** That's something you can start to consider when you're listening to all sorts of things. What is the nature of the orchestration here? Am I meant to be able to pick out individual instruments? Is it more of a wall of sound? Is it more synthetic? In terms of creating your own pieces, that might then be something that you would want to think about. Of course, it comes down to what your work is like. The way you orchestrate is going to depend on that to a large extent.
- 7.17** If you want crystalline textures where you can hear every note, you can hear every line, you can hear everything that's going on, it's transparent, then you probably want to have that leanness. You want to be able to hear every instrument in your orchestra. If, on the other hand, you want an overwhelming expanse of sound, something more rich, let's say, lots of figuration, you can't necessarily tell what's what, then you're going to go with another orchestration style.
- 7.18** Likewise, we spoke previously about dynamism in musical forms and how dynamic you want something to be, how predictable, how directional, or how static you want it to be. These are also things, these are qualities that you can amplify to a great extent with your orchestration. We're going to look at some possibilities of how you can do that, but the first thing to do is to lay out precisely: What is the constitution of a symphonic orchestra, and how can you work with these components?
- 7.19** The first thing is that the orchestra as an entity, as we know it today, is largely a 19th-century creation. The form that it exists in today is only very slightly different than what it would have been 150 years ago. To a large extent, this is a late 19th-century phenomenon. But the orchestra, up to that point, went through many periods of evolution. There were instruments that were invented, others that fell to the wayside for technical reasons or reasons of fashion.
- 7.20** The Baroque orchestra is very different in its constitution from the Classical orchestra, which is also very different in its constitution from the Romantic orchestra. You see different priorities; composers have different needs in terms of the sizes of venues that they want to have their pieces played in—remember that in music, a sound never just exists in a vacuum. A sound exists within an acoustic, within a space.

- 7.21** When you're writing a piece, you have to think in terms of the space that your piece is likely to be played in. Is it going to be played in a cathedral with a 30-second-long resonance? Is it going to be played in a small room where there's no possibility to back up, and everyone's going to be at most three or four meters from the performers? You write differently depending on the acoustic.
- 7.22** This is reflected in orchestration as well. As concert halls got larger throughout the 19th century and orchestral music became more and more popular, orchestras got larger. The number of instruments increased. The sonic power of the orchestra increased as well. We were left with the current state of the orchestra, which is a substantial thing. A professional orchestra today is on the order of a minimum of 60 and more likely 80 players, but those players can be divided up neatly into four basic categories.
- 7.23** The categories in the modern orchestra are woodwind, brass, strings, and percussion. On top of that, you have some auxiliary instruments that sometimes are part of the orchestra, sometimes not, such as keyboard instruments and harps. In addition, you have a timpani player, usually at least one. The timpani, although it's technically a percussion instrument, is played by a timpanist as opposed to a percussionist. The percussionist will play whatever is required by the score—it could be a xylophone, a marimba, a vibraphone, castanets, cymbals, wood blocks, etc. But the timpanist only plays the timpani, and it's a very special role within the orchestra.
- 7.24** Those are the four basic instrumental categories within the orchestra: the winds, the brass, the strings, the percussion, and occasional auxiliary instruments as needed. Sometimes in a particular piece, if you're writing a *Rapsodie espagnole*, a Spanish rhapsody, you might include a guitar in the orchestra for color, for example. Or you might ask for a celesta, which is a keyboard instrument that produces a silvery and bell-like tone. You might do that for reasons of color or to evoke a particular landscape or particular image. But that's the basic constitution of the orchestra.
- 7.25** How do you work with all of these elements together and turn them into something that's coherent? Well, in an orchestra, the strings are playing something like 80% of the time. The strings are really the backbone of the orchestra, and they're the largest group by number.
- 7.26** If you've ever seen an orchestra performance, you'll have noticed that the strings are often playing in lockstep with each other. You'll have maybe 20 first violins and maybe 16 second violins, and they're all bowing at the same time in the same direction. They're doubling their parts. Even though if you open the score up, you might only see one staff for the first violins and one staff for the second violins. You may have as many as 20 players to a part, all doubling the same thing.
- 7.27** What's the reason for that? Why do we need so many violins to play a single part, whereas you might have only one tuba player, you might have only one timpanist and so on? The reason is to balance out the brass and the winds, which inherently have

more sonic power, and to create a sound that has a homogenous projection. You need a lot of violins to do that. If you have one violin against an entire brass section, you simply won't be able to hear them.

- 7.28** So the strings are the backbone of the symphonic orchestra. They're the instruments that are playing by far the most often. And the strings as a group have a massive range of tones that they can produce. A double bass, for example, can play notes as low as this [plays low C]. That's the lowest C on the piano [plays low C]. And the violin can easily play way up into the top range of the piano [plays high C, then high E]. So that's a gigantic range. That's at least six octaves, possibly more.
- 7.29** So in terms of register, the strings pretty much have it covered. You can play just about any note within the audible frequency spectrum. I'm exaggerating somewhat, but it's not far off with the string instruments. But there are some things that strings can't do. They don't have the sonic power, for example, of the brass. The brass are, along with the percussion, the group within the orchestra that has the most projection, the most power.
- 7.30** The winds are used often in a soloistic capacity in the orchestra as a counterpoint to the string section. They'll have little solos, sometimes duos, sometimes trios. Sometimes they'll be playing together as a chorus. But they tend to have more distinct individual timbres than the string instruments do.
- 7.31** A layperson might find it difficult to distinguish between the sound of a violin, let's say, and the sound of a viola. The viola is a little bit larger and a little bit deeper than the violin. But if you're not accustomed to it, it's easy to mix the two up. Whereas the difference between a flute and a clarinet is much more obvious.
- 7.32** In the woodwind section, there are four basic categories or families of instruments that you're going to see in most cases: the flutes, the clarinets, the oboe, and the bassoons. Now, each of these woodwind families has a set of auxiliary instruments associated with them, that is to say, higher and lower versions of the standard instruments.
- 7.33** With the flute family, there's also the piccolo, and the piccolo is one octave higher than the standard flute. It's a tiny little flute that can play incredibly high notes. There are also lower flutes, the alto flute and the bass flute, although they're quite a bit more rare in the symphonic orchestra.
- 7.34** Likewise, in the clarinets, you have the bass clarinet, which is an octave lower than the standard B $\flat$ clarinet. Then you have the piccolo clarinets, the E $\flat$ clarinet, for example, which is pitched a perfect fourth higher than the standard B $\flat$ clarinet.
- 7.35** With all of these woodwinds, you have a family, or as was called in the Renaissance, a consort of different instruments all with the same timbre. That allows you quite a broad range of different possibilities. The oboes have a lower version, which is called the English horn or the cor anglais.

- 7.36** Finally, the bassoon has a very deep big brother called the contrabassoon. The contrabassoon is this huge, gigantic, coiled instrument with a bell pointing up that you can see in the back of the wind section. It can play very deep tones. Deeper, in fact, than what the double bass can play [plays low B $\flat$]. You have to imagine [plays low B $\flat$] what a sound like that is like sustained on a wind instrument [plays low B $\flat$ -B $\flat$]. It's quite something. So that's the contrabassoon. And if you take, again, if you take the full complement of winds from the piccolo all the way up to the contrabassoon, you can pretty much span the entire range of the keyboard [plays low B $\flat$, then high B $\flat$]. From there to there. That's a pretty amazing sonic resource.
- 7.37** Then on top of that, you have the brass. The brass typically don't play as often as the winds or the strings. And in a lot of symphonic compositions, they're present mainly in what are called tuttis. And tuttis are passages in which, as the name indicates, everybody is playing together. And tutti means "everybody." So when you want the orchestra to blossom sonically and to be at its maximum capacity, you'll introduce the brass. The brass consists of horns or French horns, trumpets, trombones, and tuba.
- 7.38** The percussion section is the most recent addition to the symphony orchestra. If you listen to older pieces, let's say from the first half of the 19th century, there will be occasional percussion instruments in them. Beethoven would write for instruments like the triangle, for example, which appears in the Ninth Symphony. Occasionally, you'll see markings for tambourines and things like that or cymbals. But at that point, the percussion, at best, was used for a little bit of color every now and then.
- 7.39** Throughout the development of the 19th century and into the 20th century, especially, the percussion section took on increasingly significant dimensions until it got to the point where with some early 20th century pieces like *The Rite of Spring* by Igor Stravinsky or *Am rique* by the French composer, Edgard Var se, you would see as many as ten or even more percussionists at the back of the stage, each of them playing a whole battery of different instruments.
- 7.40** The percussion, unlike the other instrumental groups, doesn't have a fixed constitution because pretty much anything can be a percussion instrument, including, and not limited to, anvils, which appear in many orchestral compositions, actual anvils that you strike with a hammer. It gives a very nice sound. But there's a very, very broad range. It's hundreds and hundreds of percussion instruments.
- 7.41** If you go to the offices of a symphony orchestra where they have their headquarters, they'll typically have a room where they store all their percussion instruments. It's very fun to go in there because they have shelves and shelves of them, hundreds and hundreds of instruments, a very conceivable description. Then they, of course, use them as required for the pieces that they're playing.
- 7.42** So those are the four sections of the orchestra. But the question of how you blend them together and how you work with them as a composer is a very complex one.

What we're going to do is we're going to look at a couple of excerpts of scores. I'm going to take you through them and show you how the composer has taken a chord or a line or an idea of some kind and scored it for orchestra.

- 7.43** Bear in mind, first of all, that, of course, one of the things you can do in an orchestra is you can have very sparse textures where it might be just one violin playing, one flute, one clarinet. It's very transparent. Or you might have multiple instruments all playing the same part. You might have the oboes and the clarinets doubling, in other words, playing the same thing at the same time, and then reinforced an octave lower by the bassoons with the horns in the middle, and then the cellos down here and the basses, a large number of instruments all working together to create a single sound. We're going to look at how that actually works in a piece of music.
- 7.44** This here on the screen is the beginning of a score by the 19th century French composer, Hector Berlioz. Berlioz was one of the great exponents of a new art of orchestration with his pieces. The *Symphonie Fantastique*, of which this is an excerpt, is a brilliant demonstration of that. Berlioz had an instinctive sense of drama and of color. Orchestral color and timbre form an integral component of his musical language. You can't imagine these pieces without it. It makes them extraordinarily variegated and thrilling to listen to.
- 7.45** We're going to listen to the opening measures of this piece, and then I've got the score of the first four bars here, and we're going to look at how they're put together. Here's the opening of the *Symphonie Fantastique* of Berlioz.
- 7.46** [plays *Symphonie Fantastique* of Berlioz]
- 7.47** Notice how in that excerpt, most of what you heard was the strings, probably 90% of it. The opening bars are just the winds, the rising figure following with a chord. Then there's a pause, and then the strings start playing. In this case, the strings start playing this beautiful slow theme, but without any accompaniment from the rest of the orchestra. That allows Berlioz immediately to set up a simple sonic opposition between the wind group and the strings.
- 7.48** Orchestral composers are doing this all the time. They'll pit one group against the other, and they'll use that to create an opposition. That opposition can intensify. You can have completely different materials on the winds and brass, for example, and then something else entirely on the strings. You can alternate them, you can keep them completely separate, or you can eventually bring them together. So in this case, there's this opening gesture that's uncannily like falling into a dream. There's something very poetic about it. Indeed, the first movement of the *Symphonie Fantastique* is called *Rêveries – Passions*, so a dreaming ambiance there.
- 7.49** Let's take a look at the score and see how he creates that effect. We have the flutes and the clarinets playing in octaves, so the flutes on top and the clarinets on bottom. In this case, in fact, he starts with only one flute, one clarinet, and as the passage

progresses, he adds more instruments. And that, incidentally, is one of the most basic techniques of orchestration that students have to learn, which is that if you want to have a change in the sound, and let's imagine that you want a dynamic envelope where the sound is getting progressively more intense or where it's waning, then you can't just put a crescendo marking.

- 7.50** You can't just do it with dynamics alone. You have to actually add instruments if you want to create a change in the volume of sound. If you do that, if you go from two woodwind instruments to four and then to eight, then you'll have a natural increase in the volume. It will almost play itself. It'll be very easy to do.
- 7.51** That's what he does here. As the phrase progresses, he adds more instruments. The bassoons only appear right at the end of the bar on the fourth beat. That's how he gradually changes that texture. He starts with just the flute and the clarinet. When you're listening to it at first, and we'll hear it again because it's important to hear this passage more than once, the instruments are blended together such that you're not paying extremely close attention. You might not notice who's playing what. You might not be able to tell that there's a flute and a clarinet playing together.
- 7.52** One of the things that you want to do if you're trying to get a grip on the art of orchestration is to listen to a lot of scores, with preferably the score itself, and test your ears. Listen to a passage and see if you can pick out all of the instruments that are playing, and see if you can identify them. That takes a lot of practice at first because some of the instruments, again, are more difficult to distinguish between than others. The difference between a French horn playing in its middle register and a trombone playing the same note can be subtle. But with practice, you learn to distinguish between them. You learn to immediately recognize the timbre of the flute, the timbre of the clarinet, and so on.
- 7.53** That's, in a sense, step one for somebody coming to grips with orchestration is understanding what all the instruments are, how the orchestra is constituted, and understanding the ranges of these instruments, not only in terms of the tones that they're able to play, but also in terms of their dynamic characteristics.
- 7.54** For example, the flute, which is again one of the instruments we hear right at the outset of this piece, has a fairly large range. The flute can play from middle C here [plays C] sometimes to the B below it, depending on the instrument [plays B], but more commonly [plays C] middle C, up to the C [plays C], three octaves higher. So that's about the range of the flute [plays C-C]. That's a pretty big range for a woodwind instrument. But the quality of sound throughout that range is not homogenous.
- 7.55** Now, if you're coming from the perspective of a pianist or a guitarist or an organist, then you're probably accustomed to the fact that no matter what instrument you play, no matter what pitch you play on the piano, there's no particular difference in effort needed to play a low A [plays A] than to play a high C [plays C]. It's not mechanically all

that different. You can play anywhere throughout the range of the piano with equal facility, but it's not the case on woodwind instruments, and it's not the case on brass.

- 7.56** On top of that, the timbral and dynamic characteristics of these instruments are quite heterogeneous. If you're in the bottom fifth or sixth of the flute, for example, you're going to find this range here [plays a few notes in the middle register]. The flute, towards the bottom of its range, doesn't have much power; it doesn't have much projection. So it's quite useless to write a passage for the flute in the bottom of its range in a massive orchestral tutti where everybody's playing fortissimo. You're not going to hear anything.
- 7.57** If you want to be able to hear the flutes, you have to put them in a register where they have enough projection and enough power to cut through the orchestra. In that case, you have to place them in a higher register. You might have to put them here [plays a few notes in the upper register]. Every instrument in the orchestra has these different characteristics, even the strings.
- 7.58** You'll find that the strings don't have equal power throughout their entire range. They have typically, or with any instrument, like the human voice, you'll have a range around the middle, which is very strong. Then, as you go out towards the extremes, the dynamic characteristics will intensify. It'll either become strident and difficult to control, as is the case with some woodwind instruments.
- 7.59** If you get up into the stratosphere on the flute, for example, then it's very, very difficult to play a soft tone, pianissimo tone. And inversely, conversely, I should say, if you go into the low range of the flute, then you can't have a lot of projection. These are some basic parameters of orchestration that have to be kept in mind.
- 7.60** In the case of this Berlioz's excerpt, to return to it, we have multiple doublings here. The clarinets and flutes, again, are doubling at the octave. And then when we come towards the end of this passage, you'll notice we see in the left-hand margin of the score these different markings. We see flutes—this is all in French, so I'll translate—flutes, oboes, clarinets, bassoons, horns, trumpets, and so on. Then next to them, you'll see where it's marked "clarinets en si bémol," which means in B $\flat$.
- 7.61** What does that mean? Now we come to the concept of transposing instruments. With instruments like the piano, the violin, the organ, and so on, generally speaking, when you play a C in the middle of the range [plays C], the note that you write down on the score and the note that you hear is the same thing. That means that the instrument sounds as written. If I write a middle C and I play a middle C on the piano, I'm going to get the same thing.
- 7.62** But there are also instruments that are called transposing instruments. With transposing instruments, the fingering that the player has to do to play the written pitch is going to produce a sound that is higher or lower than what is written. That

might seem like a perverse system. Why would you do that? Why would the note that you play, the note that's written, and the sounding note not be the same thing?

- 7.63** Well, again, there are historic reasons for this. With the clarinets, for example, these instruments were built in different sizes. In the late 18th century, for example, you could get clarinets—and you still can get clarinets— in C, in B $\flat$, in D, in E $\flat$, and so on. You would do that because the instruments at that time didn't have the complete chromatic key work of instruments today, which can allow you to play any tone, anywhere on the instrument, with reliable intonation.
- 7.64** At that time, it wasn't the case. If you look at old clarinets, old oboes, old flutes, you'll see that the key work is much simpler than on instruments today. They're mechanically simpler. The intonation didn't permit players to play any chromatic pitch anywhere in the range. You couldn't do that. You would have to correct with the embouchure or with different fingerings in order to get the right sound out of the instrument.
- 7.65** In order to make things easier for players, you would use a clarinet in a different key that would correspond to the piece you were playing. If you were playing a symphony in D, you would probably use a D clarinet. If you were playing a piece in C major, you would use a C clarinet. If the piece was in B $\flat$ or E $\flat$, you would use a B $\flat$ clarinet and so on. The advantage of this system was that the player could take any one of these clarinets and play them with exactly the same fingerings and exactly the same technique, but it would put them in a different key.
- 7.66** It was the same with the horns. Before the invention of valves in the 19th century, horns—you could change the pitch only by playing the notes of the harmonic series, as we talked about in one of the first lectures in this sequence. The horn was just a length of pipe, basically, and the player would use their embouchure, changes in lip pressure and air pressure, in order to change between the different partials that you can play on that length of tubing.
- 7.67** That meant that you were stuck with the notes that were the natural harmonics of whatever the fundamental was of the length of piping you were playing. If that was in C, then you could only play the notes of a C harmonic series. If it was B $\flat$, then you could only play those notes and so on. If you wanted to play pieces in different keys, you had to use what were called crooks. Crooks were little lengths of additional piping that you would add between the mouthpiece and the main body of the instrument to change the effective length of the tube. That would allow you to play in different keys.
- 7.68** You'll see things like the horn in E $\flat$, the horn in C, the horn in F, the horn in G and so on in these older scores. Now we have the horn in F, which has evolved to become the standard orchestral instrument, the French horn in F, which is equipped with valves, which allow the player to play any chromatic tone throughout the range of the instrument. But it wasn't the case in these older pieces. In this score, you can see the clarinets are in B $\flat$, the horns are in E $\flat$ and C, and so on. The flutes and the oboes are in

C, as is standard, even today, and the bassoons also. But the clarinets and the horns are transposing.

7.69 If you look at the second bar in the score here, you can see that there's a C minor chord that's played by 10 or 12 instruments at the same time. Why did he score it this way, because a C minor chord, you only really need three sounds at a minimum in order to play that? But he's doubled the pitches so that you hear them more than once and in more than one octave in order to give a stronger and richer sound.

7.70 Let's listen a second time to this excerpt. This time, I want you to really pay attention to see if you can distinguish between all the different sounds of these woodwind instruments and the way they blend together to create that chord. Now, that's just the opening seconds of the piece. Then pay attention to the contrast it creates when we move from the winds and the brass to just the strings playing.

7.71 Even just in that very short little excerpt, there's a remarkable range of very subtle little timbral distinctions that help bring the music to life. But it's a case of blink and you'll miss it because the opening wind section there is just a few seconds long. Nevertheless, it's minutely calibrated to produce exactly the timbre and the distribution of weights and tones that he wants to create this effect.

7.72 I'd like to look at another example now. This time, it's from a symphony by Schubert. The orchestra of Schubert is quite a lot earlier than that of Berlioz. It's not as large. Typically, orchestras of his time had a much leaner constitution. In the case of this piece, which is the Menuetto from the Third Symphony, you're going to see that Schubert creates strong dramatic oppositions in his orchestras between tutti passages where everybody's playing together and much lighter passages that emphasize the strings. He rapidly alternates them in order to create a strongly dramatic texture.

7.73 Let's listen to the opening bars of the Menuetto from the Third Symphony of Schubert.

7.74 [plays Menuetto from the Third Symphony of Schubert]

7.75 What you see in an excerpt like that is there are all these different sonic planes, and he's able to move between them to create these dynamic effects. The opening couple of bars are a tutti passage where everybody's playing together, and that's immediately followed by just the strings. So we have this theme here [plays the opening theme] played by everybody in unison all throughout the orchestra [repeats the opening theme]. That's played twice. Then we have this little phrase [plays the descending string phrase] just on the strings, this little descending phrase. You could present that as a straight piano piece, I suppose, but you would lose all of that subtle dynamism, all of those different coloristic shadings that are possible in the orchestra.

7.76 That's a very succinct introduction to some of the possibilities of orchestration. You have a huge collection of instruments that you can treat either synthetically or analytically. You can use them to create gigantic dynamic contrasts, you can blend

them together, and you can create oppositions between different groups that can help animate and bring to life the dramatic oppositions that are inherent within the music itself.

- 7.77** The topic is extraordinarily vast, but those are some of the broad parameters that you're going to be dealing with as you start thinking about orchestration. Of course, today, the economics of orchestral playing are not what they used to be. Orchestras are colossally expensive. Composers are, with some exceptions, generally tending to move away from the orchestra medium.
- 7.78** There's also the question of the constraints involved in rehearsing with orchestras. Because of the severe financial pressures of working with them, you tend to get a very small amount of rehearsal time, which then requires you to write pieces that can work after maybe one rehearsal. You can't necessarily write the same way as you might like to because there are very significant constraints involved in that.
- 7.79** Of course, a lot of people involved in media scoring and movie soundtracks are using VSTs. They're using virtual instruments, which are rapidly becoming extraordinarily good. Of course, they don't replace a symphonic orchestra, but a lot of what you hear now in movies and soundtracks is in fact, produced on virtual instruments. We'll stop there for orchestration and I'm happy to answer any questions you might have.
- 7.80** Explaining the composition of classical music, you would often come to rather pop music to exemplify. Would you say that some genres of music can be classified as better or lesser, the same way a single song can be classified better or lesser than another?
- 7.81** They're not made for the same purpose. You can't compare them directly. They don't do the same thing. There are people who have that attitude, but the problem is: art and commerce are intimately intertwined. It's very, very difficult to separate them.
- 7.82** There is this idea that in an ideal world, artists would be motivated only by the pursuit of their work and nothing else, and there would be no external forces brought to bear upon them. But in fact, that almost never happens, especially in a medium like music, which costs money. If you want to play an orchestral piece, an orchestral concert could easily cost in excess of \$100,000. It's very expensive to play these pieces. You can't ignore the economics of it.
- 7.83** It's a completely different set of pressures that are placed upon these different forms. There isn't one that's superior to the other. They operate differently. You can have great art in the commercial realm, just as you can have completely trivial art that is completely ephemeral in these spheres of high art. It all depends on the artist.
- 7.84** What do you do with a figure like Alfred Hitchcock, for example, who made masterpieces, bona fide masterpieces, like *Vertigo*, *Psycho*, *North by Northwest*—these are great artistic achievements, regardless of genre. He was working within a

commercial framework. These films had to be commercially successful. They had to have large audience numbers in order to be viable.

- 7.85** But if you look at the history of opera, it's the same thing. Composers of opera were also under very significant financial pressures and had to write things that were successful, that were of interest to audiences. If you were an opera composer and you were writing things that were completely incomprehensible or that completely failed to attract the interest of audiences of the day, then that would be it. You wouldn't keep getting commissions.
- 7.86** So we have this tendency to have this facile division between popular music, commercial music on the one hand, and then high art music. It's not that those distinctions mean nothing. They do mean something, but they're much more porous and much more interrelated than we think. The idea that people working in popular or commercial fields are strictly in it for money, and that the people working in the high art fields are completely divorced from any market pressures or commercial considerations is utterly false.
- 7.87** I think it's probably better to look at it a little bit differently. There are differences in mode of expression and in style, let's say, and in the ways, the circumstances in which these artworks would be heard, but there isn't an automatic distinction in terms of quality. By no means is that the case.
- 7.88** It's not hard to—you can go to any library in Europe, and they'll have thousands of manuscripts of Baroque pieces by composers that have been forgotten. And a great deal of what was produced—there may have been 10,000 trio sonatas written during the Baroque. There may have been, who knows, maybe 30,000 symphonies written. Not every single one of them is a masterpiece, I can tell you that. A lot of them were written hastily, and they're not great pieces. They deserve to be forgotten.
- 7.89** It's the same in every musical genre. There are things that we retain as a culture, and there are things that, for one reason or another, sometimes unjustly, are not retained. That's how it goes.

Lecture 8: Composing Consistently

At a glance — Composing consistently.

- 8.1** We've taken a tour through the composers' workshop. We've looked at all of the different skills that are involved in this art form. We've looked at some of the problems that composers will inevitably face as they pursue their work, the most difficult of which tends to be how you create a continuity in a piece of music, how you transform and extend your ideas. We've looked at some of the traditional techniques that composers will have to confront, such as coming to terms with counterpoint, with harmony, with rhythm, with orchestration, with form, and so on.
- 8.2** Now we come to the set skills that, although they're typically not taught at all, are perhaps in some respects the most crucial of all, and that is: How do you work? That's a problem that is actually extraordinarily difficult for a lot of young creative people to sort out. I've helped many composition students deal with this question, which can be extraordinarily complicated, particularly if you have multiple commitments in your life, if you're pulled in a lot of different directions—maybe you're a student and you also have a full-time job, maybe you've got kids, maybe you're balancing lots of different responsibilities.
- 8.3** The question is really: How do you sustain a creative practice over months, years, and decades? This isn't just a problem that is put before professional composers at a later stage in their life when things get really complicated, but it's also something you have to deal with when you're a student. Another thing that makes this question even more difficult is the fact that it's an intensely personal question. There is no one workflow that will suit everybody. We're going to deal with that question today.
- 8.4** I'm going to start with a quote from the composer Morton Feldman, F-E-L-D-M-A-N, a truly great American composer who had the virtue of being remarkably witty. He's a very funny guy. I'm always quoting him all the time. He once said, it sounds like a joke, but he was actually being serious, I think, "If only I could find the right chair, I would rival Mozart." What does that mean? It sounds facetious, but it was dead serious.
- 8.5** What he meant was all of the little contingencies, the material questions that you have to deal with if you're going to pursue this line of work are a lot more important than you might think they are. Simple things like setting up a work environment that's conducive to doing what you want to do, getting the tools that you need to do it properly, and choosing those things carefully, meeting your minimum needs in order to sustain your creative practice, are extremely important, and you have to give due attention to them.
- 8.6** When I was a student, all throughout my student years, I lived in a succession of very small apartments in Paris, and I was constantly confronted with this problem of having to balance my classwork, the pieces that I had to write, and personal obligations, and

all sorts of things, and also find a way to work in extremely confined spaces and environments. We're going to talk about some of the practical considerations that composers are inevitably going to face.

- 8.7** There are art forms, and this is where it gets a little bit unfair, where the overhead required and the material required is extremely minimal. For example, if your ambition is to be a poet, then you're going to need a lot less to get started than if you want to be a concert pianist. A poet will typically need some books, some paper, a pen. The outlay is not going to be that significant. When you're a musician, it can be a little bit different, and it can be daunting at first because you have to put all this together and have a workspace where you can be reasonably sure that you won't be interrupted every five minutes.
- 8.8** The other question is that there's a certain amount of discipline that goes into pursuing a creative field. There again, the particularity of musical composition is that most of the time, you're not going to have anybody breathing down your neck and checking to see if you've written your new song, or if you've made any progress with your new composition, or if your whatever project it is is coming along smoothly. You're going to have to impose that discipline upon yourself, which is almost impossible to do because one of the most powerful incentives for artists is deadlines.
- 8.9** The ideal situation is that you can set up a situation where you have commitments to write pieces. They don't necessarily need to be paid opportunities at first, but they need to be something where there's a date on the calendar where you have to finish a piece by a certain date. I've worked with students, for example, who, in order to get the necessary creative volition in place, would sign up to composition competitions where you have to write a piece for a given ensemble by a certain date in order to have a deadline on the calendar.
- 8.10** The thing with these competitions is that by definition, you're very unlikely to win. That's just how they're set up. You'll have a large number of entrants, and there will be one prize given out of maybe 300. So the odds are definitely not in your favor. But nevertheless, if you pursue a competition like that, then at least you have a deadline that you're working towards, and that allows you to make concrete progress.
- 8.11** Of course, if you're in a composition program somewhere, then to some extent, that question is going to be taken care of for you because you'll have assignments and projects that you'll have to finish on time. But if you're in a situation where you're not in a program, you're trying to make progress by yourself independently, and there are no particular external pressures, then unfortunately, those pressures are going to have to come from you. What that means is you're going to have to set up a sustainable routine in your creative practice that you will actually follow through on.
- 8.12** A lot of people, when they get started, get very excited, and they imagine that they'll be able to compose three hours a day indefinitely. They'll imagine that they can do

those three hours at any time. Sometimes they'll think, "Oh, I've got the entire day free. I have no appointments, so I'm going to compose from 9:00 in the morning until 9:00 PM." That doesn't work either.

- 8.13** What you quickly discover is that any creative process like that, especially in the early stages when you're starting to plan a piece, you're starting to put together your ideas, is intensely cognitively demanding. It's very difficult work if you're really focused. What I've found through discussions with my students and also through observing my own working habits, is that if I'm doing really difficult creative work, the absolute most that I can sustain that for, except in extreme circumstances, is about three hours a day.
- 8.14** Now, when I say three hours a day, I don't mean three hours where you sit down at the piano, you do 15 minutes, then you get up and go for a walk, then you check your emails, then you're on your phone. I mean where you're really focused with complete intensity and complete presence, three hours is really the upper limit, and that's hard to do. Most people probably can't do three hours.
- 8.15** The reason it's important to know that is it allows you to set up reasonable, realistic expectations about what you're going to get done. Because if you have a vague or a completely unrealistic goal, then you won't meet it, and then you'll become discouraged, and then you'll start to associate the creative process with something burdensome, unpleasant, and daunting, which it doesn't need to be.
- 8.16** One of the experiments you can do, if you're in the early stages of some creative discipline is to work out and be ruthlessly honest with yourself, "How long can I do this for before I start to get really distracted?" You don't want to push it until you become so distracted that it becomes painful. You want to stop before that point. You want to stop when it's going well. What a lot of creative people do is they'll figure out what that limit is for them, and they'll stop when they're red hot; they won't keep going.
- 8.17** The reason for that is when they start the next day, they'll be looking forward to getting back into it. They'll be excited to get back to their desk. They'll be excited to work, and they won't be burnt out. They won't associate it with some unpleasant, painful experience. As counterintuitive as that might sound, it's very useful when you're trying to make something to stop when it's going well rather than to stop when you're dead tired and you're banging away at the thing and it's getting worse and worse and you're getting more and more frustrated. So don't do that.
- 8.18** The other thing is, nearly everybody I've worked with has been at their most productive in the morning. That seems to be a physiological constant. If you're like most people, then you're going to be at your sharpest when you wake up or shortly after breakfast. Now, obviously, not everybody can work this out because we all have multiple constraints to deal with. You might have a job to get to, you might have children to take care of, and you might not be able to compose or pursue your creative work in the morning. But if you can make it work, in my experience, that's the ideal.

- 8.19** People will find remarkable solutions to make that happen. I have a friend in Paris who's now 98 years old, a composer called Betsy Jolas, born in 1996— sorry, born in 1926. She began her career in France at a time when there were no female composers. Really, it was very uncommon. She had children, of course. She had three children of her own and a husband with a busy career. I was impressed to see that she had a very long worklist of impressive pieces dating all the way back to when she was a teenager with basically no interruptions. She's still writing today at 98 years old.
- 8.20** I had to ask her, "How did you manage this, even when you were raising three kids? She told me, "In those days I would get up at 4:00 in the morning. I would get up at 4:00 in the morning and make sure that I had a couple of hours of composition before the kids would wake up." Obviously, it's not reasonable to expect everybody to do that. Maybe you need to have some extraordinary physical capacity in order to do that. Maybe she's physically constituted differently from the rest of us. I don't know. But she made that commitment because at that time, that was the only way that she could maintain some measure of consistency in her output.
- 8.21** It's also partly a question of: How badly do you want to do it? There are a lot of for people who give up when it gets to be inconvenient, which it inevitably will. The reason that it becomes inconvenient is that any long-term pursuit like that, where this is something you would ideally like to do over years, over decades, it's very easy to say, "It's not a big deal if I don't do it today. It's not a big deal if I don't do it this week. It's not a big deal if I take a month off," because you can still pick it up and you can do it later.
- 8.22** There are going to be countless things that are going to be a higher priority than your creative work because the creative work might not pay anything, it might take away from other responsibilities that you have in your life. You might be overwhelmed with responsibilities, and you might be really struggling to find a place in your life to maintain it. That's where the subtle art of negotiation comes into it. You really do have to figure out, "Given my situation, given what I'd like to do, what are my minimum needs in order to sustain a creative practice?"
- 8.23** For some people, you may find that 15 minutes a day is enough. You can take 15 minutes a day, but keep to it religiously, do it every day—if you do something 15 minutes a day for 30 years, you can actually create something monumental. Maybe 15 minutes isn't enough. Maybe you need more than that. But whatever it is, you have to figure out a way that you can integrate it into your life in a way that works and that is iterable. That's where it gets difficult for a lot of people.
- 8.24** What are some of the things that you have to bear in mind if you're going to pursue this successfully? Well, the first is to recognize that you are inevitably going to come against roadblocks. There are going to be days when you don't want to do it. There are going to be days when you're hitting your head against the wall because you can't find

the solution to some rather creative difficulty. Then there are those who say that they suffer from either procrastination or writer's block.

- 8.25** Both of those are problems that you don't need to have. In both cases, I think what's actually happening is that you are dealing with an insufficient number of constraints and you become paralyzed. People who are terrified of the blank page or who say they suffer from writer's block or they have no ideas, simply need to voluntarily adopt more constraints. When you do that, again, paradoxically, it frees you up because it reduces the number of questions you have to deal with when you sit down to work. That is really critical, because if you're sitting there and absolutely anything goes, and there are no external or internal pressures brought to bear upon the creative situation, you won't be able to produce anything. It's not possible.
- 8.26** So you have to impose them. You have to invent constraints of some kind. Now, those constraints can be very simple. It could be as simple as saying, "I'm only going to write something that I can finish in one sitting." That's a really powerful one because then that takes away the whole question of having to sustain interest in a project over months and months and months.
- 8.27** If you're having difficulties with concentration, for example, or maybe in confidence, you're not sure that what you're producing is going to be good enough, then you can adopt that mode of working where whatever you do, when you sit down to work, you can't get up until you've finished something, some finished standalone work. Maybe it's a little short poem. Maybe it's a very short composition. Maybe it's a one-minute piece. But do something that you can finish in one sitting.
- 8.28** Again, if you do that over time, what will happen is you'll build up patterns. You'll find that you can accomplish that task. It becomes easier. You become conditioned to it in the same way that athletes can become conditioned through training. Then it becomes easier, and you can then modify the constraint as you need to, lightly at first, you don't want to go from writing a one-minute piece to writing a two-hour piece, obviously, but you can calibrate the degree of constraint as you go along.
- 8.29** So you need to impose limitations, and you need to design projects for yourself that allow you to make progress as an artist, that stretch your abilities, but that don't put you in a situation that's far beyond what you can handle. I've seen innumerable times young composers coming to me with scores that are way beyond the level of training that they have. They've never studied orchestration. They've never written for anything larger than one or two instruments, and suddenly they're writing a 35-minute long cello concerto. That doesn't work. You can't do that. That would be akin to a first-year architecture student wanting to build a 110-story skyscraper. You have to learn a lot of different skills first before you can take something to take that on.
- 8.30** Now, that doesn't necessarily mean that you need to scale back your artistic ambitions or change anything about your expressive project. It simply means that the vehicle for

your creativity, in that case, is completely disproportionate to the range of skills that you've already acquired. If you're finding that the project is far too difficult and you're getting lost and you can't stick to your original plan, then there's a good chance that you've created a project that simply isn't coherent with your level of skill.

- 8.31** What you need to do in that case is to scale it back until it's something that you can do comfortably, but that gives you just a bit of a push to expand your range of competence, whatever that is. Then if you do that ritualistically and carefully, then you can expand your skills organically and naturally over time. But this is, again, something that is counterintuitive for a lot of people because they get very ambitious and have all these ideas about what they want to do, and they want to move forward quickly. But unfortunately, it doesn't have that effect. It actually slows you down. Start with simple tasks.
- 8.32** Also, when you're sitting down to work, another thing that's very helpful is to conceptualize to yourself verbally what it is that you're about to do. If you have a vague ambition, like, "I want to sit down and be a composer," then it's probably not precise enough for a lot of people to be properly motivated to do it. You want to narrow down the task to something much more specific than that. It can't just be, "I'm going to sit down and write a piece." Maybe if you have decades of experience and you know exactly what you're doing, you can do that, but otherwise, it's not advisable.
- 8.33** What you can instead say is something like—and remember, you have to bargain a little bit with yourself in order to stick to the schedule that you're trying to maintain, because what you want is a situation where you can do this regularly. It can be a part of your life. It can be a sustainable practice as opposed to something you pick up every now and then and then forget about.
- 8.34** What does that look like? Well, you have to narrow down the task. You have to say to yourself, not, "I'm going to sit down and compose," but "I'm going to sit down for 10 minutes and try to put two chords together." Now, that's a really specific thing. "I'm going to find two chords that I really love that sound fantastic. I'm only going to do that for 10 minutes. If I don't find anything in those 10 minutes, that's fine. I'll still stop."
- 8.35** What you're doing there is you're setting yourself a very specific, realistic task, and you're giving yourself a time frame within which to do it. The deal is: if you want to stop after 10 minutes, you can, and you can go do something else. If you sit down—this is the little trick that actually is unbelievably effective. If you sit down, what you'll find is that most of the time the procrastination is only to do with actually getting started. But once you're actually there, sitting down and working, often that will just disappear. The hardest thing is to get yourself to the table or to the instrument.
- 8.36** But by lowering the bar like that and making it so that there's a very low barrier to entry, so to speak. It's the thing where you can't possibly say no because 10 minutes is not a very big commitment; it's not a big deal. So you do it, you start working, and you

might find that it's actually going well. "Maybe I want to do 15 minutes." If you want to, go a little bit longer, you can. But you can also stop.

- 8.37** So you've set yourself a specific task. The idea is by the end of the working session, you'll have two chords. Then the next day, you sit down, or the next time you sit down, if it's later that day, you can come up with another idea. These little plans that you're making or these goals, these little micro-goals that you're making in the creative process, are not straight jackets. They're not meant to force you to do something you don't want. That doesn't even work. It's not realistic. What they are is they're tools to make it likelier that you will do what you want to do and that you'll actually carry it out.
- 8.38** You're not going to stick to any of these plans 100% of the time. That's totally unrealistic. If you're lucky, you'll be able to do it between 60 and 80% of the time. That's if you've really got things together, you're pretty disciplined, and you're pretty careful about how you organize your time. The rest is all the aleatoric insanity that you have to deal with as a result of being alive. But if you can manage to keep your practice going with reasonable consistency, 75-80% of the time, then you're doing extremely well.
- 8.39** Most creative people never get there. That's the thing. They give up, they can't focus, they don't have a place to work, they lose interest, they get tired, they get distracted. Again, in very many cases, it's a result of not having a clear goal, having completely disproportionate ambitions, and not being realistic about what they can do.
- 8.40** It's the same with artists who have ambitions of becoming very famous or very successful. A rock band might have visions of selling out a big theater somewhere, selling out Madison Square Garden or selling out whatever it might be. What they forget is that you don't actually need that to have a sustainable creative practice. You don't need to write huge orchestral pieces to have a sustainable creative practice either. You need to work out what the minimum requirement is for you for that to be viable. What you want is something viable and sustainable. That's the key thing.
- 8.41** If you look at the work of great composers like Chopin or Liszt, there are virtually no pieces in their output that aren't for piano or that involve piano in some respect or not. They were both pianists, and they both wrote substantial quantities of solo piano music. Why did they do that? Well, because they were performers, they were pianists, so they could play their own pieces in private salons. That was enough. They didn't have to write orchestral pieces. They didn't have to write operas. They didn't have to write anything else in order to express what they had to say. They found a medium for their particular form of expression that was adequate and that was sustainable and that allowed them to get their work out into the world.
- 8.42** That's a form of pragmatism that artists also have to consider: What is the medium that actually is realistic that allows you to get your ideas out? If you're thinking, "It has to be a 45-minute orchestral piece that costs \$100,000 to perform," then you're going to find

that very challenging. Not that those opportunities don't exist, but they're far harder to come by, and it's not even the key thing. Ultimately, it doesn't matter.

- 8.43** The piano works of Chopin are some of the most influential of the 19th century, they continue to be played all over the world. Nobody is saying, "Oh, but he only wrote piano music. He's not a serious composer." Just as you can find innumerable 19th-century composers who churned out operas that are never played, completely forgotten. It doesn't matter. What you need then is a practice that's sustainable, and you need a medium that allows you to get out the work in the way that you want to.
- 8.44** Of course, that depends on a whole host of factors that will inevitably vary from composer to composer. I've seen this up close through the people that I work with. Some of them have lots of friends who are performers, and they can write pieces for their friends to play. That's brilliant. That solves the problem. If you know a talented flutist or a talented pianist or two or three players in your midst, you write pieces for them, and you arrange concerts. They could be private concerts or whatever. You could just get together and play through them and record them.
- 8.45** If you don't know anybody who plays an instrument and you don't play yourself, then there's always a solution. You can work with synthesizers, you can work with virtual instruments, you can work with MIDI. These technologies are virtually free now anyway. It costs almost nothing to get in on these different tools. The key thing is to not obsess over the medium that you're going to be working with. Make sustainable goals for yourself and find something that is coherent with what you want to do on whatever scale that is. It's very, very powerful to do that over decades rather than to get stuck, have a burst of activity for a couple of years, and then you stop, which is very, very common.
- 8.46** Some of the great artists of the 20th century, their entire bodies of work are limited to one medium or one process or one form that was coherent with their life, with their life circumstances. I have a friend here in London, a painter, very interesting character. He's a people person. He's very extroverted. He likes to be around people all the time when he's working, and he likes to work in public places. He doesn't like to be cooped up in his flat.
- 8.47** He's developed a method of working where he takes these pieces of Japanese rice paper and folds them up and uses pigment pens. He can work this way on a clipboard anywhere. He gets his materials and puts them in a backpack and goes into central London. He has all these places where there's people that he knows, and he's around with them, and he can sit and do his work. He's been doing this for decades now, and he's produced an amazing body of paintings, working on them one panel at a time—the panel that fits on the clipboard, and then he'll turn it over and unfold it and do the next panel. This is his working method. It's brilliant because that suits his personality. It suits the nature of the work he's doing, and it's sustainable, and he enjoys it, and so he can keep doing it. That's really crucial.

- 8.48** You need to systematically sit down and figure out, "If I'm not doing this to my fullest potential, I'm getting stuck creatively. It's not going well," sit down and analyze your situation: What are the roadblocks? What's the thing that's not working? Then you try to remove them. It could be, "I don't have a place set up to work." It could be that you don't have a specific idea of what you're aiming for. It could be all sorts of things, but you have to sit down and have a conversation with yourself about what's not working and try to remove those roadblocks bit by bit. There's almost always a solution.
- 8.49** The American photographer Vivian Maier is another example that I like to cite because she was somebody who was also a very eccentric person who was a nanny professionally for her entire life in Chicago. She would live with different families for years at a time and take care of their kids. But she often had the afternoons free to go out. She would take a Rolleiflex camera with her and go wandering through the streets of Chicago.
- 8.50** She did that thousands of times for years, creating this completely secret body of work that she didn't show to anybody. Often, she didn't even have prints made from the negatives. In her attic flats, she had these piles and piles of negatives, tens of thousands of them that she accumulated over the years. Through that process, she became one of the great photographers of the 20th century. She just got better at it by continuously doing it every day. Remember, she was a nanny, but this was a way that she could have a creative practice that was sustainable. She got better and better and better.
- 8.51** Fortunately, this trove of work was discovered almost by accident by a photography historian, and now it's exhibited all over the world. She's considered to be one of the greats. It's an amazing story. Now, obviously, that's not typical. You can't assume that that's going to happen. The odds of it happening are almost infinitesimal. But the point of the story is not that she became well known or the work eventually found its audience, but that whatever your circumstances, whatever the particular set of constraints that you have to deal with is, there's a way to fit a creative practice into your life if you really want to.
- 8.52** I'm going to take questions because this is a subject where you have to get into individual particularities. It can be difficult to speak in general terms about how to set up a practice as an artist. But I'd like to hear from anybody who has any specific questions or any particular difficulties that they may be facing.
- 8.53** Firstly, I'd just like to say thank you for the content of these lectures. Some of the points, especially, were very relatable, very personal. I found the topic of constraints the most beneficial, something I'm going to be taking away from this and thinking about a lot. But the question I have that I keep coming back to throughout these lectures is: any students that you've had across the years that have maybe had deficit, neurodivergent, any suggestions, advice that you would have given to them, any anecdotes that you want to share, anything like that? Thank you.

- 8.54** Yeah, absolutely. I have, in fact, had a number of students who have been diagnosed with ADHD and other disorders who have come to me. Very often, these are very talented individuals who are deeply frustrated because they can't finish anything. They have problems with concentration. They have problems sustaining their interest or their attention in a project.
- 8.55** There's a few things that I've had them do. One of them is you have to design projects of a scope that is manageable. It's probably not—depending on your circumstances and the nature of your particular personality, you might find that you can't necessarily write a 20-minute piece; that's not going to work for you. You have to figure out how far can you stretch it? How far can you stretch that?
- 8.56** It might be the case that you need to work within forms that are set ahead of time so that you have a much more specific goal when you're sitting down to write, and you might need to work on pieces that are shorter. Having ultra-specific goals really helps in that case. Again, and this comes back to what I said earlier, saying to yourself, "I'm going to sit down and compose," is not going to do it in that case. It could be something as simple as, "I'm going to sit down and look at this sketch page for five minutes." You're not even asking yourself necessarily to sit and write.
- 8.57** Or it could be, maybe, "I need to clean up my desk so that I don't feel horrified every time I sit down. Maybe my desk is such a catastrophe that I can't even contemplate sitting there and writing." So it might be something like that. You take it step by step. Then it might be, "What can I manage to do?" It might be, "Can I sit down and write one bar? Can I sit down and write five notes? Can I sit down and play the guitar for three minutes?" Then you check with yourself and you see, "How was that? Was that unpleasant? Was it uncomfortable? Was it okay? Maybe I could stretch it to five."
- 8.58** You start with the smallest goal that would still allow you to do something, and then you try to expand it beyond that. I've had success with my students in getting them to do things like this. I've often given them very, very specific, constraining exercises as well to do as composers. One of the exercises that my first composition teacher gave me— and I'll probably never forgive him for this because it was really uncomfortable, but I was really excited about being a composer, and I wanted to do all kinds of things and had all sorts of ideas. He said, "Okay, great. I want you to write me a piece for snare drum." I was like, "A piece for snare drum? What am I going to do with one snare drum?"
- 8.59** I went home and wrote a little snare drum piece, brought it to him. He tore it apart and told me 50 things that were wrong with it. He said, "Do another one." I went back, wrote another snare drum piece, tried to make it a little more interesting. There were problems with form, and it was too repetitive, et cetera. I tried to write one that was a little better, and the same thing happened. This went on for weeks. I was having to write all these stupid snare drum pieces.

- 8.60** Finally, I understood the point of the exercise, which was, first of all, to make sure that I kept writing. It was an exercise that was so tightly constrained and so specific that you would just do it because it wasn't particularly difficult. Secondly, to recognize that even within that ridiculous constraint, there was still the possibility of expression, and there was still the possibility of doing something meaningful, and you had to take it seriously as an exercise.
- 8.61** Designing those little projects, checking with yourself on a regular basis to see what the edge of your concentration is, trying to work within that, and then seeing if you can push it a little bit. If you're finding yourself resisting it too much, then you scale it back.
- 8.62** When you started composing, did you have to let go of any ambitious ideas?
- 8.63** Yeah, quite a few. My own history as a composer was maybe unusual in the sense that I started out being very interested in all kinds of things. I was very open to lots of different musical influences, and I heard a very broad range of Classical music and pop music and experimental music, even. My uncle back in Canada was a noise musician, which is a microgenre that I enjoyed at the time and still enjoy, actually, although it's not for everybody. So I was interested in quite a broad range of things.
- 8.64** When I first started wanting to make music, I wasn't particularly trained. I didn't have a lot of skills, and I was trying to figure out what I could do because I had the desire to express and the desire to make things, but not the skill to make things. And so what happened was my first compositions were, of course, as you would expect, completely incompetent in every respect. I didn't know what I was doing at all. But I had a music teacher in Toronto who was sympathetic with what I was trying to do, and he would give me sheets of music staff paper at the end of the classes, and I would take them home and try to write something.
- 8.65** There are times where you're not really sure what you're doing, but you try to make it look nice on the page and things like that. My early efforts were not the output of a prodigy by any means. What I ended up doing was I had some friends in grade 6, grade 5, grade 6, who played instruments in the high school—not the high school, this is not high school, but in the school orchestra, the school band, whatever. They could play a little bit of trombone, or a little bit of flute, or a little bit of cello or whatever.
- 8.66** I'd get them to come over after school, and I'd write little things for them to play. I wrote little trios like that and songs, and I'd get them to play little parts, and I'd record them on a cassette deck and stuff like that. It was actually great in a way. It was a great musical education in a sense, because I had all this time after school to try things out. My friends were these kids who could play an instrument a little bit. They would come over, help me out and play them.
- 8.67** The work that resulted from that is not deathless, let's say that; it wasn't going to win any awards, but I learned a lot—I learned a lot very quickly by doing that. In terms of ambitions, I think that I was reasonably good at making projects that were specific

enough and on a scale that was appropriate so that I wasn't confronted with a totally unmanageable situation.

8.68 One of the things that I found quite often with my own students is that it's very difficult to learn multiple complex skills at the same time. Usually, for most people, that's not something I would advise. For example, if you want to write an orchestral piece, then you have to have a command of form. You have to have a command of harmony. You have to understand texture. You have to understand the orchestra. You have to understand harmony and all of these different things. There's a very, very broad range of skills involved in that.

8.69 If you've worked on all of those things and built them up bit by bit, then you can attempt a small orchestral piece. But if you've never done that before and you try to master 50 hard problems at the same time, then it's not going to work. It's impossible. You're going to give up after a couple of pages because you're realizing that you've made a complete mess and you don't know how to fix it. It would be like if you had no experience of architecture, electrical engineering, plumbing, and you tried to build a house from scratch. Those are all very different separate technical skills that you have to master. You can't learn them all at the same time.

8.70 In my experience, I started writing really short little pieces, writing songs, and then trying to record them. Then I would iterate that. I wrote about 200 songs over maybe a three- or four-year period, and recorded them with little cassette decks. Later I got, at the time—this was the mid-1990s. Tascam had a range of PortaStudios, which were little recordable multi-track cassette recorders, so I got one of those and could do little multi-track experiments where I would record a line and then add another line to it, another line.

8.71 You could do a thing called bouncing, where you could record three tracks and mix them to the fourth track and then erase them and add a bunch of other tracks to them. This was all before digital technologies were widespread enough and cheap enough that you could do all of this on a—you could basically do that on a cell phone now without any difficulty.

8.72 I had enough constraints to work within also that were externally imposed. That also helped. That's how I did it. It was a very long process of producing very large amounts of work all the time—lots and lots and lots of pieces. Also putting myself in as many musical situations as possible. Then gradually, you learn, you build up an arsenal of skills. That's how it was for me.

8.73 Thank you as well for all these wonderful lectures. I have more questions. How can a composition be wrong? You mentioned the professor said the piece you wrote earlier, it was wrong. Is it not the case that it's a creation. It can be just an expression, your opinion. How can it possibly be wrong? Second question: What do you think of AI? Can it play a role in musical composition? Can it be a tool? Is it a threat? What do you think

in general? What's the situation in that domain? The third one: Can you play a song with one note? Can you compose a song just with one note?

- 8.74** I'll start with the last one and work backwards. Yeah, there are pieces with one note. There's an Italian composer called Giacinto Scelsi, who was a member of the Italian nobility, which conferred upon him a certain independence, and he had the means to pursue his creative work in whatever way he wanted. Scelsi, as it's spelled S-C-E-L-S-I, was something of a mystic. He developed this very interesting and idiosyncratic process of composition.
- 8.75** He acquired this early keyboard instrument, an electronic keyboard instrument called an ondioline, which you could control. There was a little lever under the keyboard that you could press against with your knee to control different variables in the sound. A very cool instrument. He would get himself into this hypnotic state and sit down and improvise, and then he would record what he had come up with.
- 8.76** One of the things the instrument would allow you to do would be to bend the pitch up and down a little bit. You could have these little subtle glissando effects in the sound. He wrote a series of pieces called Quattro pezzi—Four Pieces—on a single note. These are actually really beautiful works. They are orchestral pieces where he takes a single tone and then continuously changes the timbre through orchestration and the dynamics and the texture. Then the pitch will bend a little bit up and a little bit down, and then it'll go back to where it started again.
- 8.77** It's amazing because similarly to the Wagner example, Das Rheingold, that we listened to earlier, the Quattro pezzi, in a sense, have no material at all. It's reduced to one note sustained over one or two minutes. Also, he does it four times. There's four of these pieces, and each one is different. Each one tells a different story and has different sonic characteristics. You can find that piece on YouTube. It's very easy to find it. Just look for Giacinto Scelsi, Quattro pezzi, for orchestra.
- 8.78** Your second question pertained to AI. A lot of people are worried about AI. A lot of artists are concerned about it. I'm not concerned about AI for a couple of reasons. Now, there's a lot of people that got upset when there was a photograph that was posted on Twitter of the Pope wearing a white puffy jacket, and it was an AI photograph. If you were not looking very carefully and not really paying attention, not thinking very much, you would see this ridiculous photo of the Pope wearing this Balenciaga jacket and think, "That's odd. Why would he do that?" But of course, it was AI.
- 8.79** With stuff like that, I just think—some photographers were getting upset. They were thinking, "Is this what's going to happen? Nobody's going to need photographers anymore because you can generate these images and nobody can tell the difference?" But I mean, what worries me is not that AI can make these pictures, but the fact that people are so undiscerning that they can't tell the difference between the real thing and a really third-rate pastiche. That's the same in the artistic disciplines.

- 8.80** I've interacted a lot with ChatGPT, Grok, and other large language models and tried different things out. You can have a lot of fun with them. You can give it instructions like, "Produce a poem in the style of Allen Ginsberg that tells the story of whatever, the Hindenburg blimp disaster," or something. It'll spin it out for you. It can do stuff like that. But for anything critical, it's nowhere near good enough. There's no perceivable threat that I can see at the current state that that technology is at.
- 8.81** For things that are easily generalizable, where it's very easy to copy something and make a convincing simulacrum of it, then AI can do that. But that's the genres that have the least amount of invention and the least amount of individual personality in them. For example, if you want to make some lofi beats, or if you want to make a bit of passable elevator music, then AI can do that easily. But that's not an indication so much that AI is encroaching upon human territory so much as it is a reflection that those genres contain very little that is individual or personal.
- 8.82** I think what might happen is that as AI becomes more and more capable with things that can be modeled, then the things that only we can do will become more and more clear. I think there's going to be increasingly a gap. I anticipate—and maybe I'm wrong, maybe I'm overly naive, but I anticipate that artists who can do something truly individual and distinctive are going to be more in demand than ever because I cannot conceive of how a machine could replace that process. So I'm not worried about AI.
- 8.83** As far as music goes, there have been various attempts to program AIs to spit out bits of counterpoint and harmonize to chorales and things like this. Everything I've heard has been catastrophically bad, not even slightly usable. It doesn't mean that it won't get better. But the concern, again, with me is more that we could lose as a society, the discernment and the knowledge, the capacity for appreciation to recognize the difference between a great work and a third-rate pastiche.
- 8.84** So your first question was: How can a piece of music be wrong? I talked about how I showed these pieces to my professor, and he would get me to tear them up and start over again and so on. Well, it's not that the piece is wrong exactly. There are certain technical aspects of writing music that if you don't understand them, the piece simply won't function.
- 8.85** If you're trying to write multiple superimposed lines and you don't understand how counterpoint works, then you're going to write a mess, and it's not going to work as a piece. Likewise, if you don't understand about scansion, about how words line up with stresses in poetry or in lyrics, then you're going to end up writing words for a song that simply don't fit the rhythm of the song. It's going to be very awkward. It's going to be very awkward to sing. It's not going to sound very good.
- 8.86** So there are technical aspects of music where you can say objectively that there's a problem. There's a problem with this arrangement. You've crowded a whole bunch of instruments together in a narrow frequency range. There's no bass. There's no high-

end. The instruments that are there are on top of each other and you can't tell them apart, and it's crowded, and it's not a properly laid-out texture. Those are technical things. You have to work through those because if the technique isn't allowing you to say what you want to say, then the expression just isn't going to happen.

- 8.87** Now, that doesn't mean that you have to be a technical virtuoso in order to do anything. It simply means that the technique, whatever it is, your skill, has to be adequate to what you're doing. There are so many examples of artists who had very limited ranges of things that they could do, but who used those capacities to the utmost.
- 8.88** Now, there are many singers who haven't—Leonard Cohen is a good example. He can barely sing at all. He has a range of less than an octave. But Leonard Cohen is a great singer. Those songs are powerful because he knew what he could do and he knew what he couldn't do, and he worked within what he could do. He wasn't going to get up and sing a coloratura aria; he couldn't do that. But he could do some things.
- 8.89** There was a French painter, one of the surrealist painters called Yves Tanguy, a Breton painter, who was self-taught. I was completely bewitched with his work when I was a teenager. I had a friend who showed me his paintings. I was really struck by them. He put together a technique that was limited, but that was 100% appropriate to his expressive project, and it worked.
- 8.90** He had a very limited pictorial vocabulary. He painted these strange landscapes where the sea, the ground, and the sky were blended into each other. You couldn't tell which was which. They looked like these post-apocalyptic landscapes. They're very strange paintings. That was basically what his subject matter was. He stuck with that for decades and refined it and got better at it and produced a range of work. But he had the technique that was necessary for that, and he figured out what he needed to do in order to be able to make those paintings.
- 8.91** It's not that you have to be able to do everything under the sun, that you have to be a total virtuoso, but try to figure out what you want to do, and then you work out what tools you need, what techniques you need, so that the technical side just gets out of the way, and you can concentrate on your work.